Configuration Summary

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FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-04-22 11:06:05
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lon_size........................ 128
  target_lat_size........................ 64
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics.pdf
  output_dir............................. data/processed/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 5
rollout_epoch_end...................... 50
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed/sst_final.nc
checkpoint_dir......................... checkpoints/
best_model_path........................ checkpoints/best_model.pt
training_log_path...................... results/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints/best_model.pt
  input_data_path........................ data/processed/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
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val_predictions_path..................... results/predictions_val.nc
test_predictions_path.................... results/predictions_test.nc
processed_data_path...................... data/processed/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed/std_params.nc
trend_params_path........................ data/processed/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
output_dir............................... results/
report_file.............................. results/analysis_report.txt
metrics_csv.............................. results/metrics_by_lead_step.csv
figures_dir.............................. results/figures/
analysis_pdf............................. results/forecast_analysis.pdf
generate_plots........................... True
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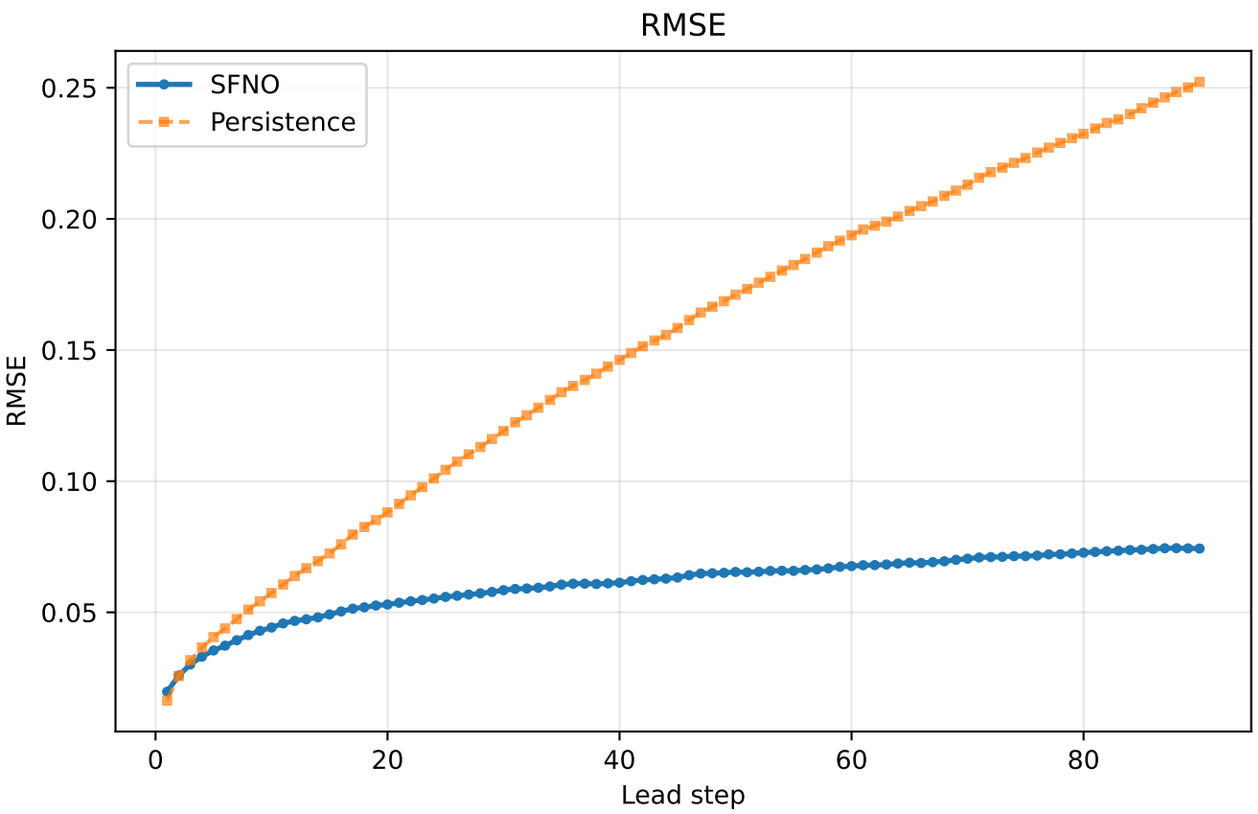
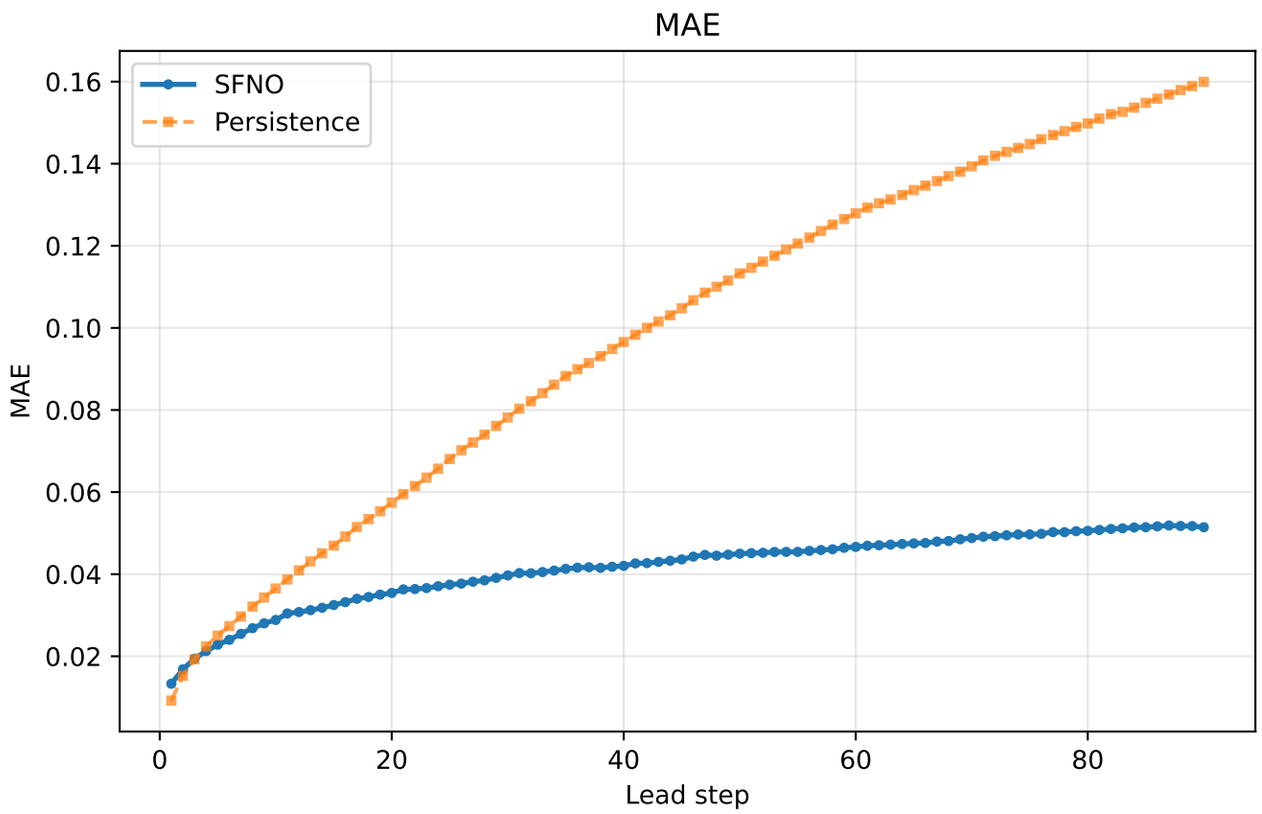
Metrics by Lead Step — Validation

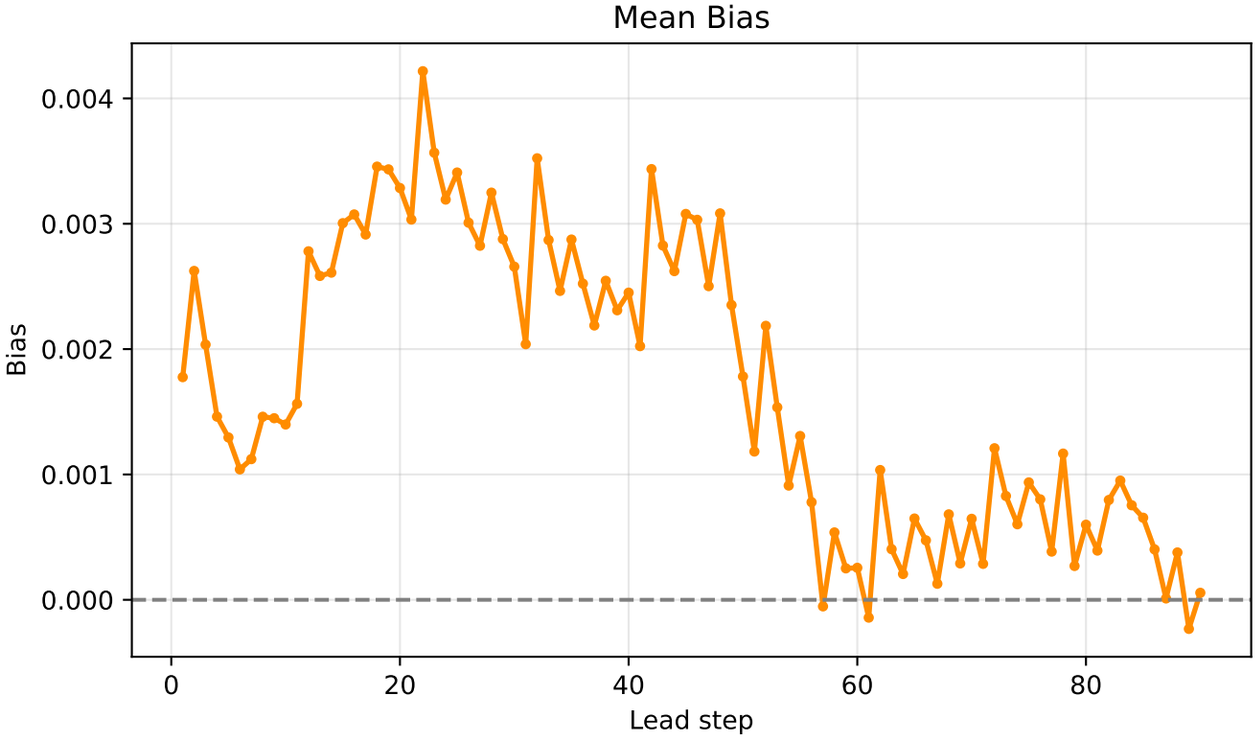
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0197	0.0164	-20.6%	0.0133	0.0018	0.9453	0.9592	0.9996
2	0.0259	0.0257	-0.6%	0.0168	0.0026	0.8990	0.8991	0.9993
3	0.0301	0.0318	5.2%	0.0194	0.0020	0.8577	0.8481	0.9991
4	0.0331	0.0366	9.5%	0.0213	0.0015	0.8248	0.8008	0.9989
5	0.0355	0.0406	12.6%	0.0228	0.0013	0.7945	0.7574	0.9988
6	0.0373	0.0439	15.0%	0.0240	0.0010	0.7690	0.7202	0.9986
7	0.0394	0.0474	17.0%	0.0255	0.0011	0.7429	0.6815	0.9985
8	0.0413	0.0510	19.0%	0.0268	0.0015	0.7175	0.6425	0.9983
9	0.0430	0.0542	20.7%	0.0280	0.0014	0.6930	0.6072	0.9982
10	0.0442	0.0574	22.9%	0.0288	0.0014	0.6713	0.5705	0.9981
11	0.0458	0.0606	24.5%	0.0304	0.0016	0.6482	0.5345	0.9980
12	0.0467	0.0639	26.8%	0.0308	0.0028	0.6280	0.4984	0.9979
13	0.0473	0.0668	29.2%	0.0312	0.0026	0.6091	0.4642	0.9978
14	0.0481	0.0695	30.9%	0.0318	0.0026	0.5903	0.4363	0.9978
15	0.0492	0.0725	32.1%	0.0325	0.0030	0.5712	0.4112	0.9976
16	0.0503	0.0759	33.7%	0.0332	0.0031	0.5521	0.3812	0.9975
17	0.0514	0.0797	35.5%	0.0340	0.0029	0.5331	0.3482	0.9974
18	0.0519	0.0825	37.1%	0.0344	0.0035	0.5139	0.3242	0.9974
19	0.0526	0.0852	38.3%	0.0350	0.0034	0.4954	0.3058	0.9973
20	0.0530	0.0881	39.9%	0.0354	0.0033	0.4788	0.2821	0.9973
21	0.0537	0.0913	41.3%	0.0363	0.0030	0.4697	0.2589	0.9972
22	0.0542	0.0946	42.7%	0.0364	0.0042	0.4583	0.2377	0.9971
23	0.0547	0.0978	44.1%	0.0366	0.0036	0.4457	0.2157	0.9971
24	0.0553	0.1011	45.3%	0.0371	0.0032	0.4310	0.1932	0.9970
25	0.0559	0.1043	46.4%	0.0374	0.0034	0.4155	0.1725	0.9970
26	0.0563	0.1075	47.6%	0.0377	0.0030	0.4028	0.1551	0.9969
27	0.0568	0.1103	48.5%	0.0382	0.0028	0.3876	0.1418	0.9969
28	0.0572	0.1130	49.4%	0.0385	0.0032	0.3733	0.1313	0.9968
29	0.0578	0.1161	50.2%	0.0391	0.0029	0.3582	0.1164	0.9968
30	0.0584	0.1191	50.9%	0.0397	0.0027	0.3420	0.1038	0.9967
31	0.0589	0.1225	51.9%	0.0403	0.0020	0.3368	0.0889	0.9966
32	0.0591	0.1251	52.8%	0.0402	0.0035	0.3303	0.0821	0.9966
33	0.0594	0.1280	53.6%	0.0405	0.0029	0.3211	0.0694	0.9966
34	0.0599	0.1310	54.3%	0.0409	0.0025	0.3136	0.0582	0.9965
35	0.0606	0.1339	54.8%	0.0413	0.0029	0.3051	0.0464	0.9964
36	0.0609	0.1363	55.3%	0.0416	0.0025	0.2964	0.0400	0.9964
37	0.0609	0.1386	56.0%	0.0417	0.0022	0.2919	0.0378	0.9964
38	0.0608	0.1410	56.9%	0.0415	0.0025	0.2844	0.0334	0.9964
39	0.0610	0.1437	57.5%	0.0418	0.0023	0.2775	0.0265	0.9964
40	0.0613	0.1462	58.1%	0.0420	0.0025	0.2664	0.0227	0.9963
41	0.0619	0.1489	58.4%	0.0426	0.0020	0.2538	0.0178	0.9963
42	0.0623	0.1514	58.8%	0.0427	0.0034	0.2455	0.0118	0.9962
43	0.0626	0.1536	59.3%	0.0430	0.0028	0.2334	0.0084	0.9962
44	0.0628	0.1558	59.7%	0.0433	0.0026	0.2245	0.0082	0.9961
45	0.0633	0.1584	60.1%	0.0436	0.0031	0.2127	-0.0016	0.9961
46	0.0642	0.1614	60.3%	0.0443	0.0030	0.1960	-0.0150	0.9960
47	0.0648	0.1643	60.6%	0.0447	0.0025	0.1857	-0.0262	0.9959
48	0.0649	0.1665	61.0%	0.0445	0.0031	0.1801	-0.0296	0.9959
49	0.0651	0.1686	61.4%	0.0448	0.0024	0.1745	-0.0316	0.9959
50	0.0654	0.1711	61.8%	0.0450	0.0018	0.1700	-0.0359	0.9958
51	0.0653	0.1733	62.3%	0.0451	0.0012	0.1639	-0.0392	0.9958
52	0.0655	0.1757	62.7%	0.0452	0.0022	0.1561	-0.0454	0.9958
53	0.0658	0.1779	63.0%	0.0454	0.0015	0.1492	-0.0488	0.9958
54	0.0659	0.1803	63.5%	0.0454	0.0009	0.1569	-0.0525	0.9958
55	0.0658	0.1824	63.9%	0.0454	0.0013	0.1552	-0.0540	0.9958
56	0.0661	0.1847	64.2%	0.0457	0.0008	0.1479	-0.0585	0.9957
57	0.0664	0.1871	64.5%	0.0459	-0.0001	0.1480	-0.0631	0.9957
58	0.0668	0.1896	64.8%	0.0461	0.0005	0.1412	-0.0696	0.9957
59	0.0673	0.1917	64.9%	0.0464	0.0003	0.1330	-0.0708	0.9956
60	0.0676	0.1938	65.1%	0.0466	0.0003	0.1297	-0.0716	0.9955
61	0.0680	0.1959	65.3%	0.0469	-0.0001	0.1268	-0.0755	0.9955
62	0.0680	0.1974	65.5%	0.0470	0.0010	0.1190	-0.0678	0.9955
63	0.0682	0.1989	65.7%	0.0472	0.0004	0.1170	-0.0586	0.9955
64	0.0686	0.2009	65.9%	0.0474	0.0002	0.1146	-0.0567	0.9954
65	0.0689	0.2030	66.1%	0.0475	0.0006	0.1077	-0.0588	0.9954
66	0.0688	0.2049	66.4%	0.0476	0.0005	0.1046	-0.0583	0.9954
67	0.0691	0.2067	66.6%	0.0480	0.0001	0.1023	-0.0556	0.9953
68	0.0695	0.2088	66.7%	0.0481	0.0007	0.0980	-0.0542	0.9953
69	0.0700	0.2108	66.8%	0.0485	0.0003	0.0926	-0.0508	0.9952
70	0.0704	0.2131	66.9%	0.0488	0.0006	0.0858	-0.0529	0.9952
71	0.0709	0.2157	67.1%	0.0491	0.0003	0.0800	-0.0598	0.9951
72	0.0711	0.2178	67.4%	0.0493	0.0012	0.0747	-0.0609	0.9951
73	0.0712	0.2196	67.6%	0.0495	0.0008	0.0746	-0.0564	0.9951
74	0.0714	0.2214	67.7%	0.0497	0.0006	0.0758	-0.0508	0.9950
75	0.0714	0.2232	68.0%	0.0497	0.0009	0.0712	-0.0483	0.9950
76	0.0716	0.2253	68.2%	0.0498	0.0008	0.0644	-0.0507	0.9950
77	0.0721	0.2272	68.3%	0.0502	0.0004	0.0582	-0.0482	0.9949
78	0.0721	0.2290	68.5%	0.0502	0.0012	0.0501	-0.0477	0.9949
79	0.0724	0.2308	68.6%	0.0505	0.0003	0.0487	-0.0476	0.9949
80	0.0727	0.2324	68.7%	0.0506	0.0006	0.0437	-0.0445	0.9948
81	0.0730	0.2345	68.9%	0.0508	0.0004	0.0433	-0.0448	0.9948
82	0.0733	0.2366	69.0%	0.0510	0.0008	0.0409	-0.0443	0.9948
83	0.0735	0.2380	69.1%	0.0512	0.0010	0.0355	-0.0426	0.9947
84	0.0737	0.2399	69.3%	0.0514	0.0008	0.0337	-0.0439	0.9947
85	0.0739	0.2422	69.5%	0.0514	0.0007	0.0376	-0.0456	0.9947
86	0.0742	0.2443	69.6%	0.0517	0.0004	0.0379	-0.0445	0.9946
87	0.0744	0.2464	69.8%	0.0519	0.0000	0.0407	-0.0446	0.9946
88	0.0745	0.2484	70.0%	0.0517	0.0004	0.0393	-0.0447	0.9946
89	0.0744	0.2501	70.3%	0.0517	-0.0002	0.0428	-0.0431	0.9946
90	0.0743	0.2522	70.5%	0.0514	0.0001	0.0446	-0.0441	0.9946

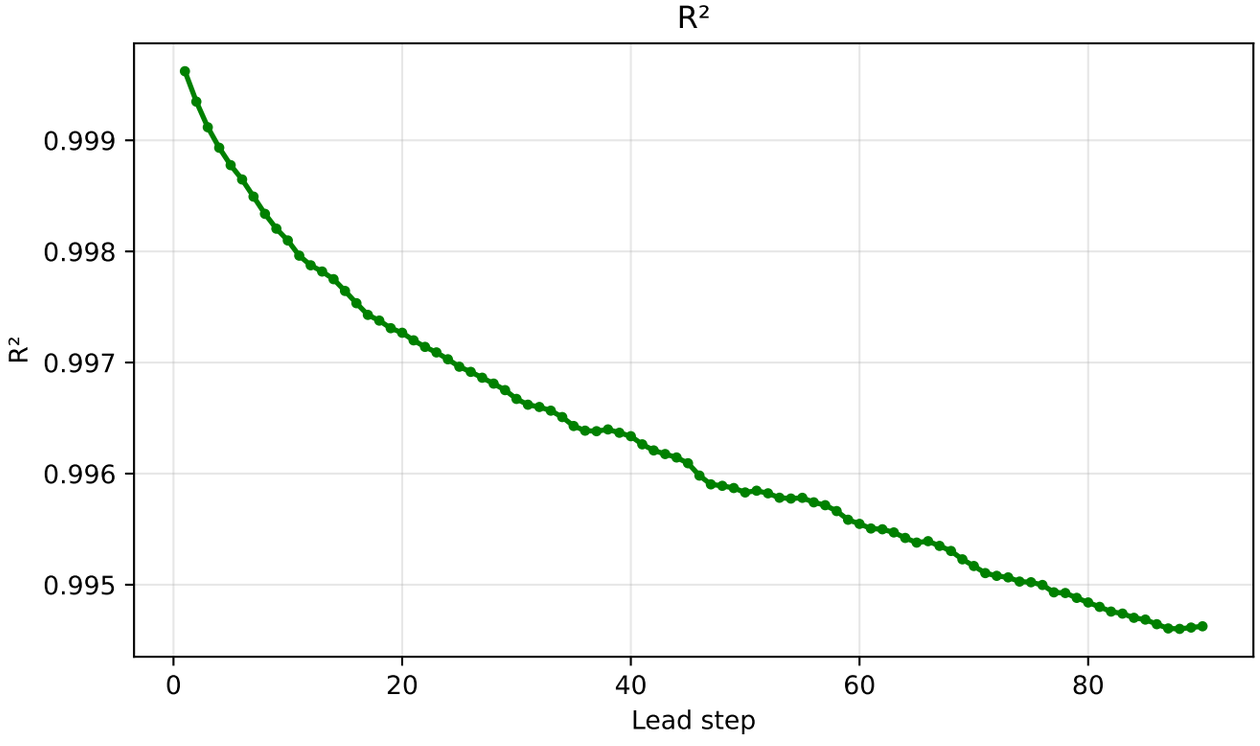
Forecast Skill vs Lead Step — Validation





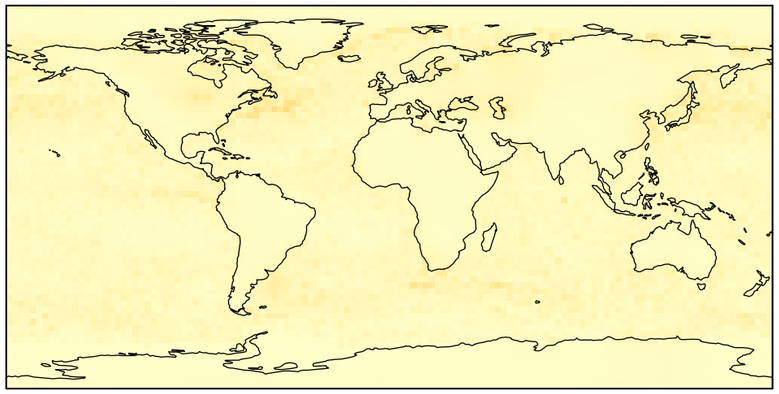
Bias and R² — Validation



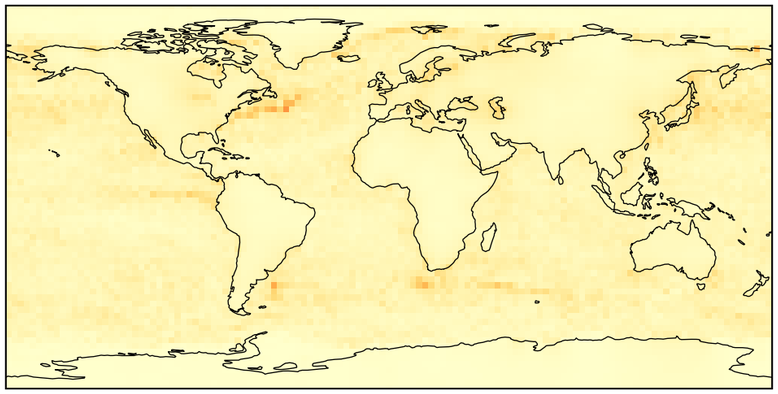


Spatial RMSE (all grid) — Validation

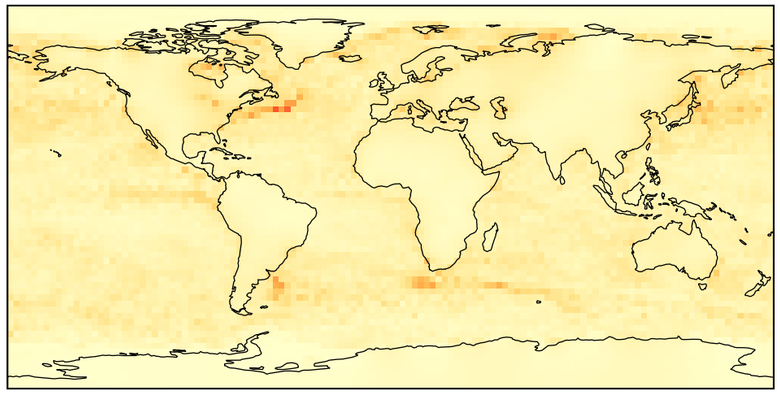
Step 1



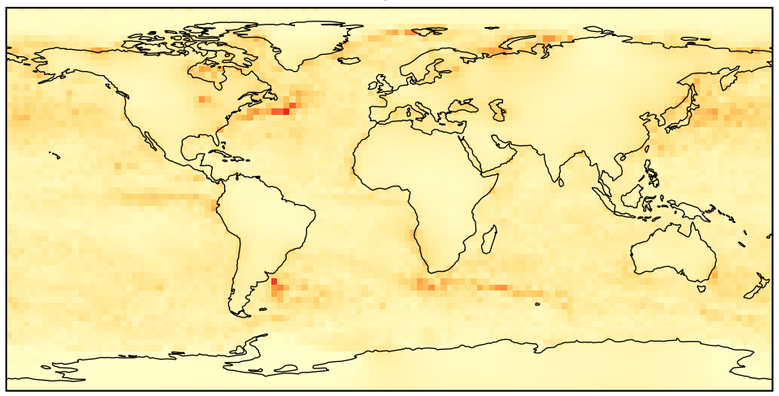
Step 3



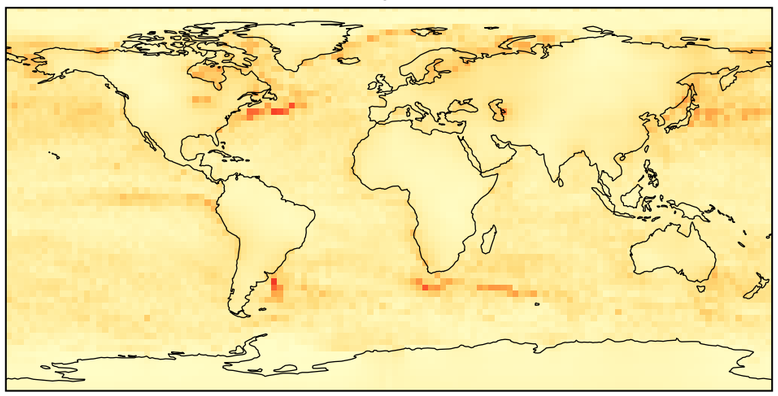
Step 7



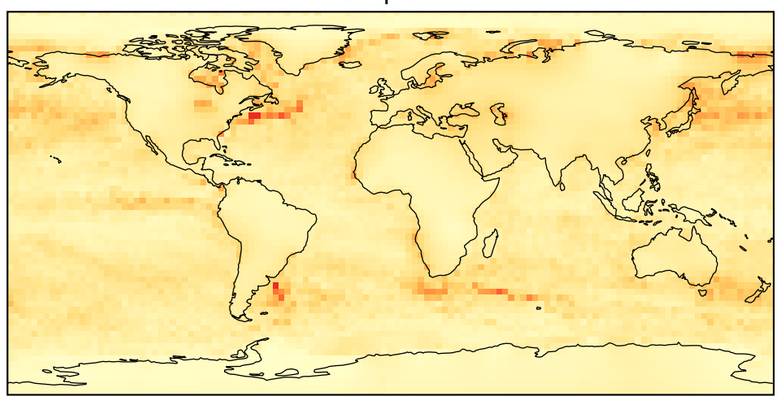
Step 14



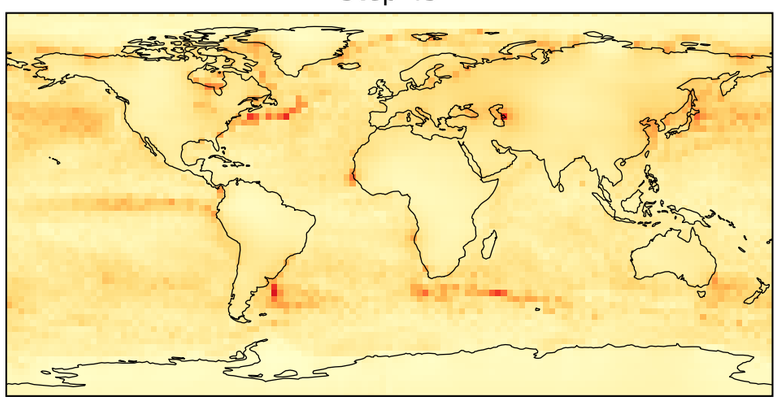
Step 21



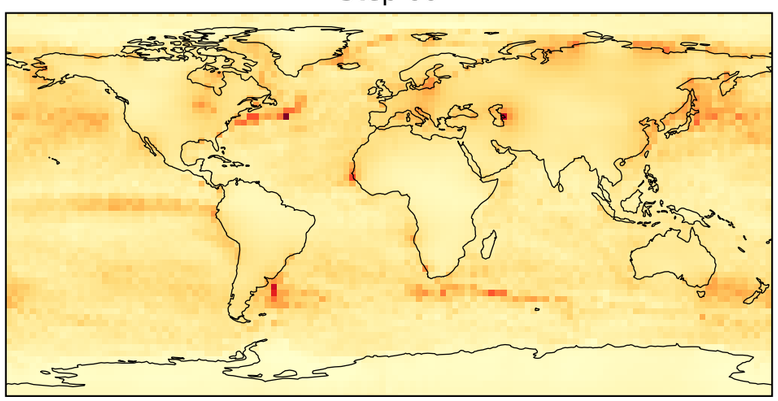
Step 30



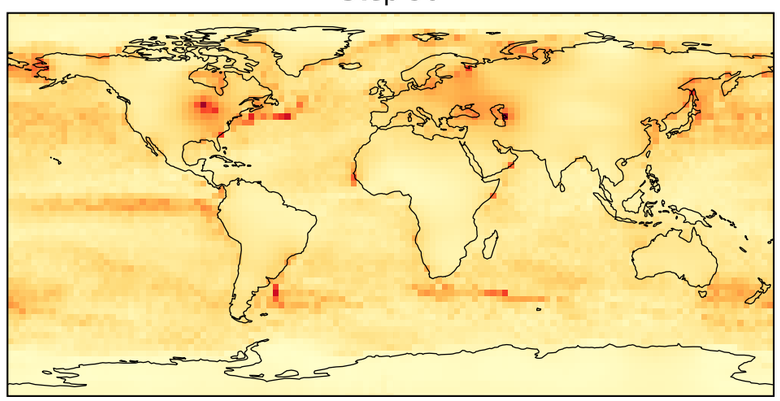
Step 45

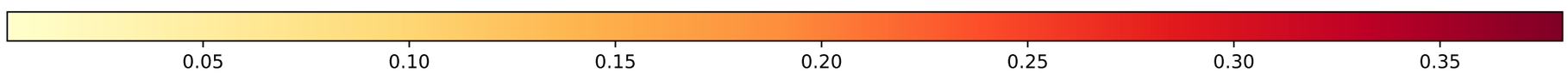


Step 60



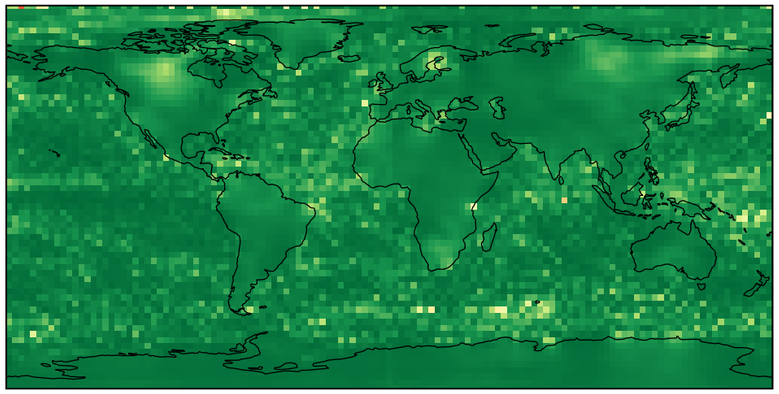
Step 90



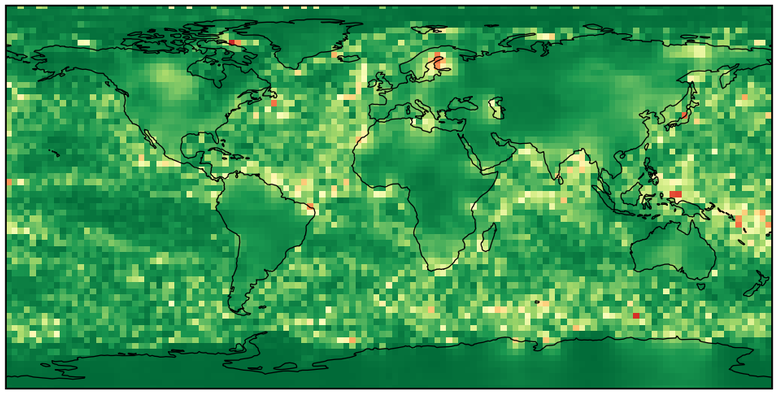


Spatial ACC (all grid) — Validation

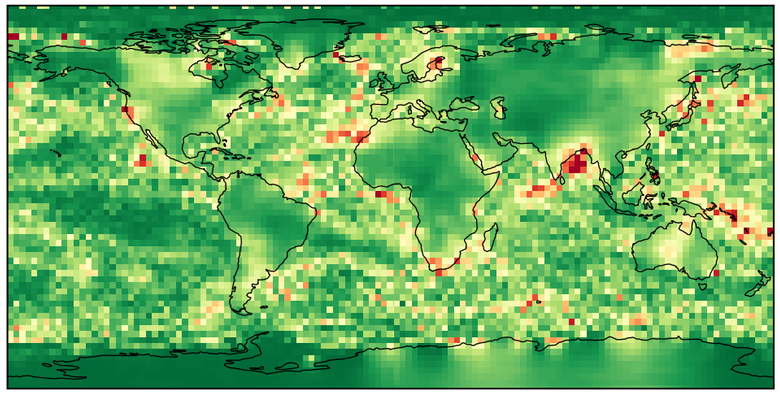
Step 1



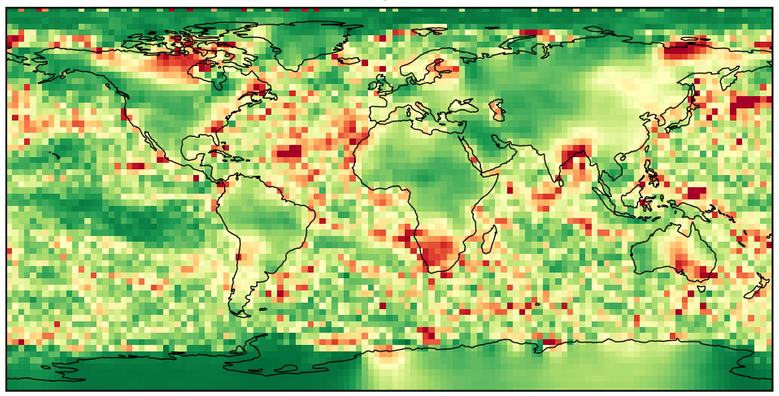
Step 3



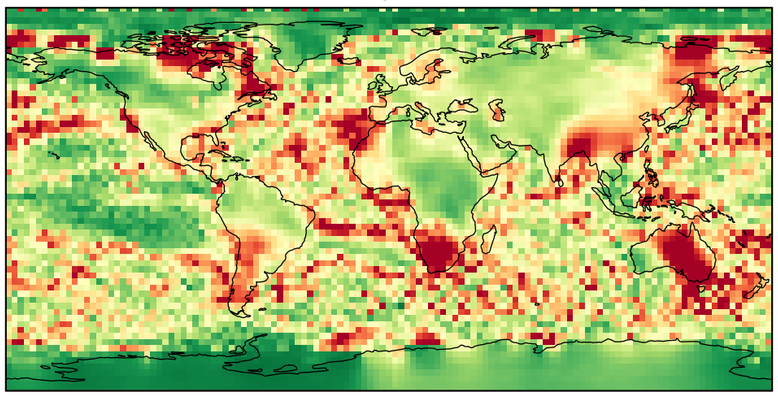
Step 7



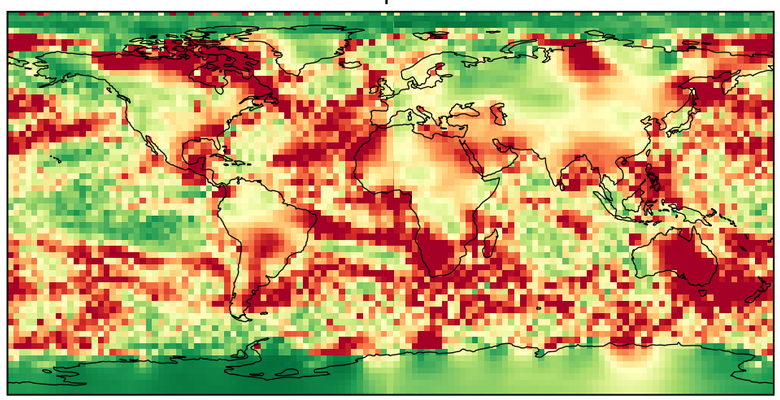
Step 14



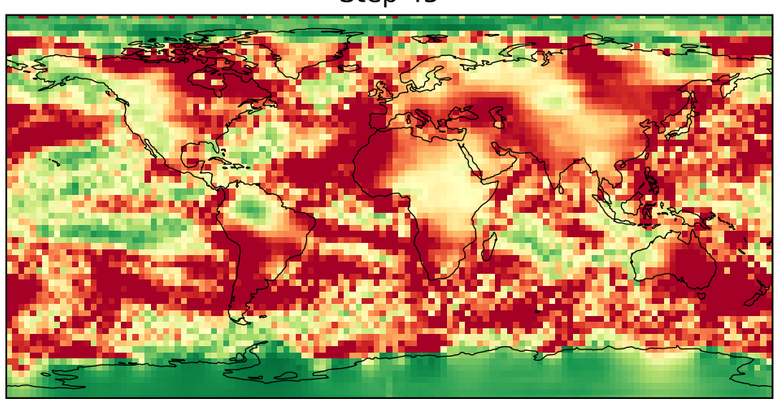
Step 21



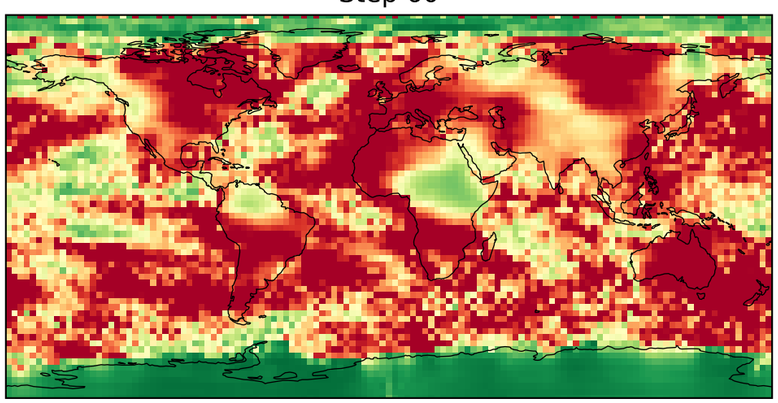
Step 30



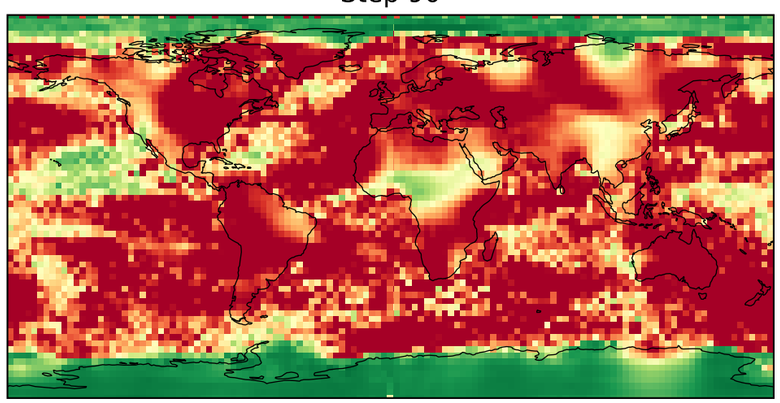
Step 45

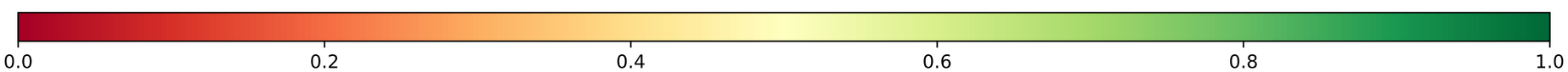


Step 60



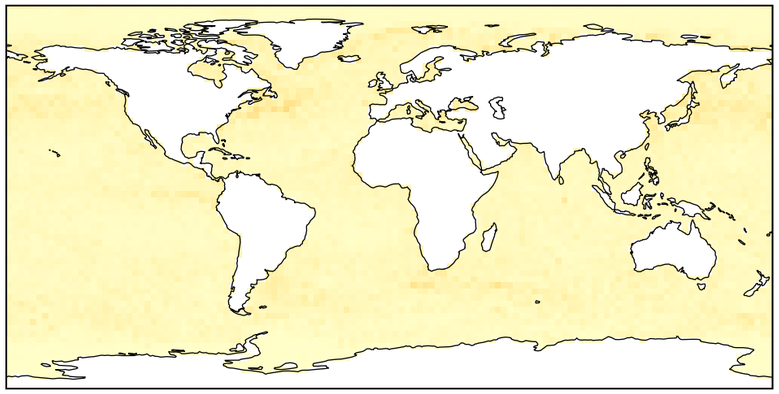
Step 90



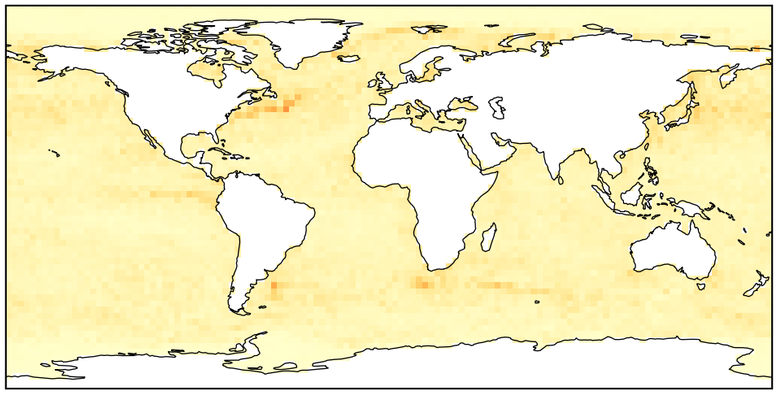


Spatial RMSE (ocean only) — Validation

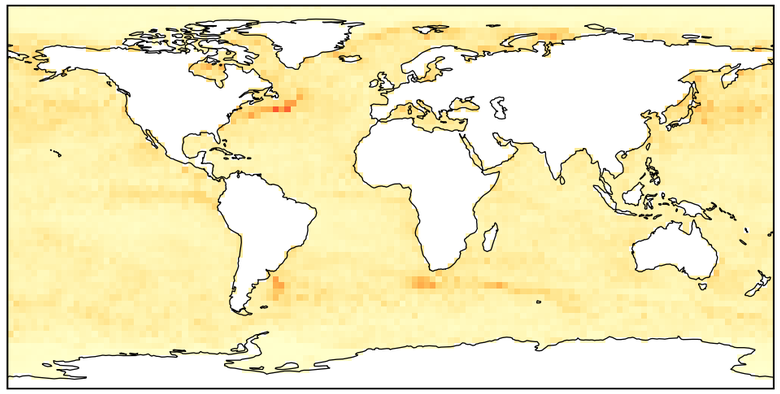
Step 1



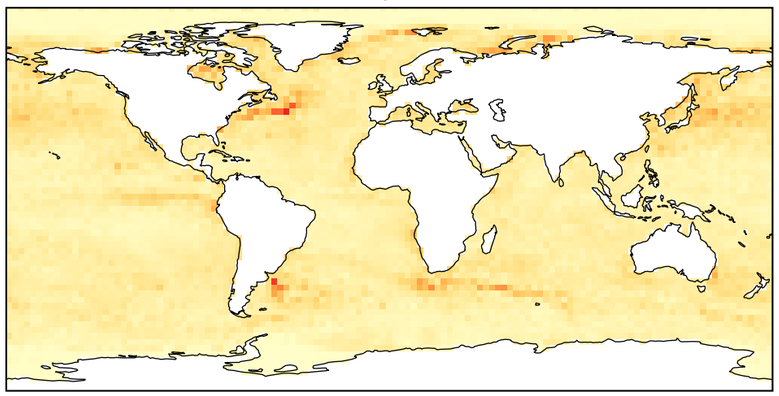
Step 3



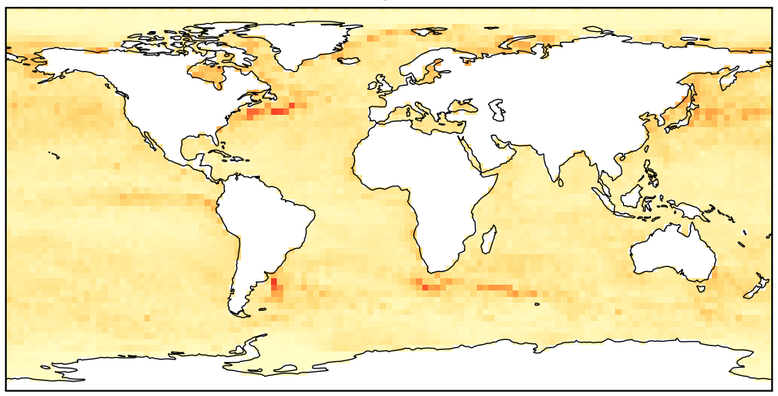
Step 7



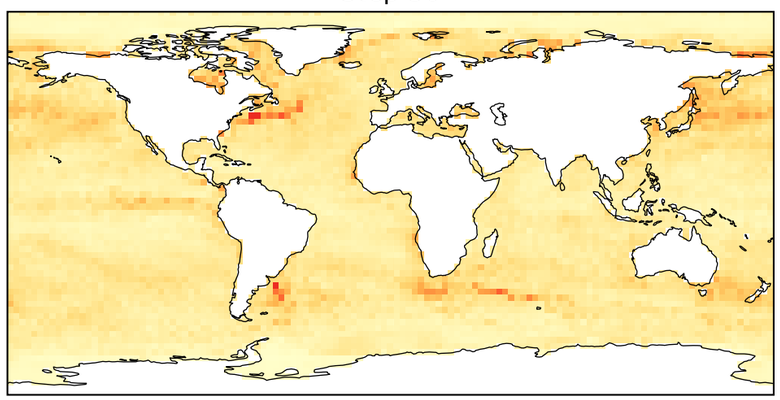
Step 14



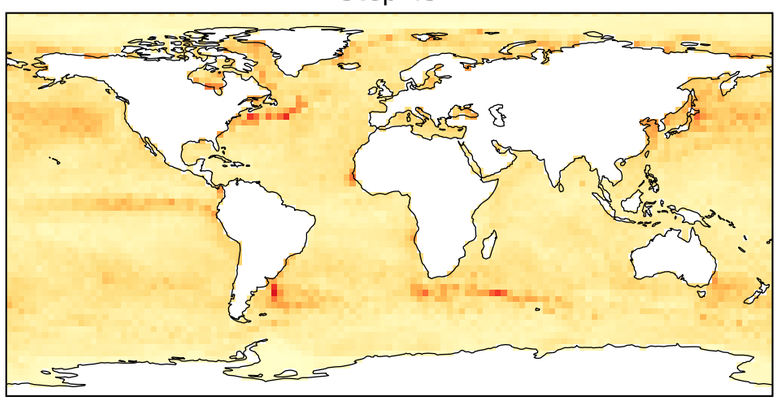
Step 21



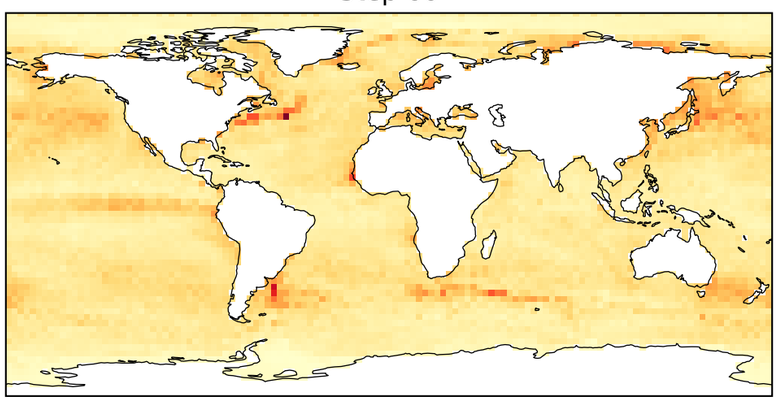
Step 30



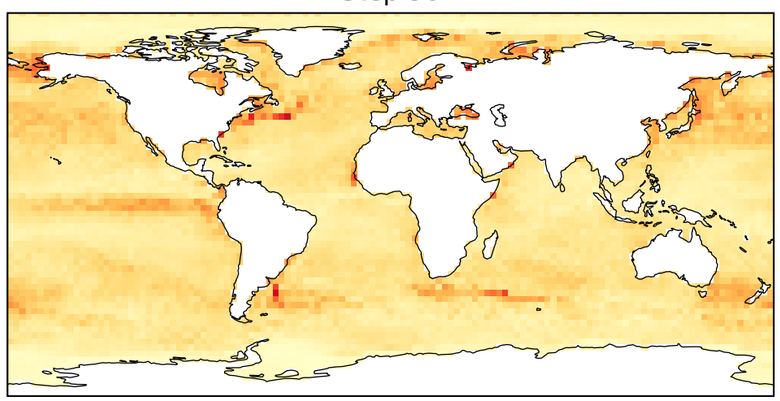
Step 45

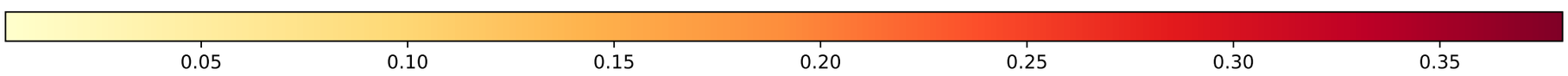


Step 60



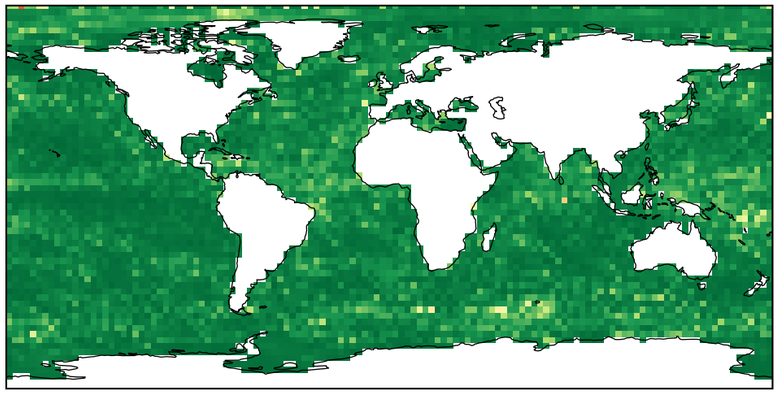
Step 90



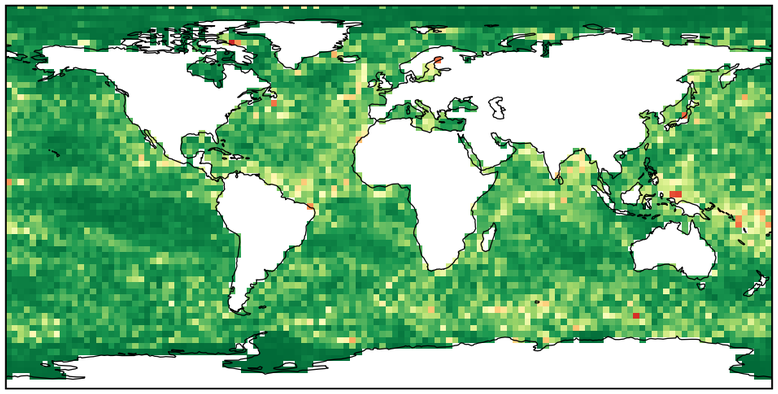


Spatial ACC (ocean only) — Validation

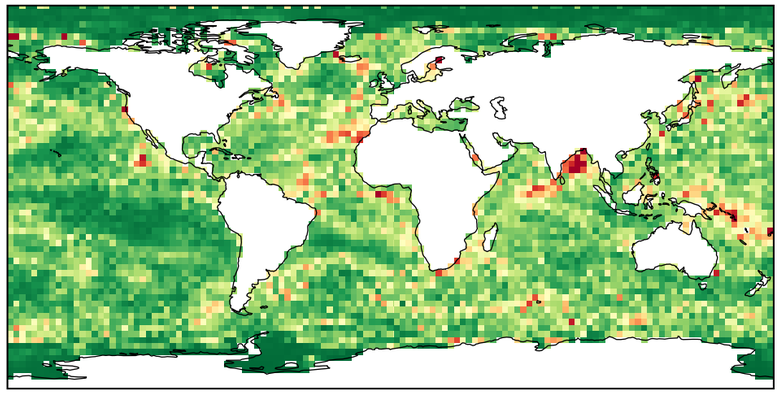
Step 1



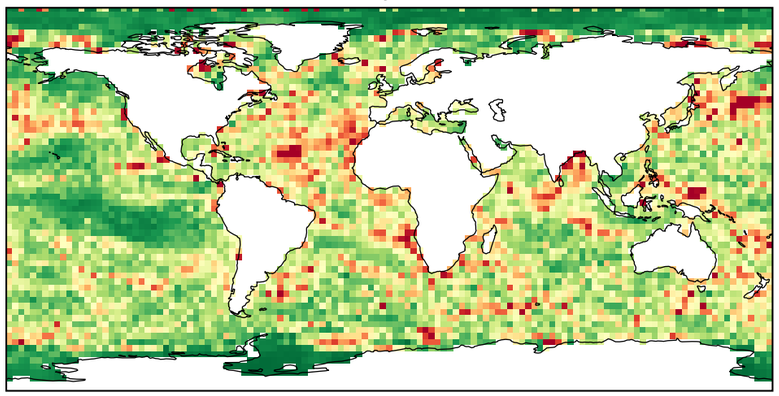
Step 3



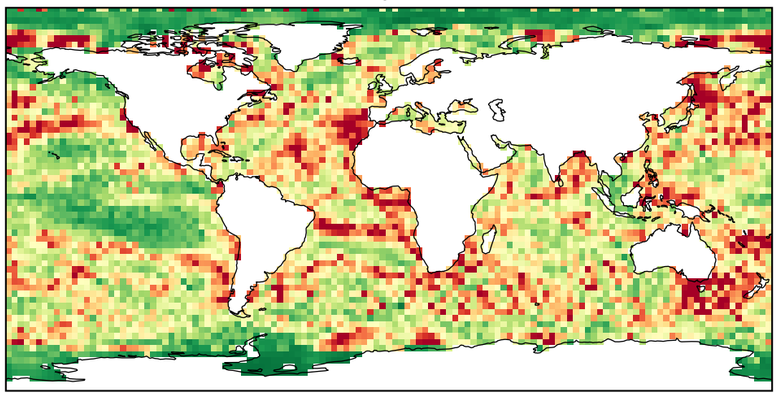
Step 7



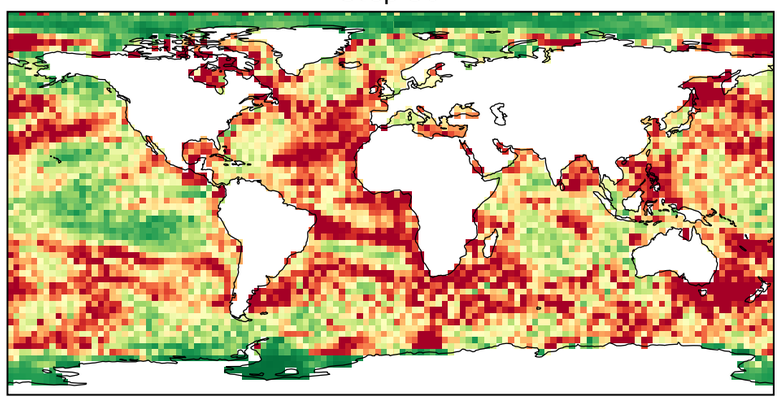
Step 14



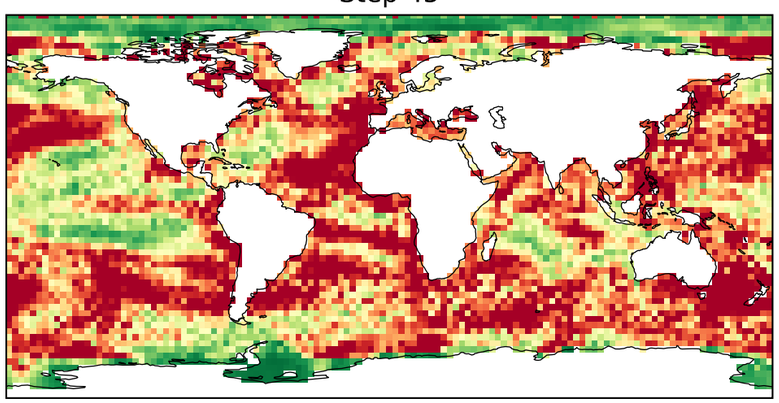
Step 21



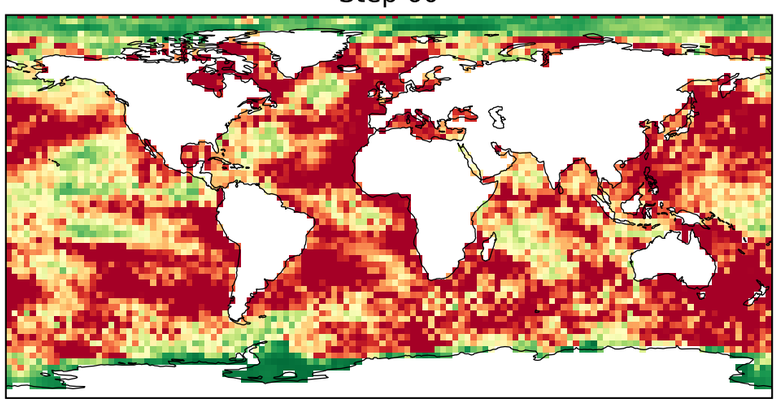
Step 30



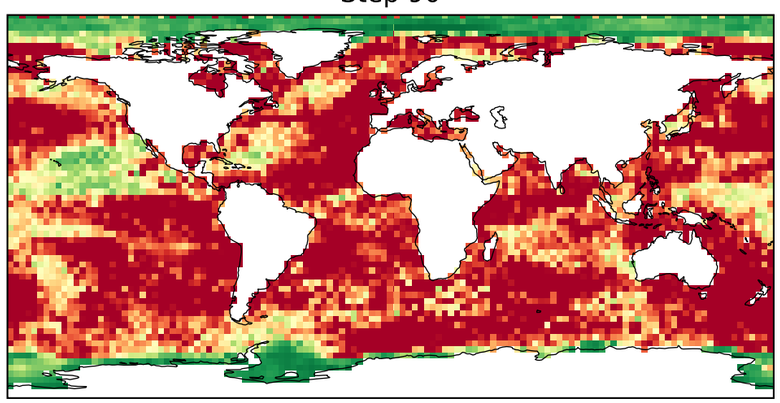
Step 45

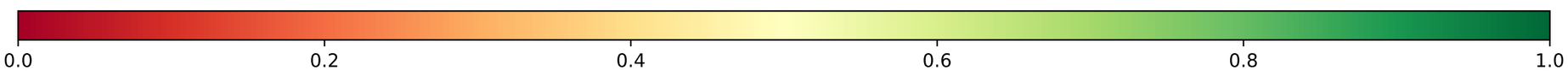


Step 60



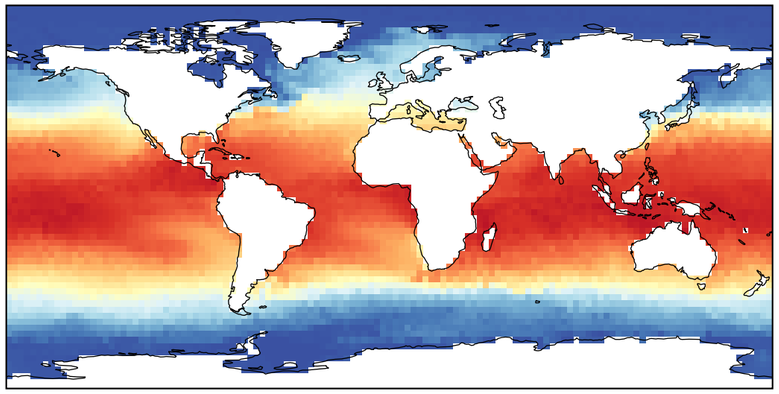
Step 90



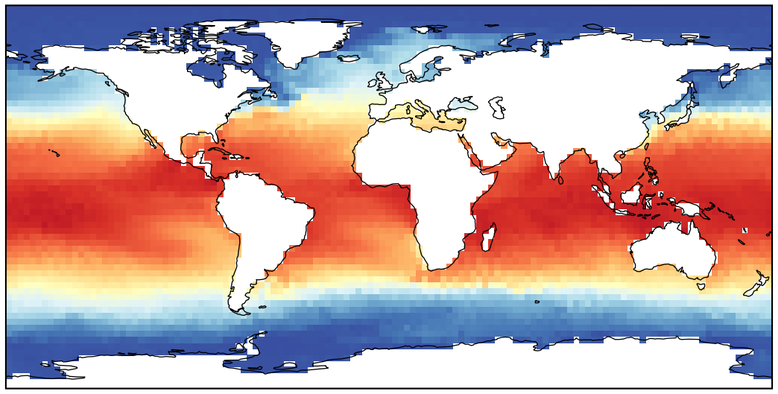


Chain init: 2016-01-11 — Validation (°C)

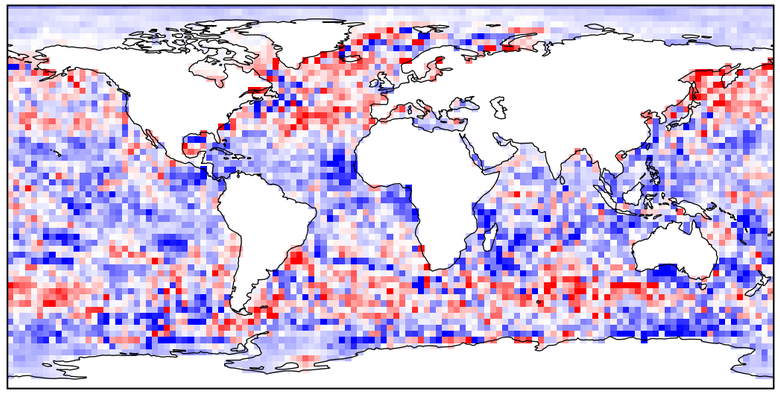
Actual — Step 1 (°C)



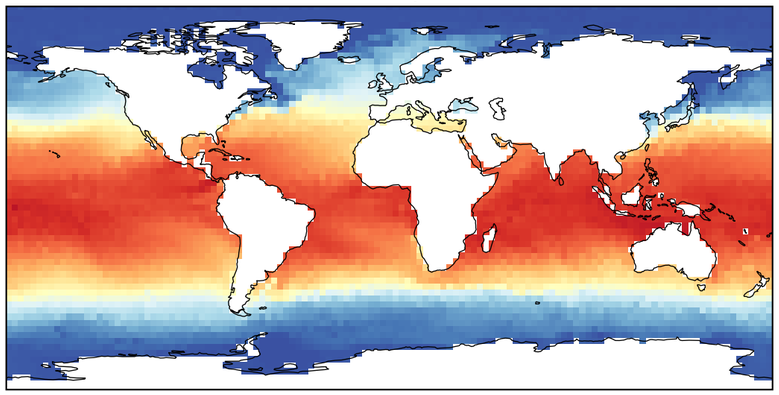
SFNO — Step 1 (°C)



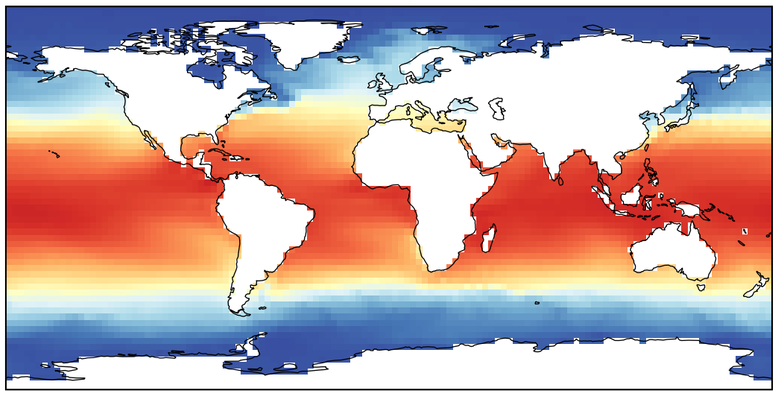
Error — Step 1 (°C)



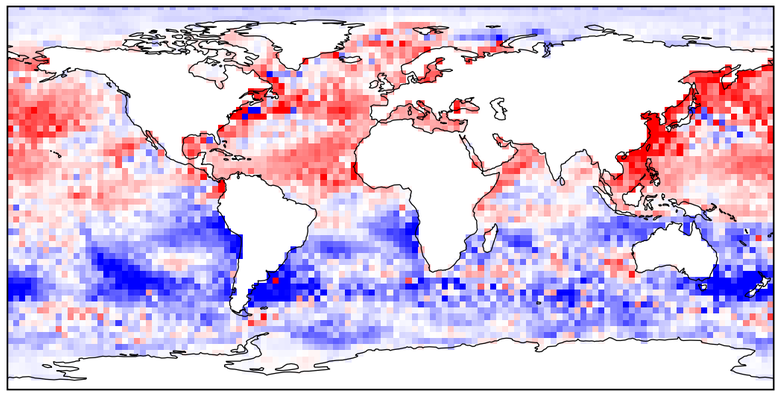
Actual — Step 46 (°C)



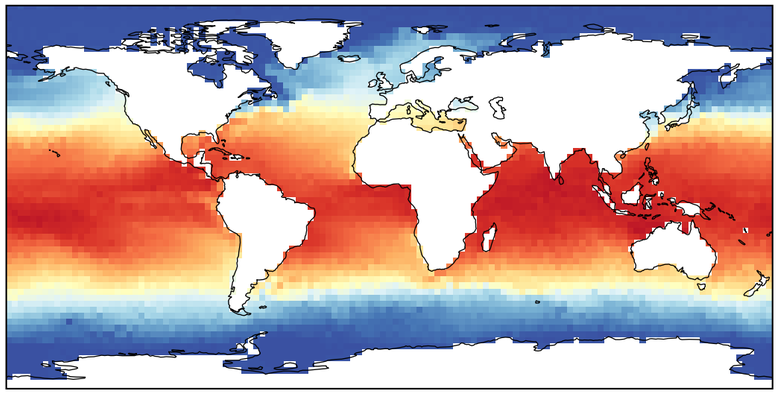
SFNO — Step 46 (°C)



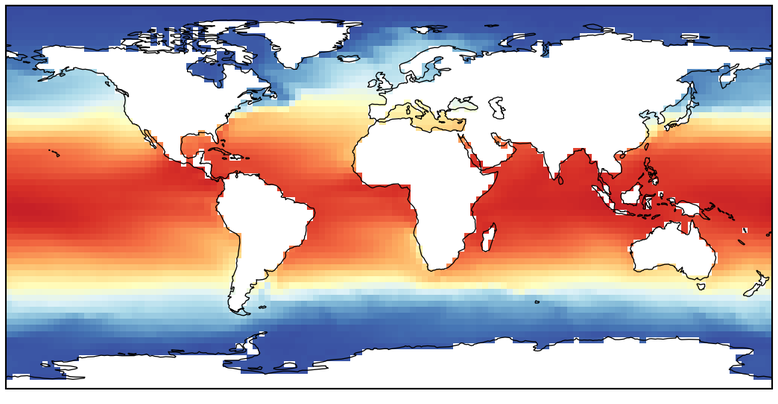
Error — Step 46 (°C)



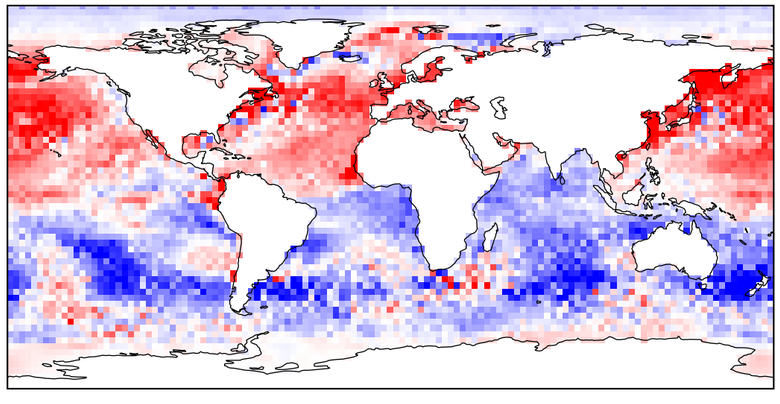
Actual — Step 90 (°C)

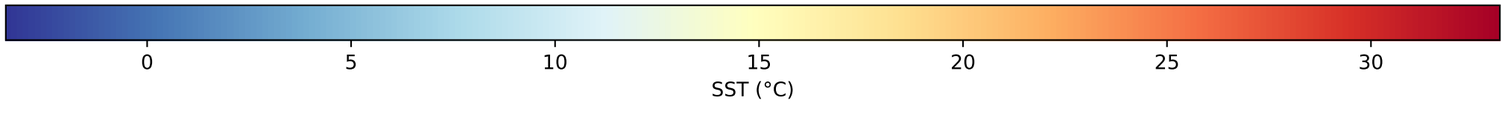


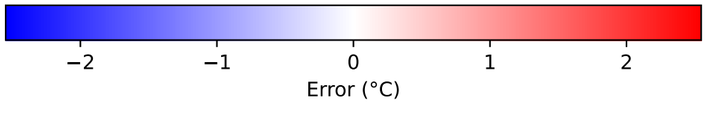
SFNO — Step 90 (°C)



Error — Step 90 (°C)

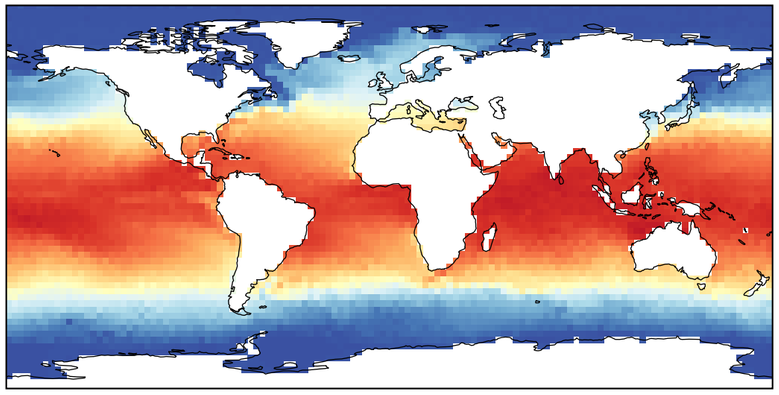




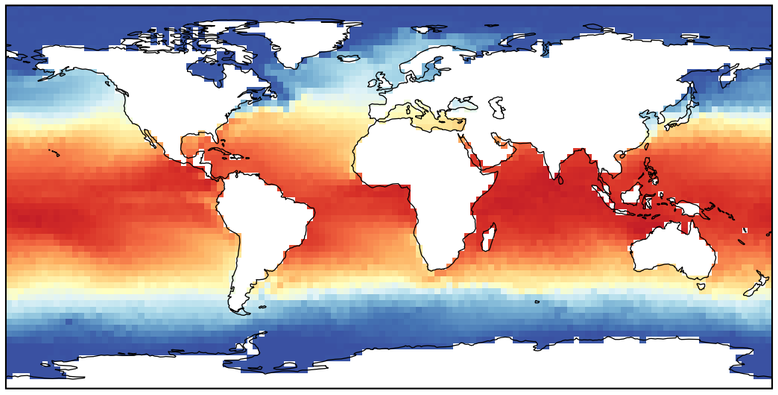


Chain init: 2016-04-10 — Validation (°C)

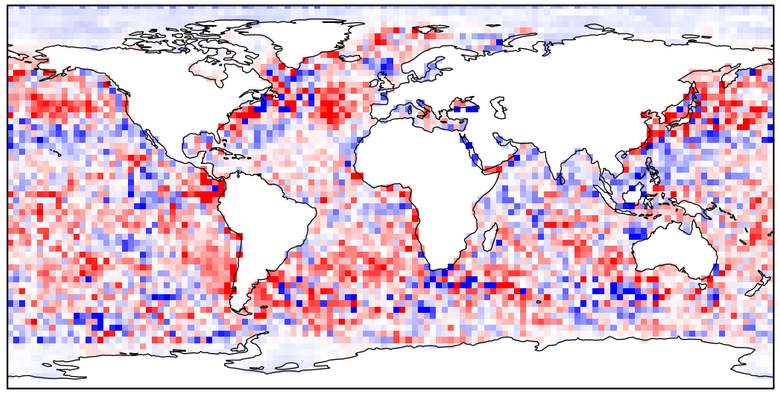
Actual — Step 1 (°C)



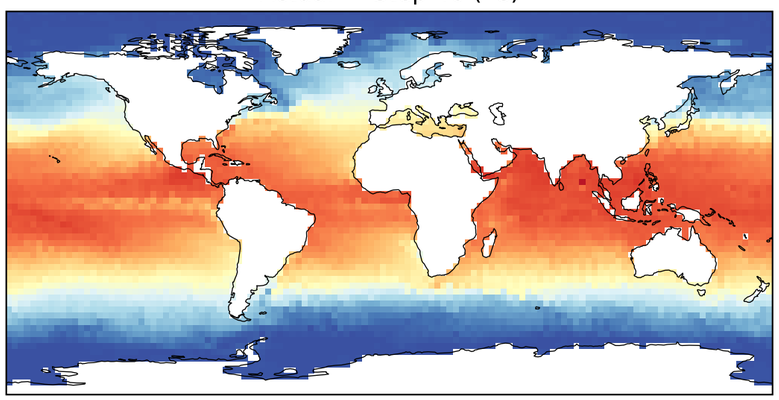
SFNO — Step 1 (°C)



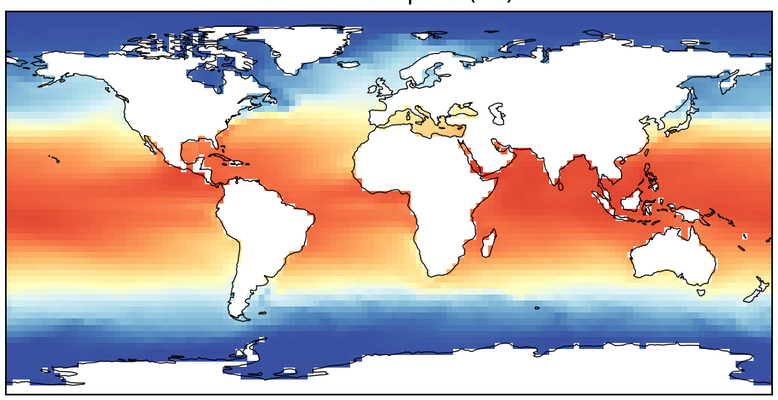
Error — Step 1 (°C)



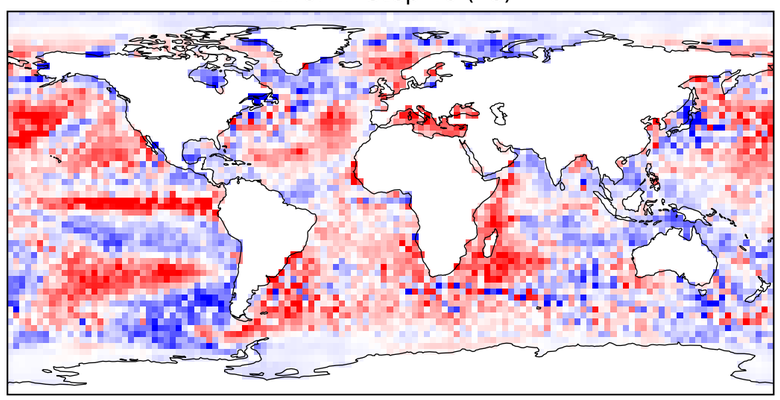
Actual — Step 46 (°C)



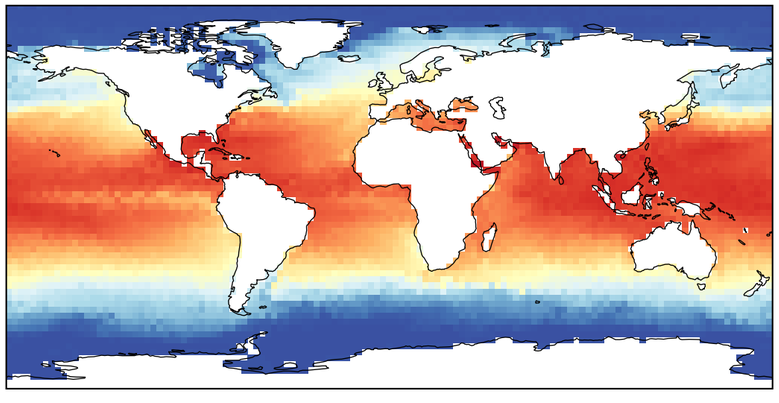
SFNO — Step 46 (°C)



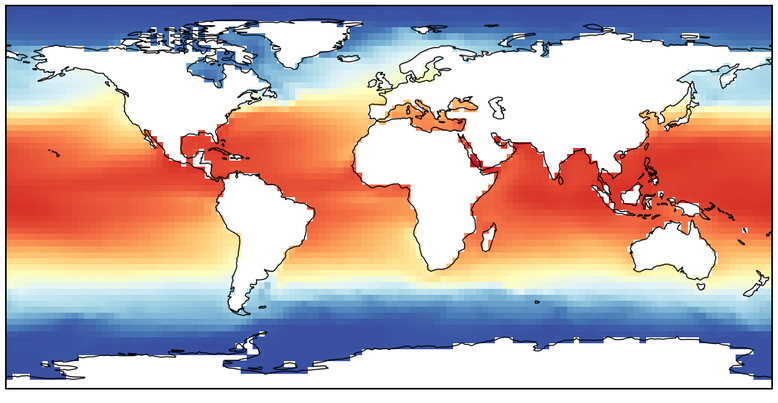
Error — Step 46 (°C)



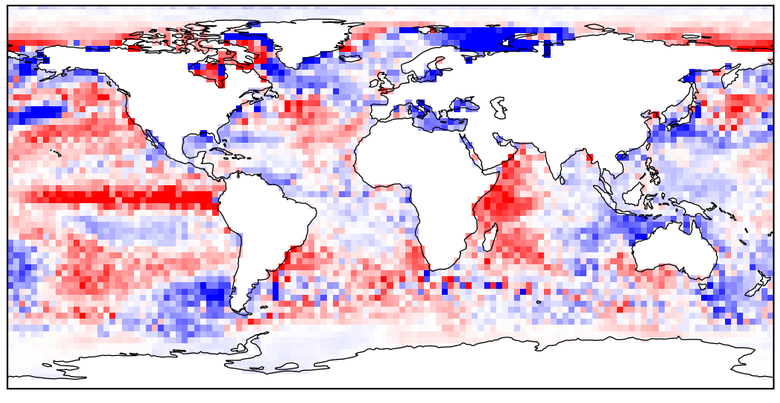
Actual — Step 90 (°C)

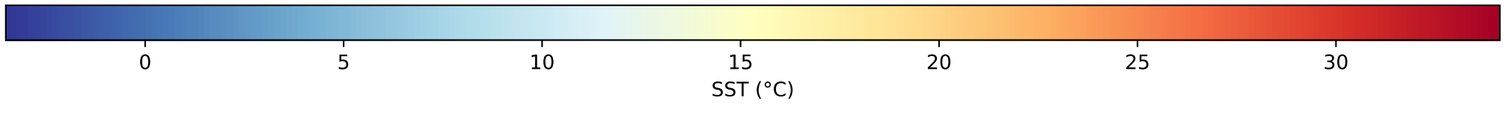


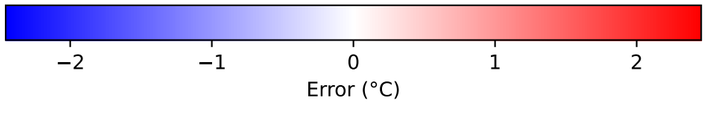
SFNO — Step 90 (°C)



Error — Step 90 (°C)

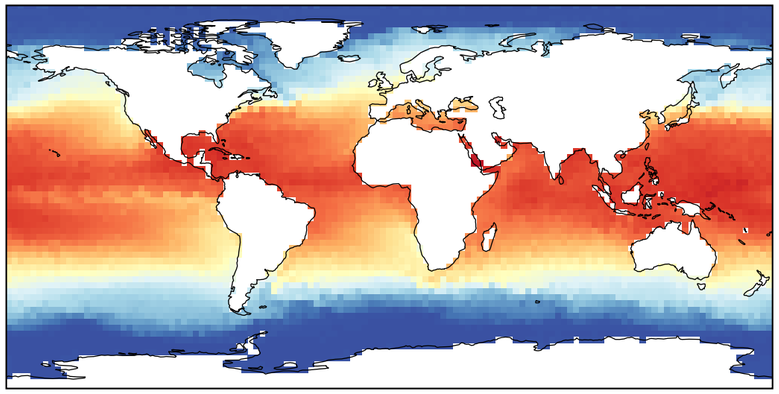




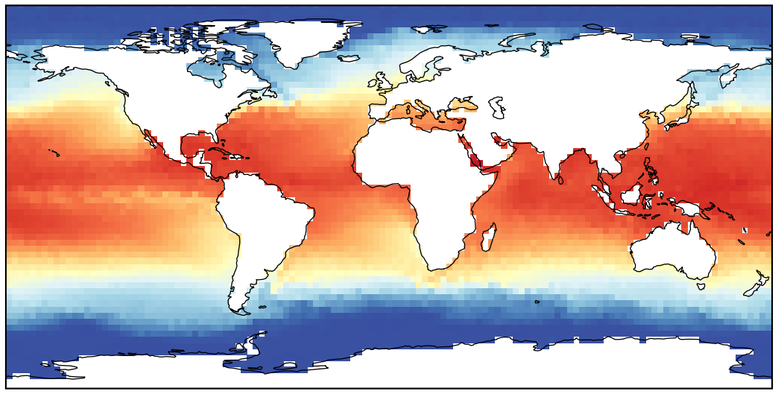


Chain init: 2016-10-07 — Validation (°C)

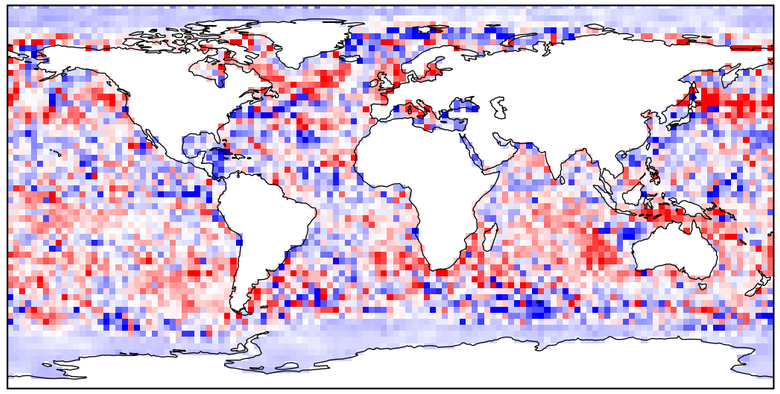
Actual — Step 1 (°C)



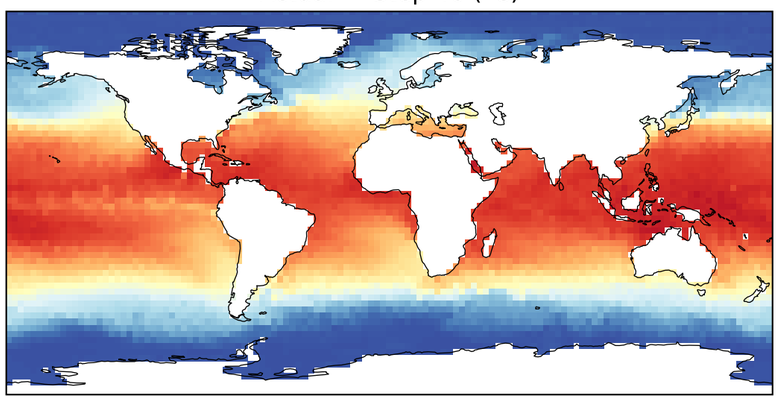
SFNO — Step 1 (°C)



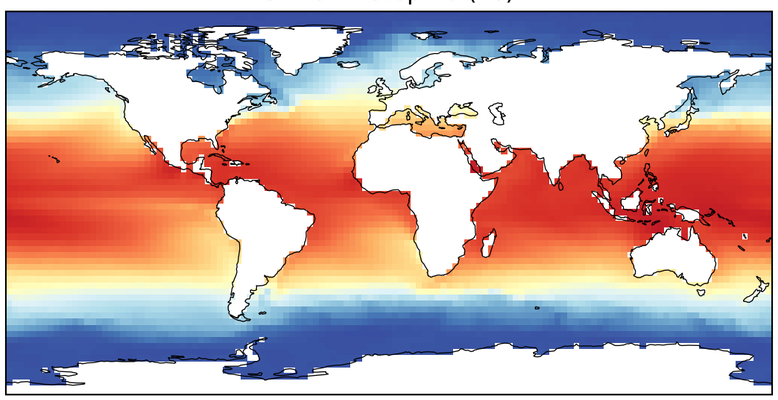
Error — Step 1 (°C)



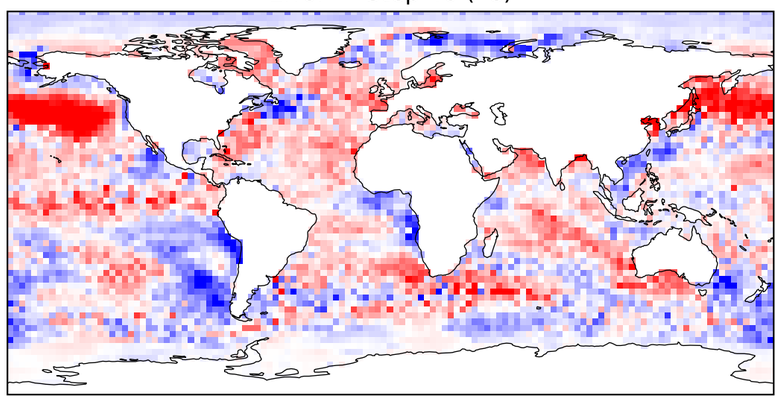
Actual — Step 46 (°C)



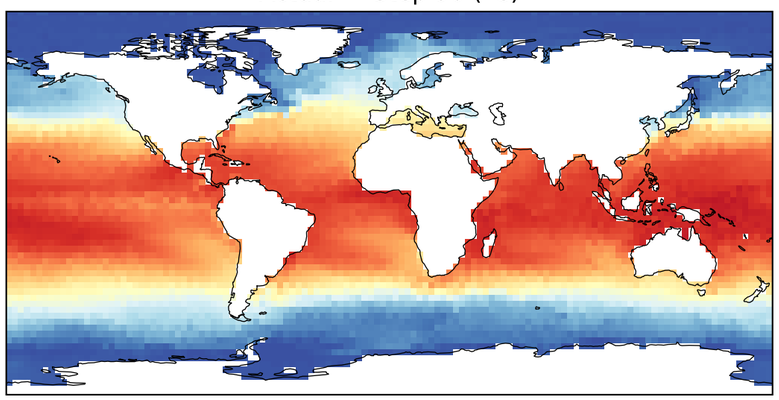
SFNO — Step 46 (°C)



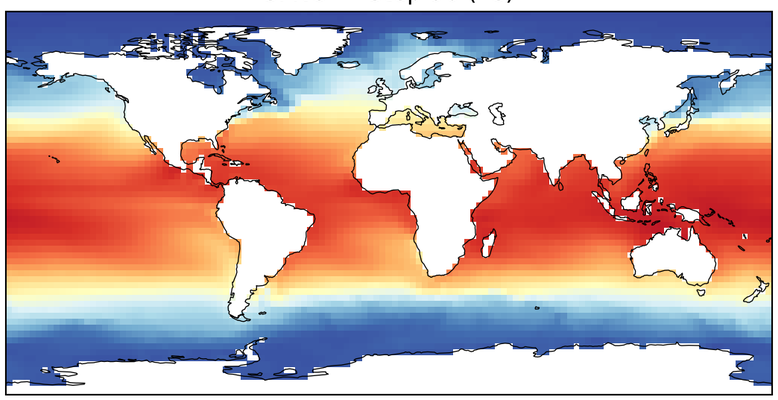
Error — Step 46 (°C)



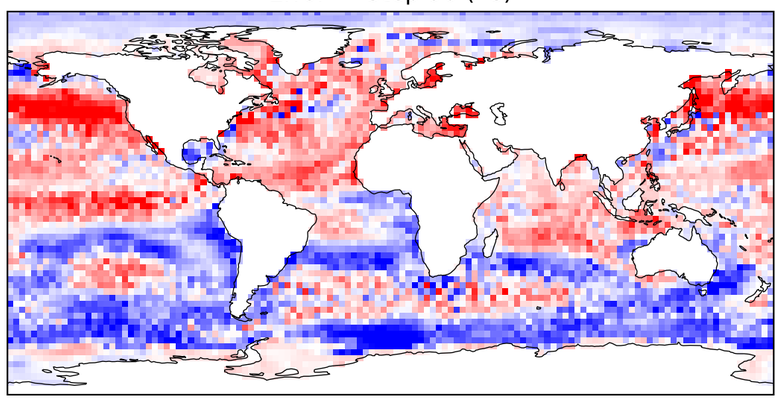
Actual — Step 90 (°C)

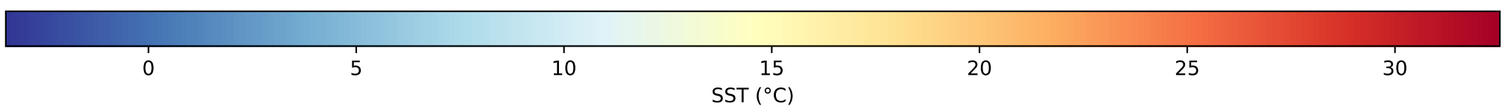


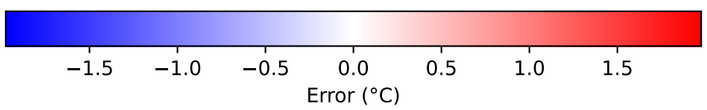
SFNO — Step 90 (°C)



Error — Step 90 (°C)

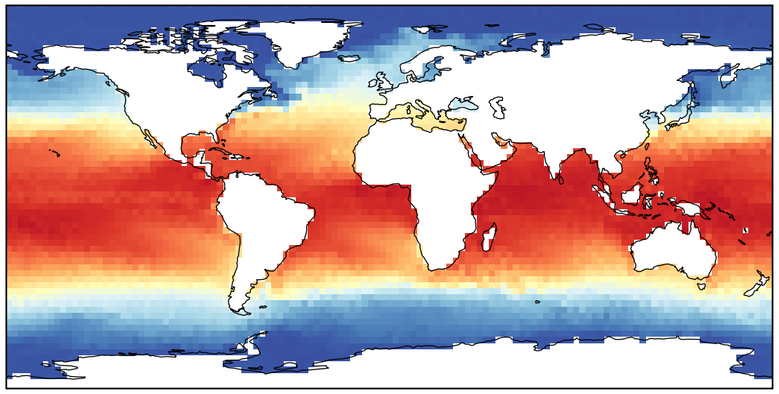




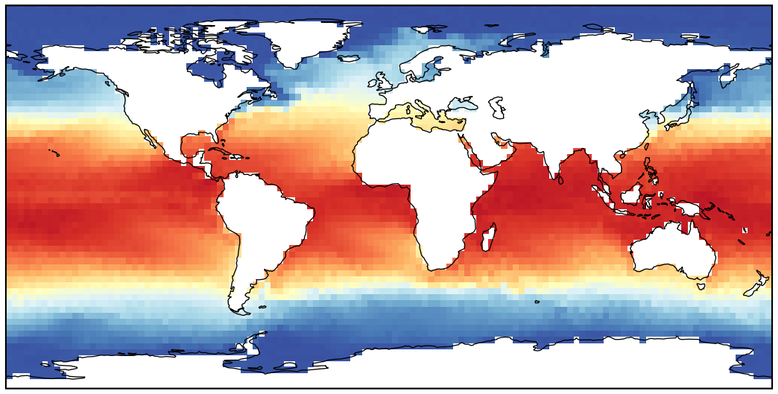


Chain init: 2017-04-05 — Validation (°C)

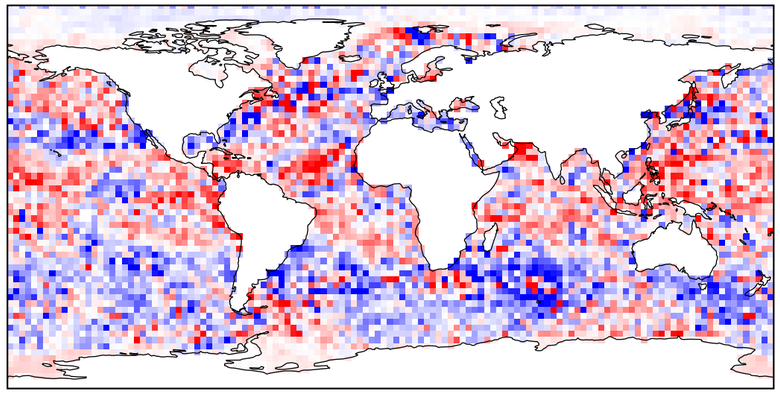
Actual — Step 1 (°C)



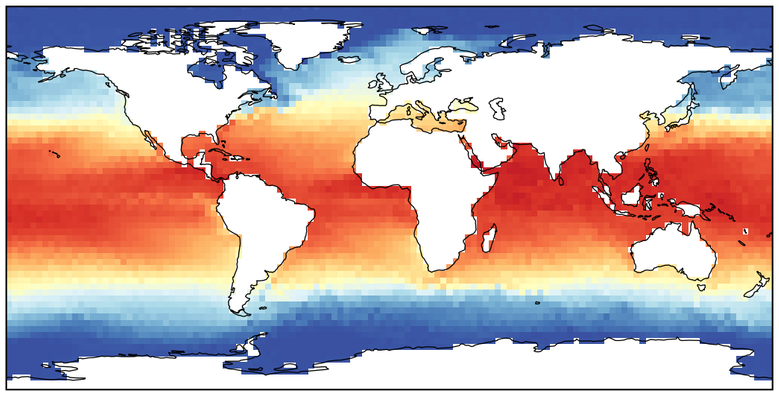
SFNO — Step 1 (°C)



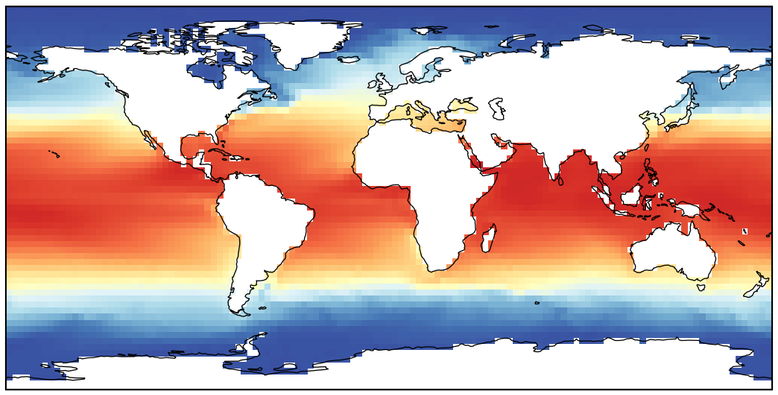
Error — Step 1 (°C)



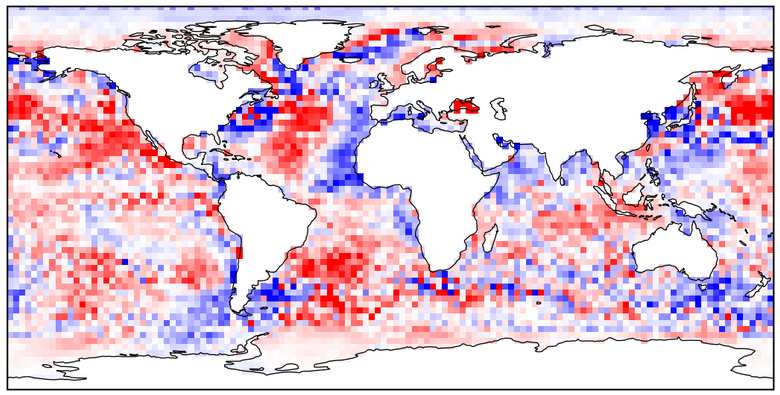
Actual — Step 46 (°C)



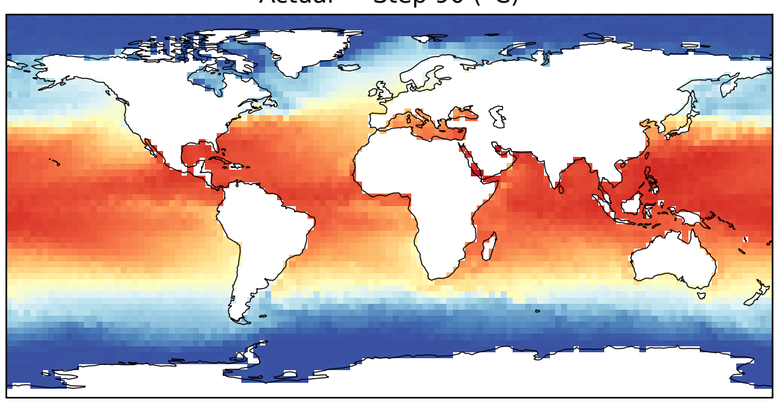
SFNO — Step 46 (°C)



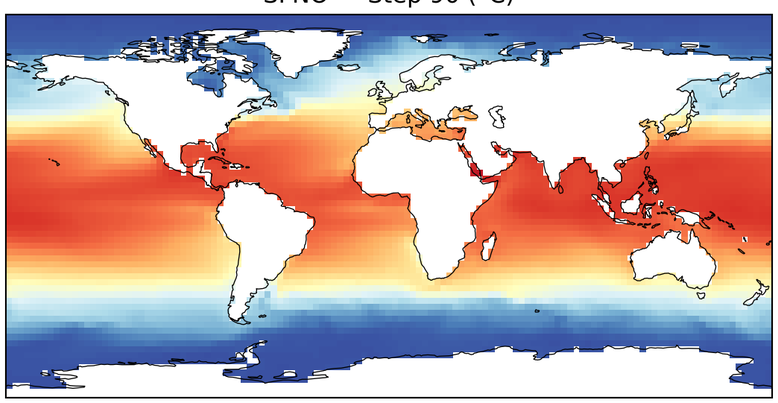
Error — Step 46 (°C)



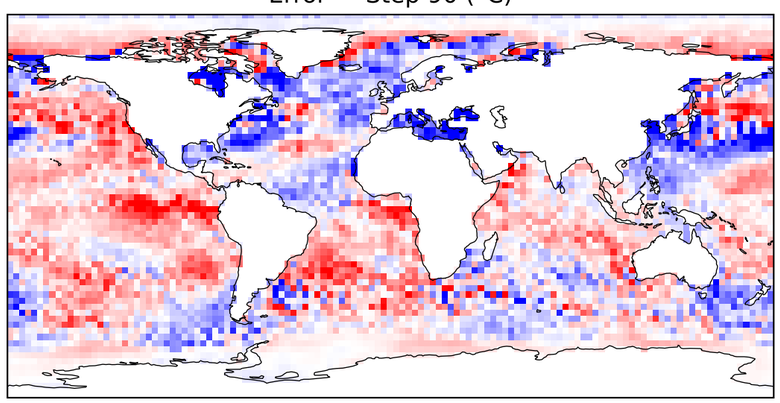
Actual — Step 90 (°C)

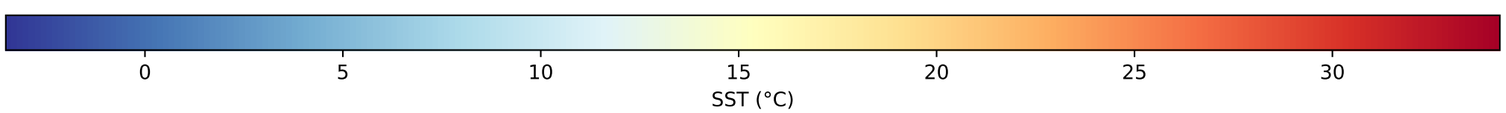


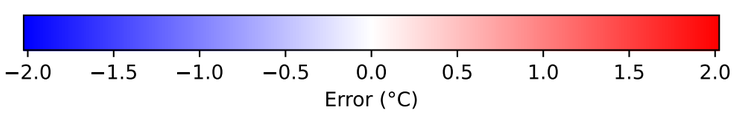
SFNO — Step 90 (°C)



Error — Step 90 (°C)

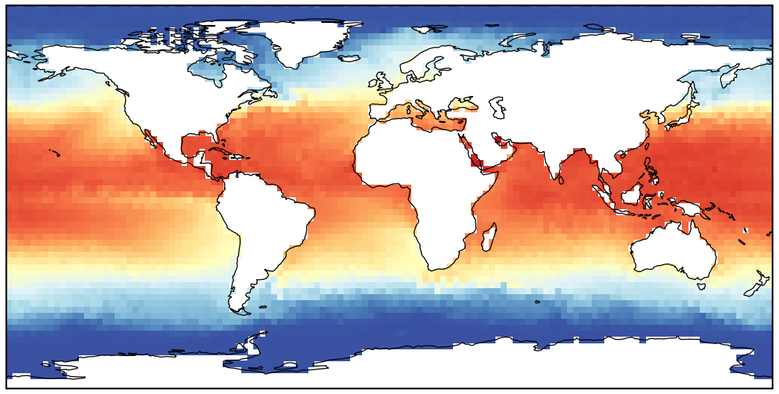




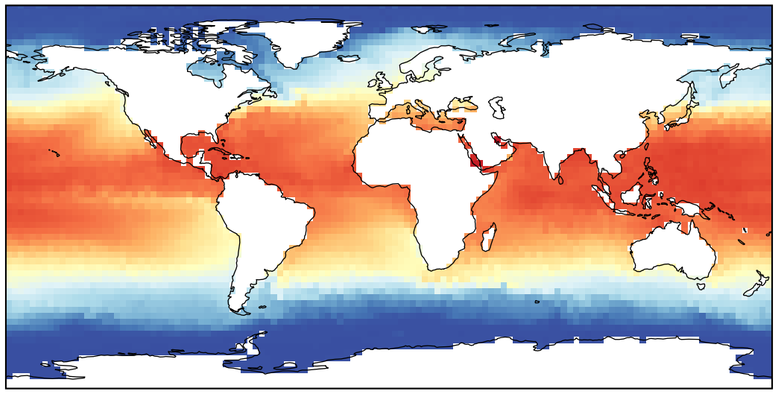


Chain init: 2017-10-02 — Validation (°C)

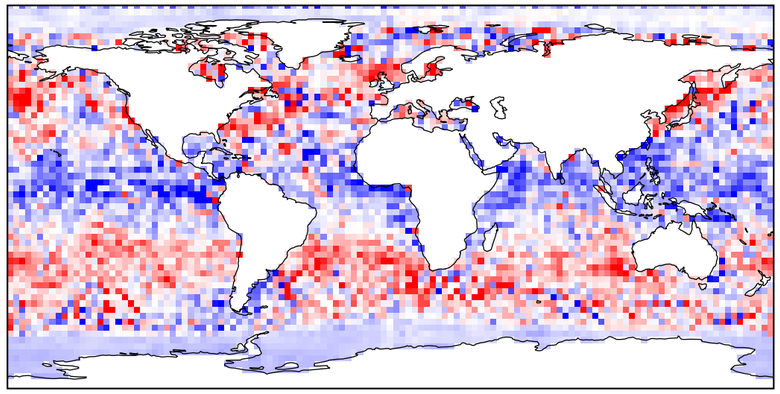
Actual — Step 1 (°C)



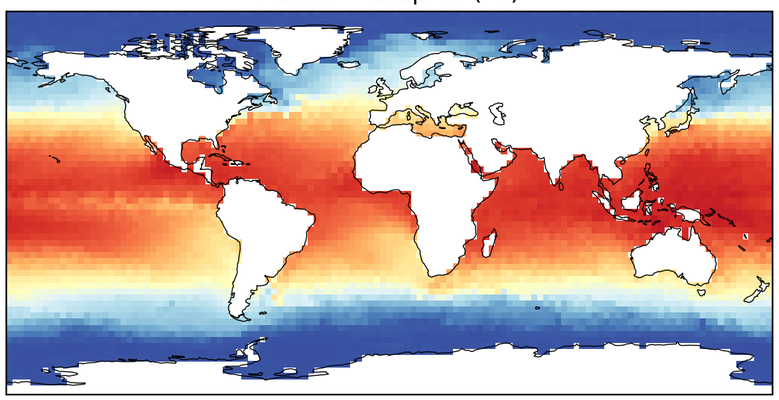
SFNO — Step 1 (°C)



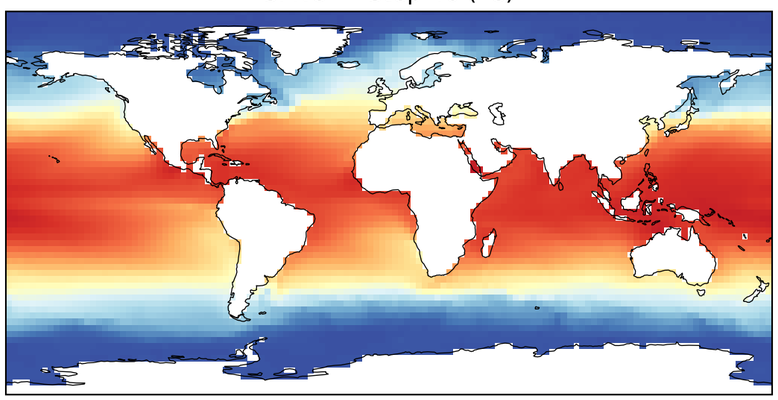
Error — Step 1 (°C)



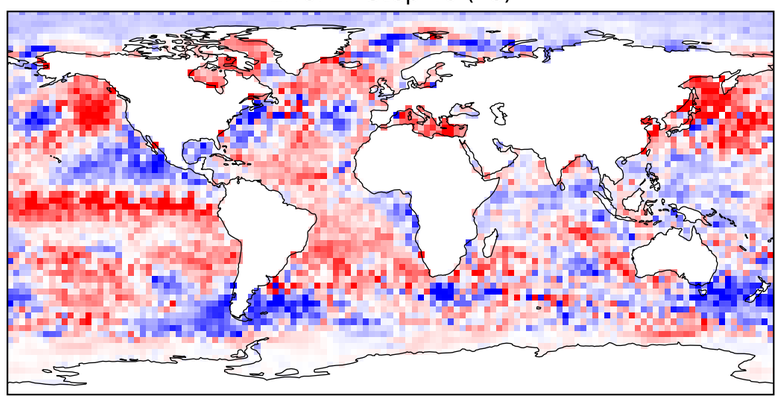
Actual — Step 46 (°C)



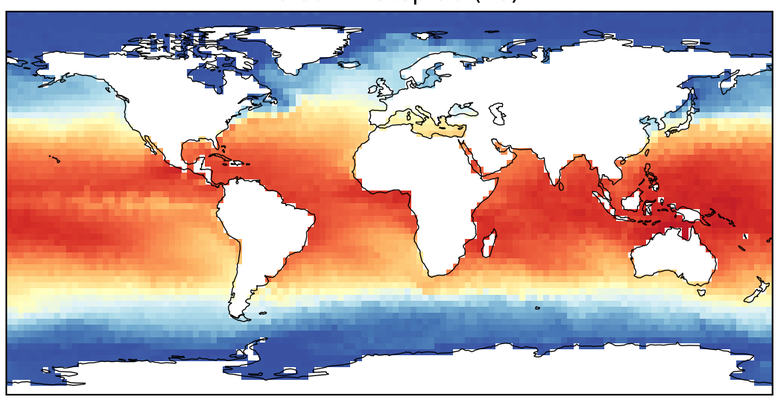
SFNO — Step 46 (°C)



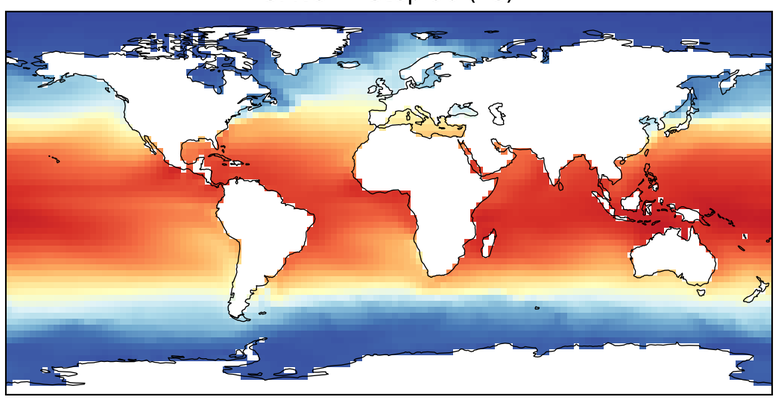
Error — Step 46 (°C)



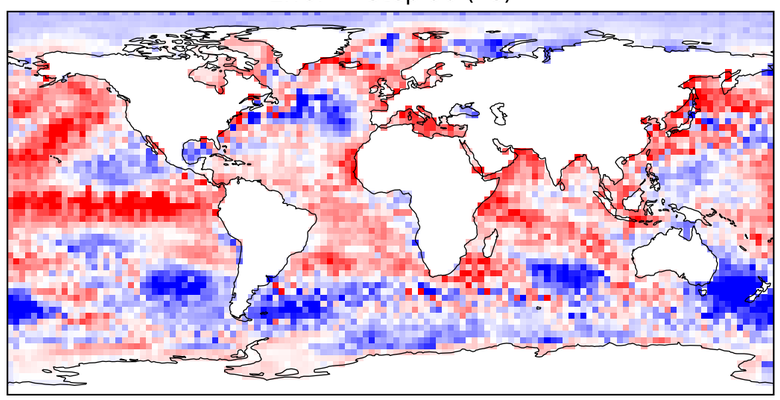
Actual — Step 90 (°C)

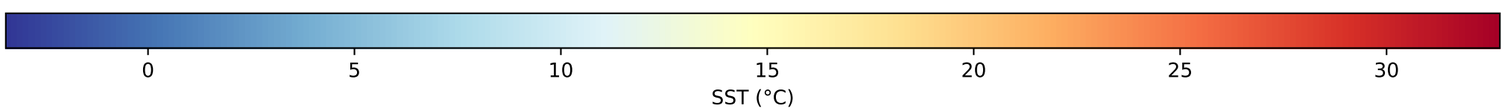


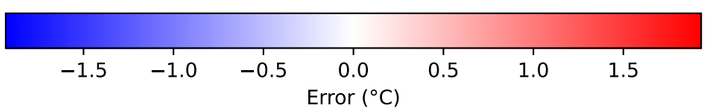
SFNO — Step 90 (°C)



Error — Step 90 (°C)

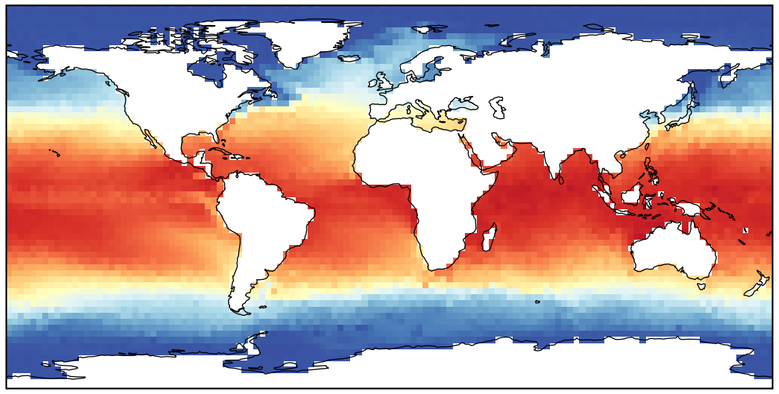




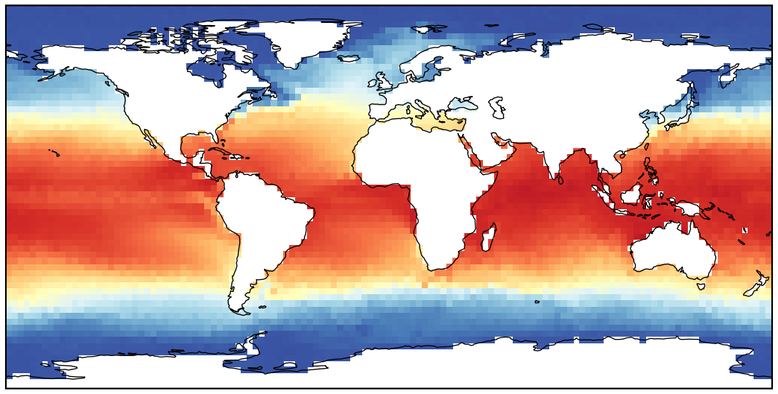


Chain init: 2018-03-31 — Validation (°C)

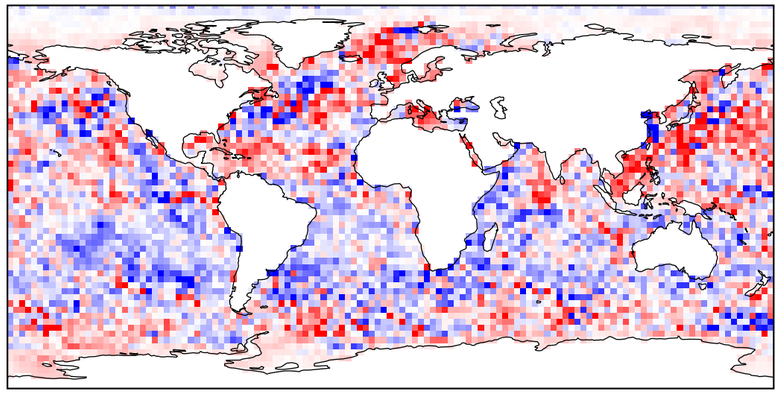
Actual — Step 1 (°C)



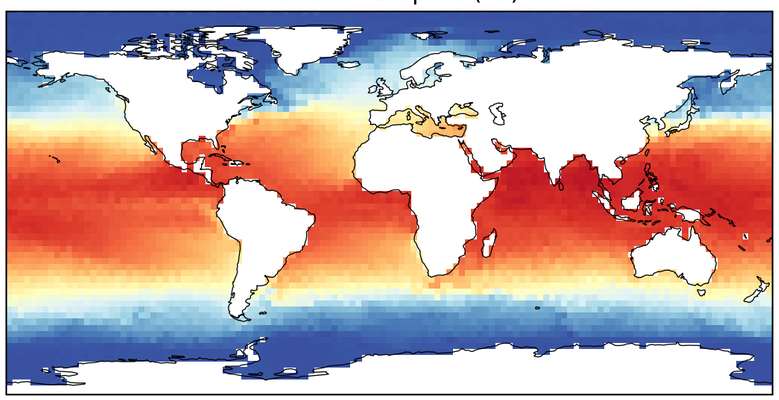
SFNO — Step 1 (°C)



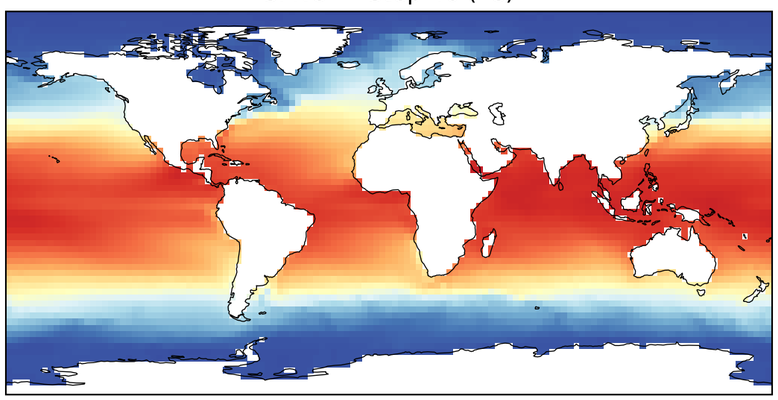
Error — Step 1 (°C)



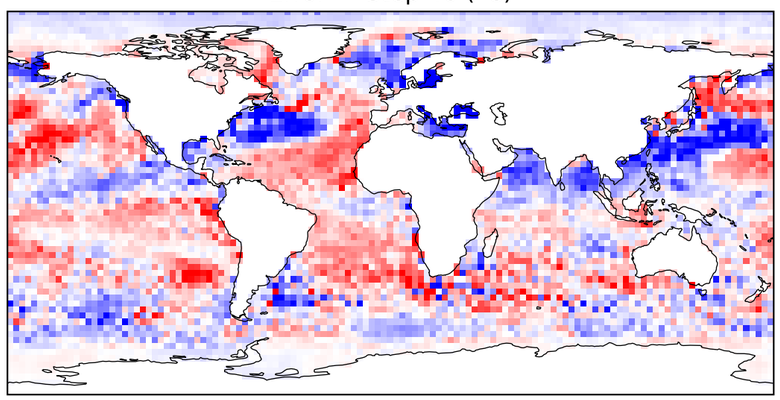
Actual — Step 46 (°C)



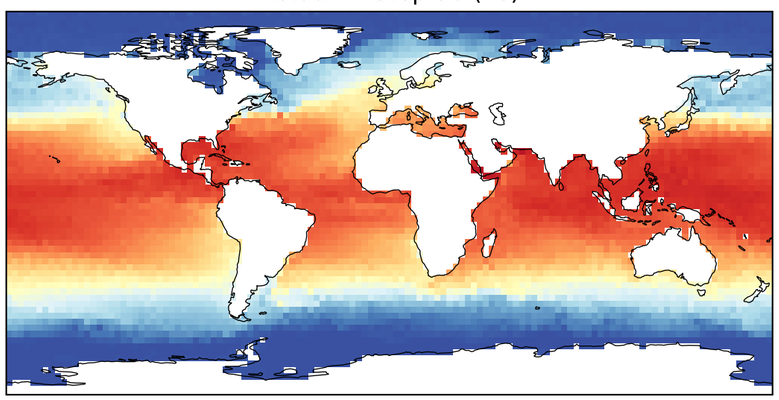
SFNO — Step 46 (°C)



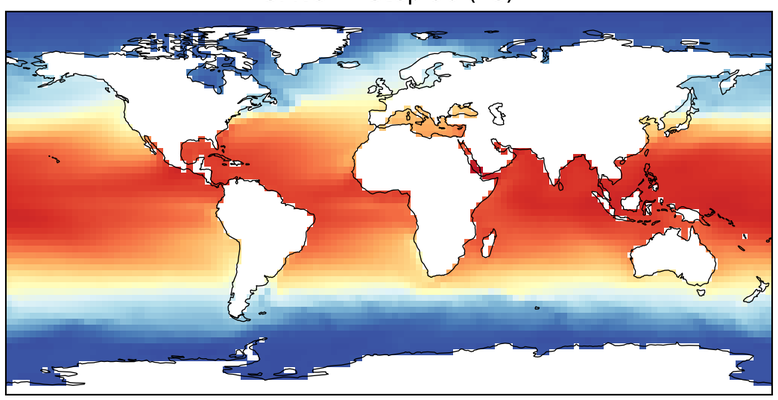
Error — Step 46 (°C)



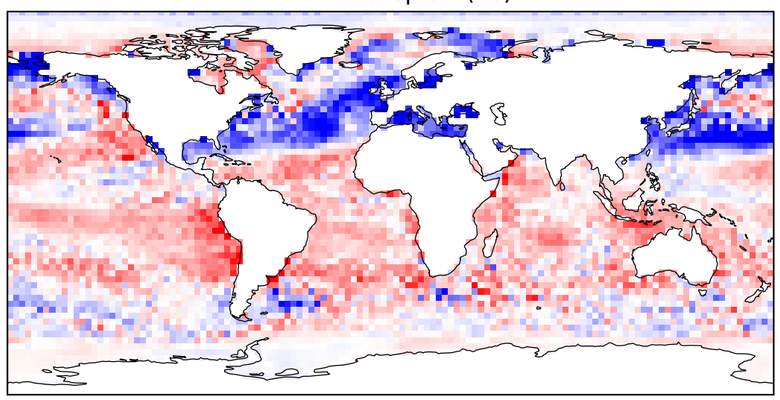
Actual — Step 90 (°C)

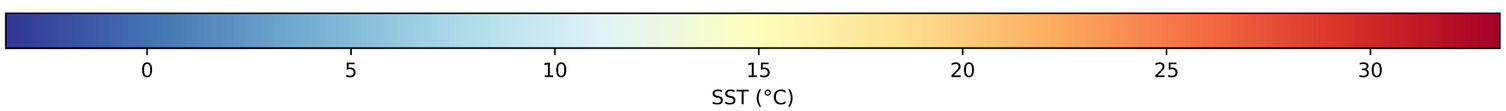


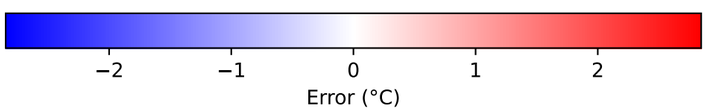
SFNO — Step 90 (°C)



Error — Step 90 (°C)

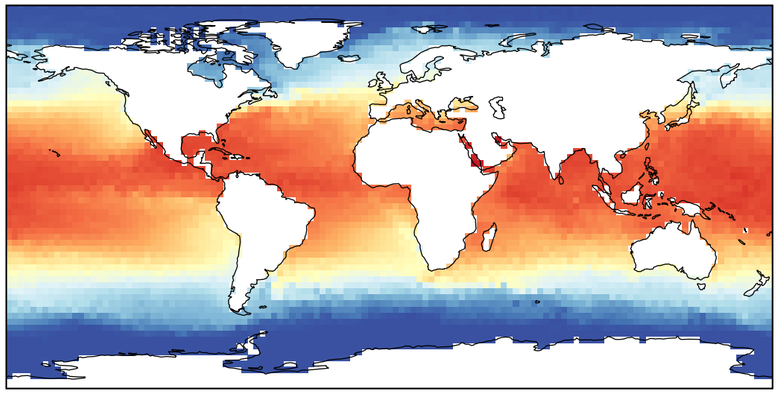




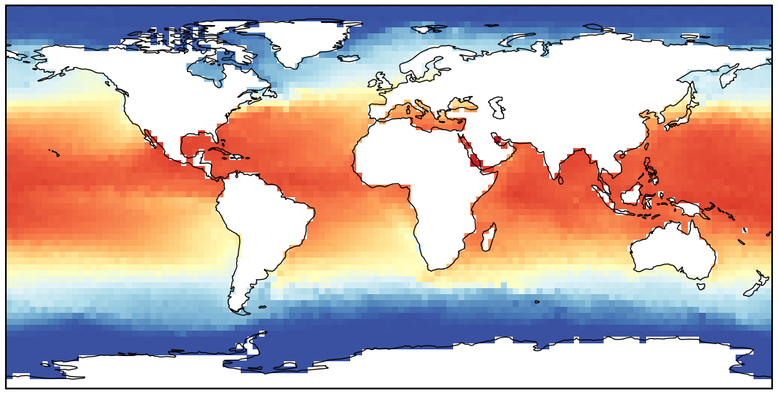


Chain init: 2018-09-27 — Validation (°C)

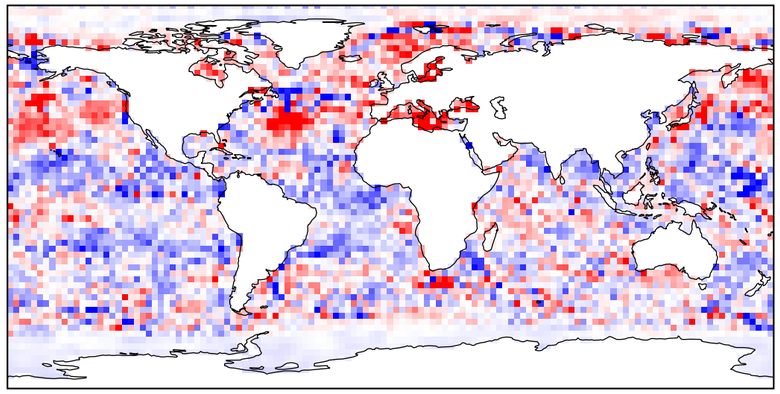
Actual — Step 1 (°C)



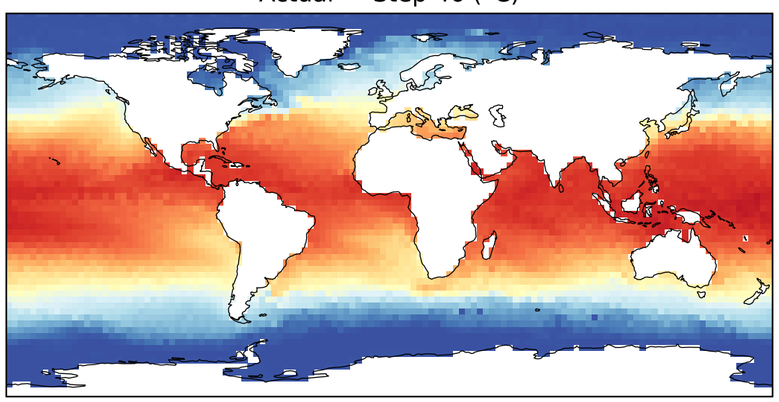
SFNO — Step 1 (°C)



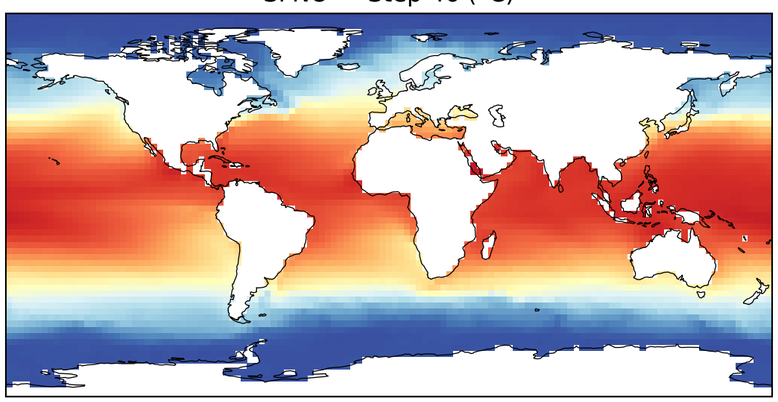
Error — Step 1 (°C)



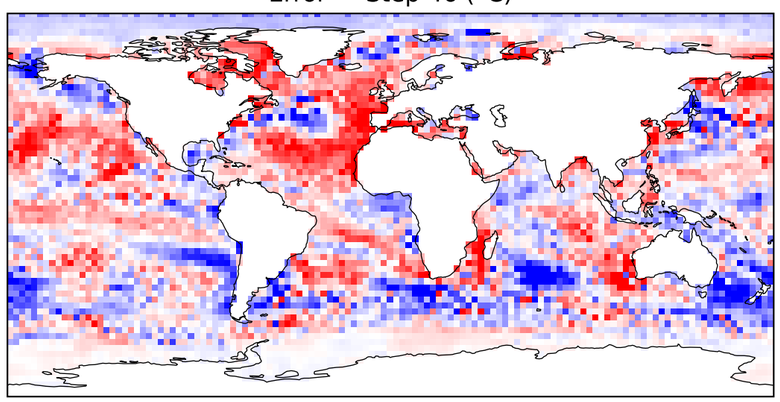
Actual — Step 46 (°C)



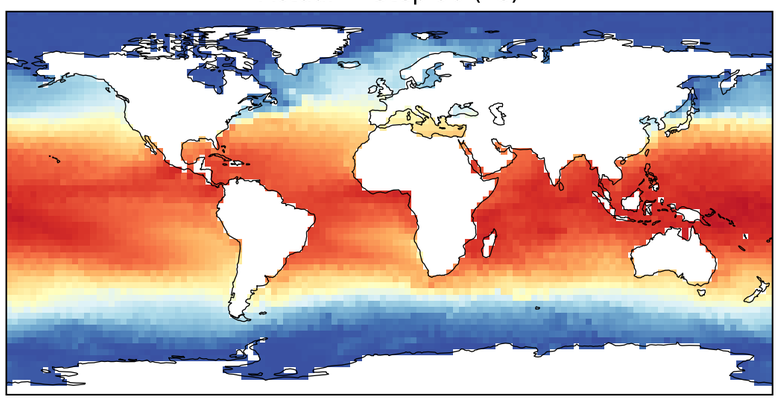
SFNO — Step 46 (°C)



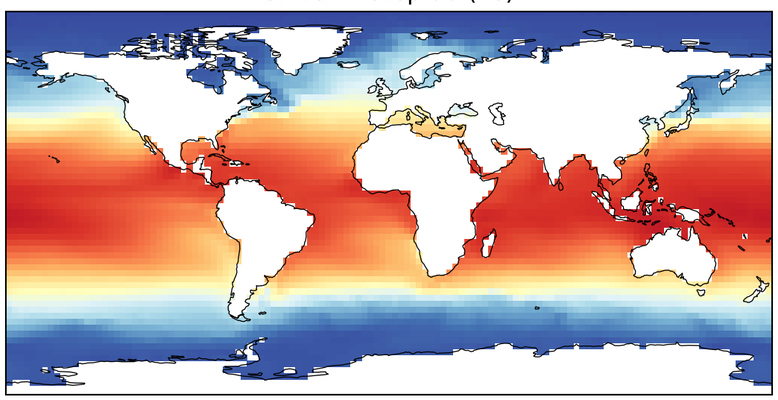
Error — Step 46 (°C)



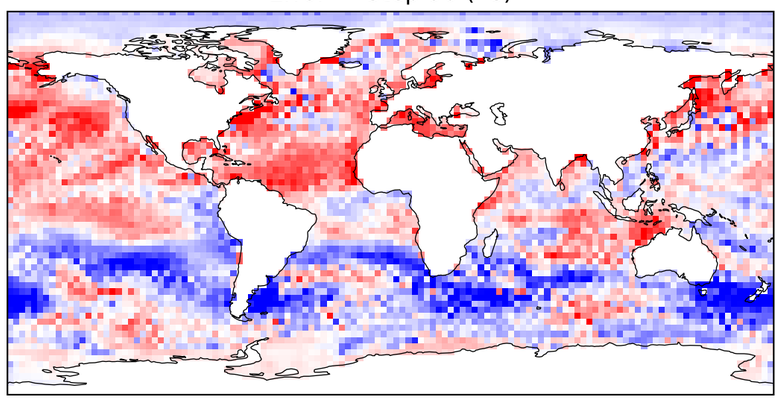
Actual — Step 90 (°C)

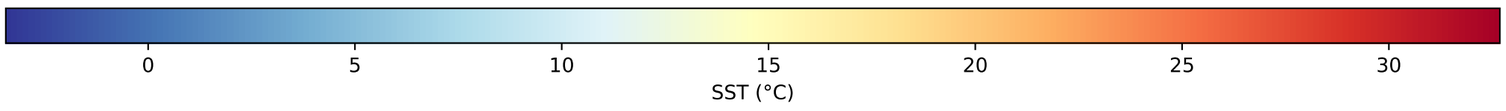


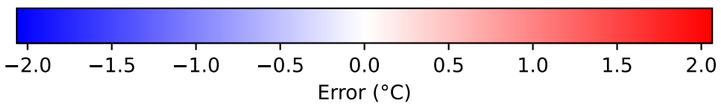
SFNO — Step 90 (°C)



Error — Step 90 (°C)

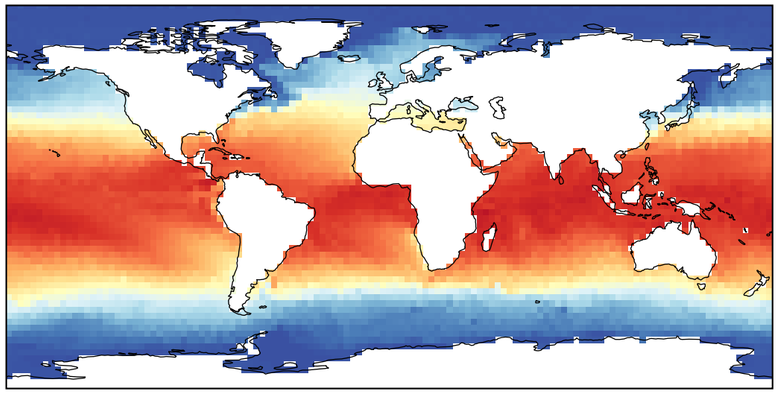




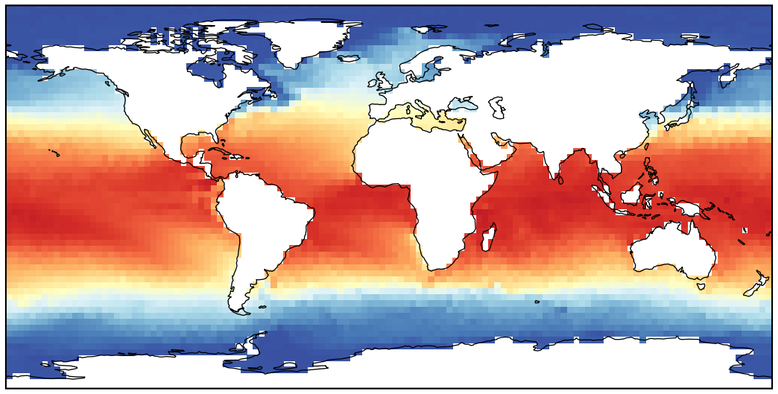


Chain init: 2019-03-26 — Validation (°C)

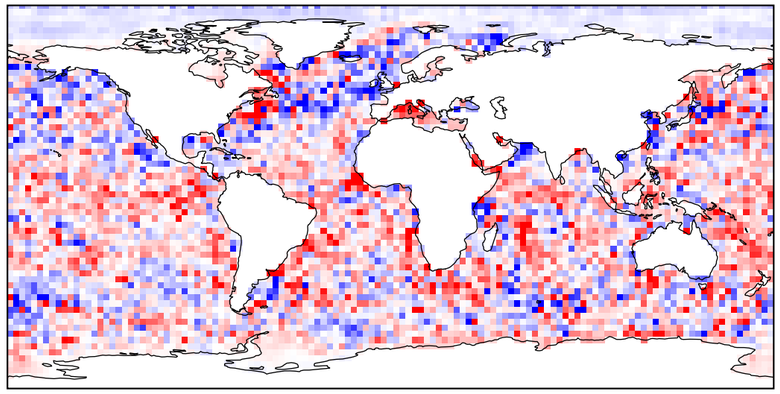
Actual — Step 1 (°C)



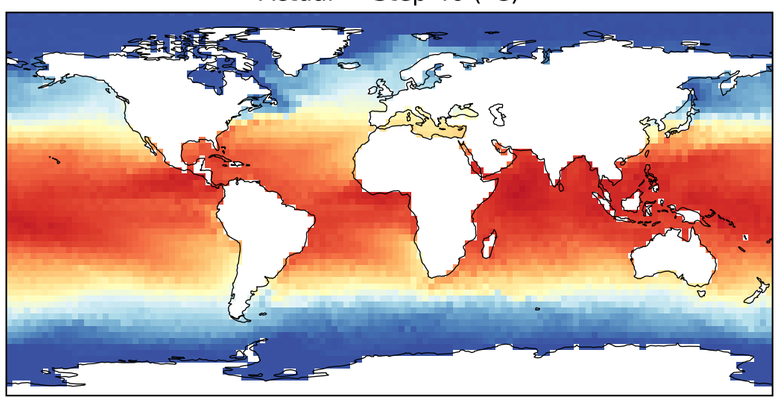
SFNO — Step 1 (°C)



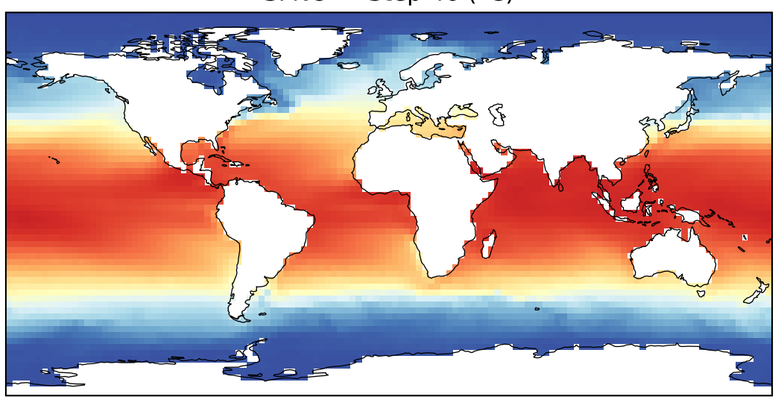
Error — Step 1 (°C)



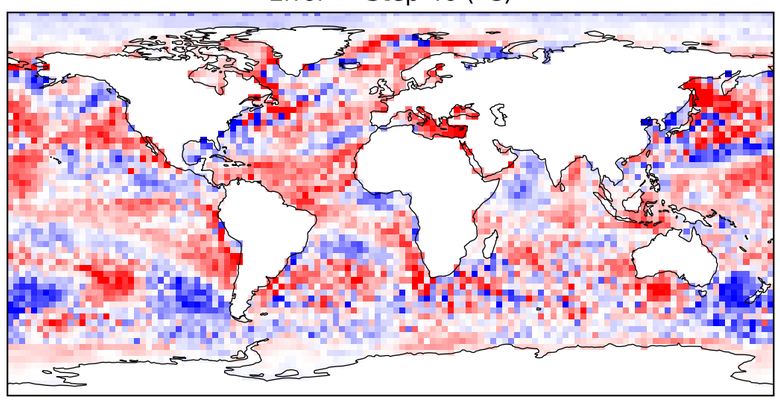
Actual — Step 46 (°C)



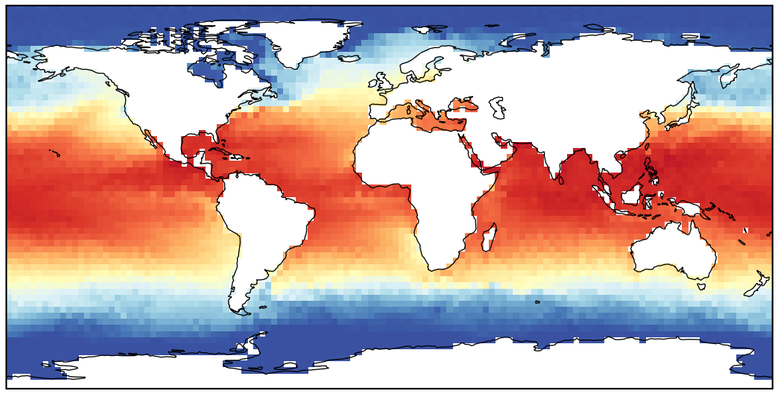
SFNO — Step 46 (°C)



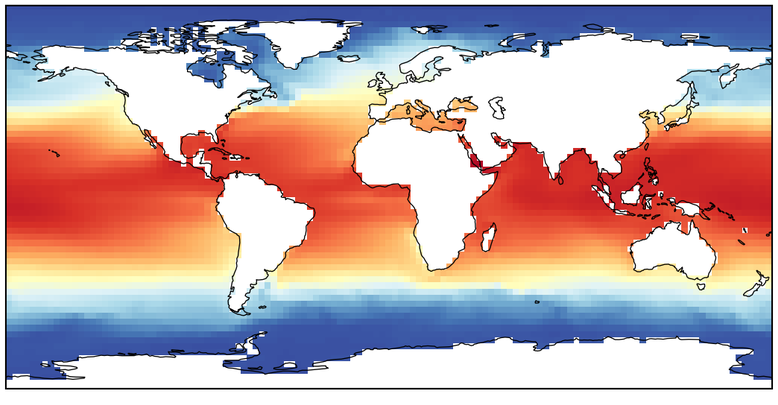
Error — Step 46 (°C)



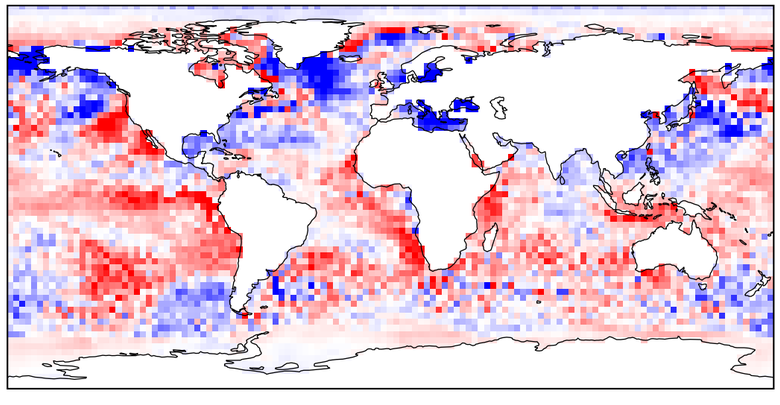
Actual — Step 90 (°C)

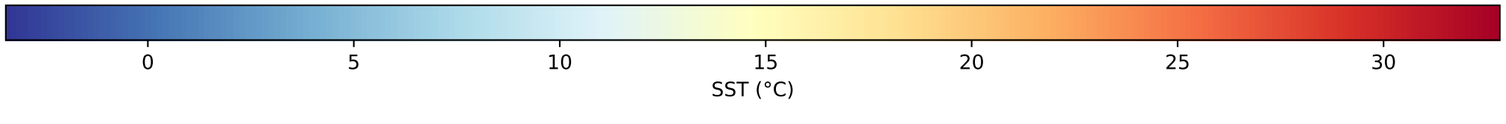


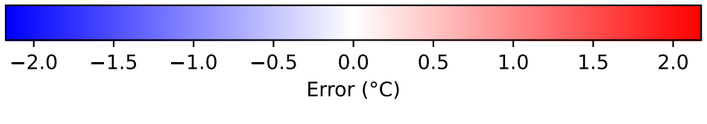
SFNO — Step 90 (°C)



Error — Step 90 (°C)

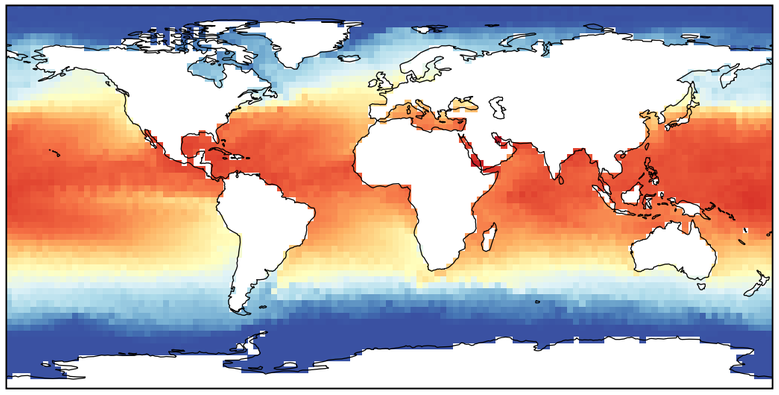




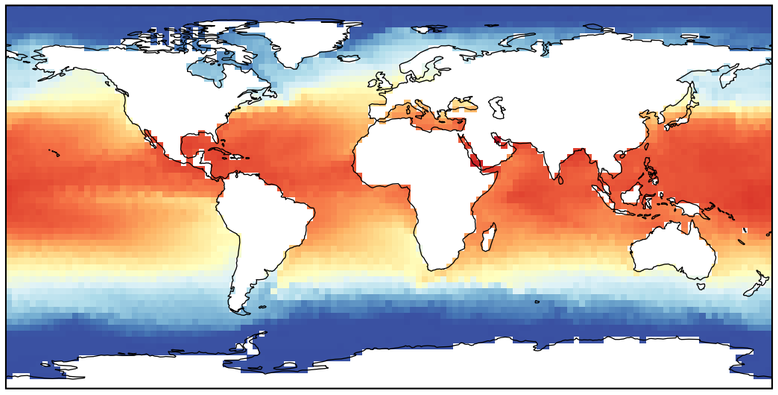


Chain init: 2019-09-22 — Validation (°C)

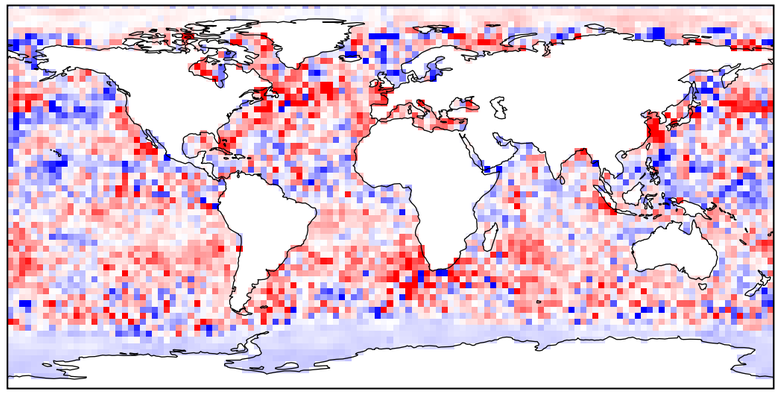
Actual — Step 1 (°C)



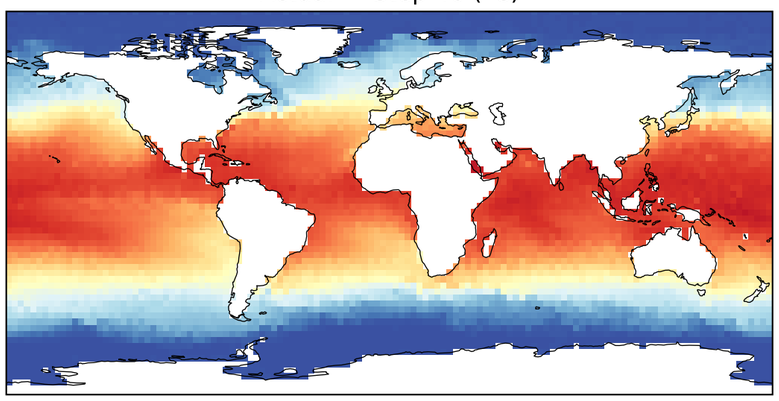
SFNO — Step 1 (°C)



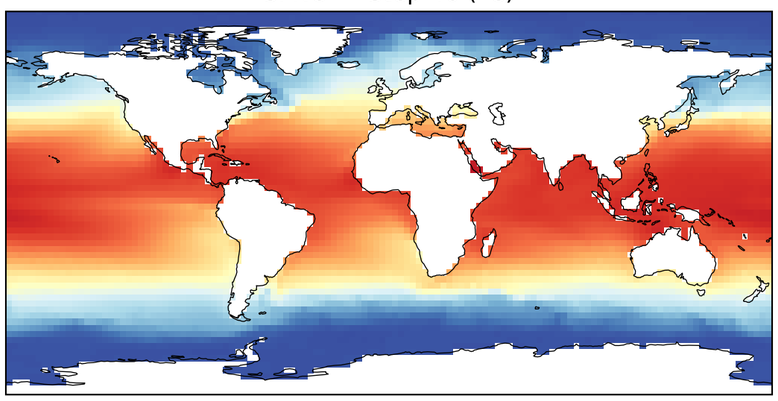
Error — Step 1 (°C)



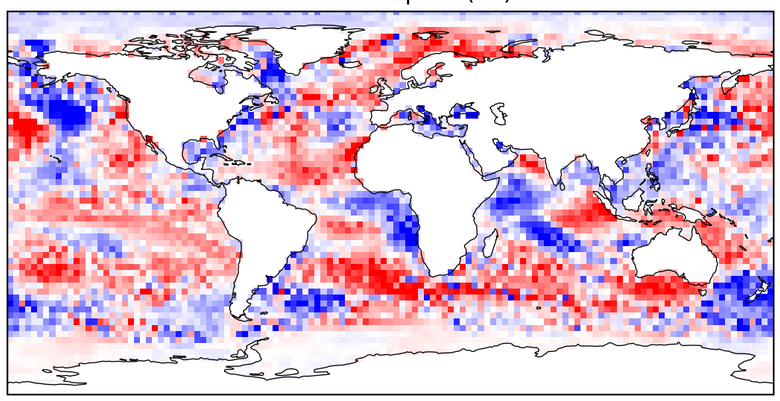
Actual — Step 46 (°C)



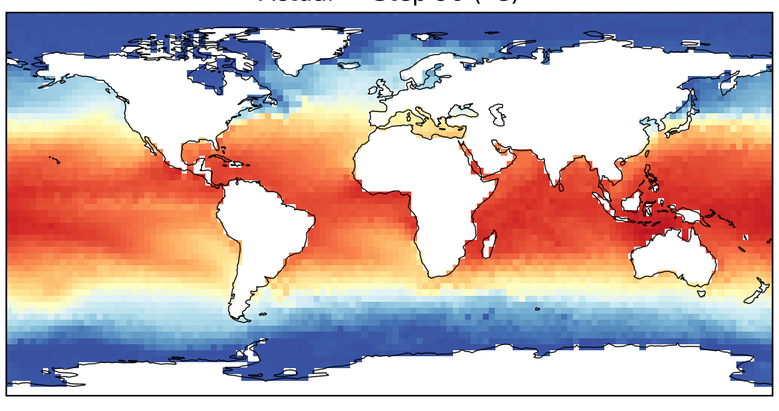
SFNO — Step 46 (°C)



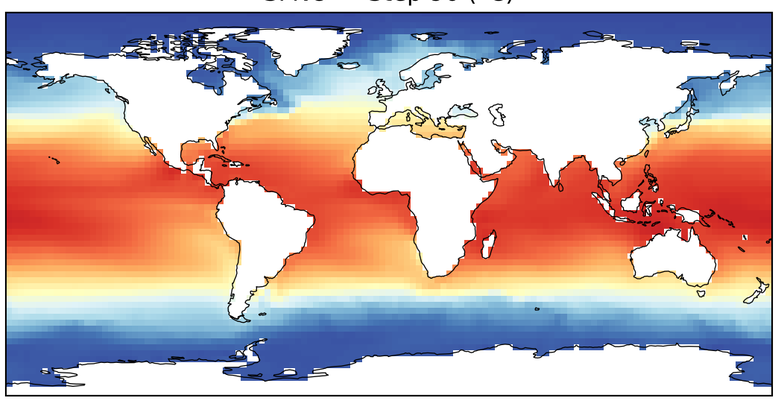
Error — Step 46 (°C)



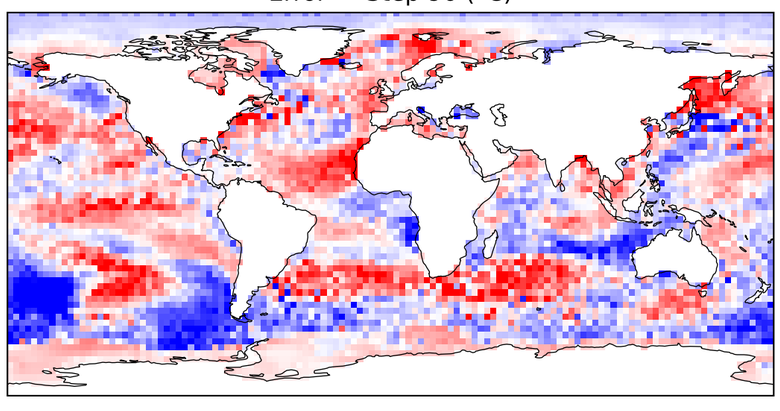
Actual — Step 90 (°C)

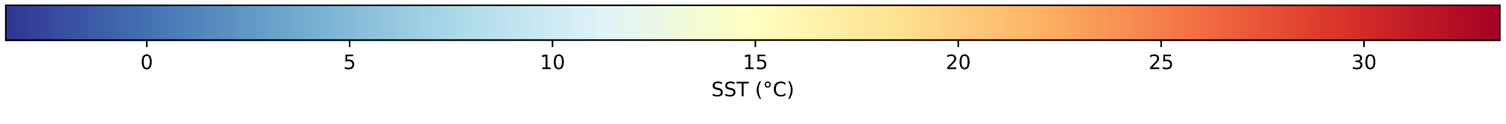


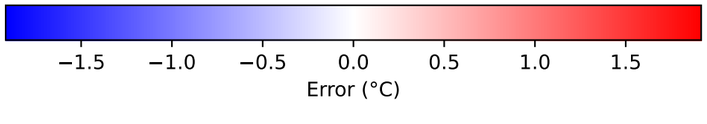
SFNO — Step 90 (°C)



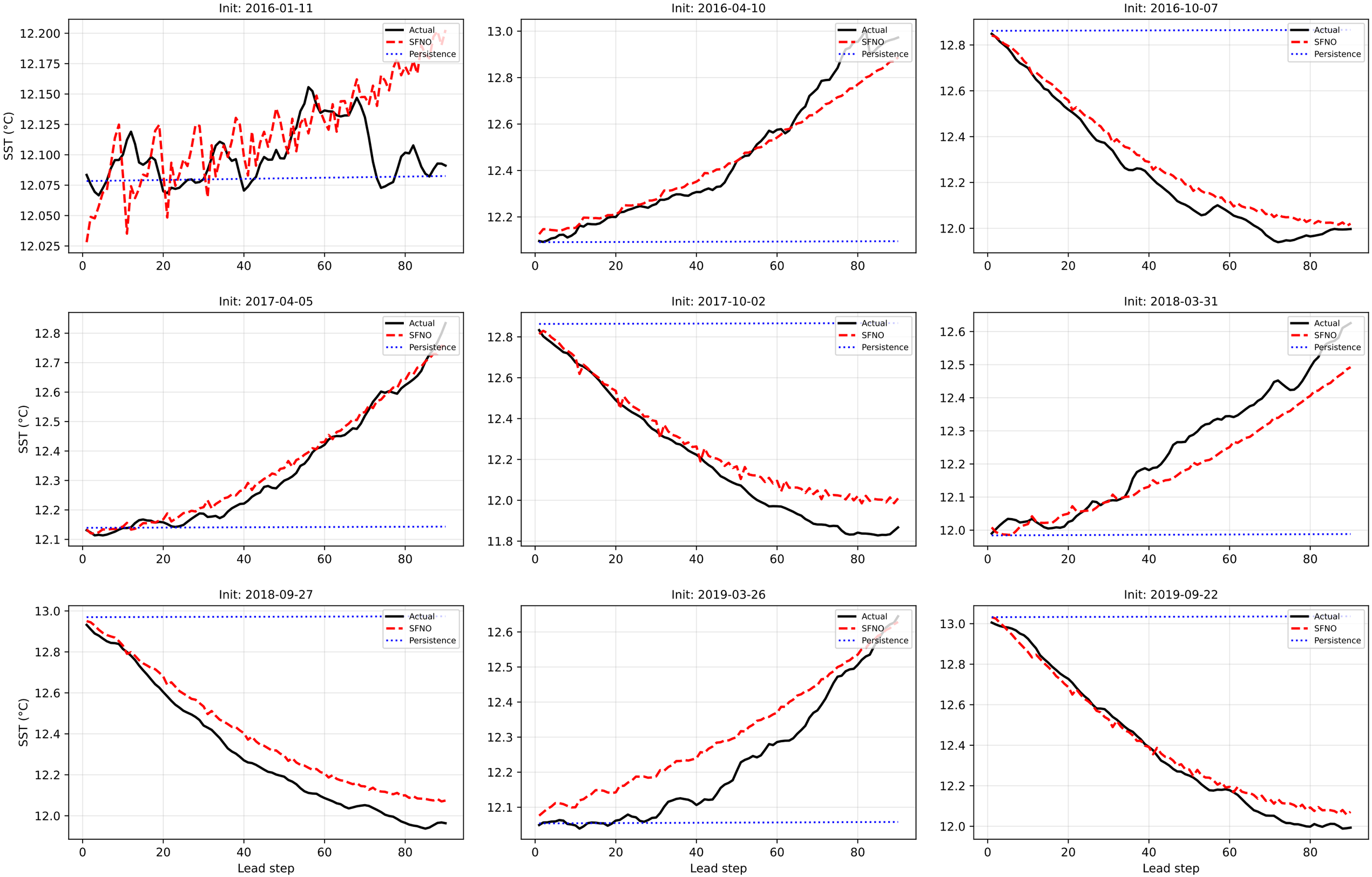
Error — Step 90 (°C)





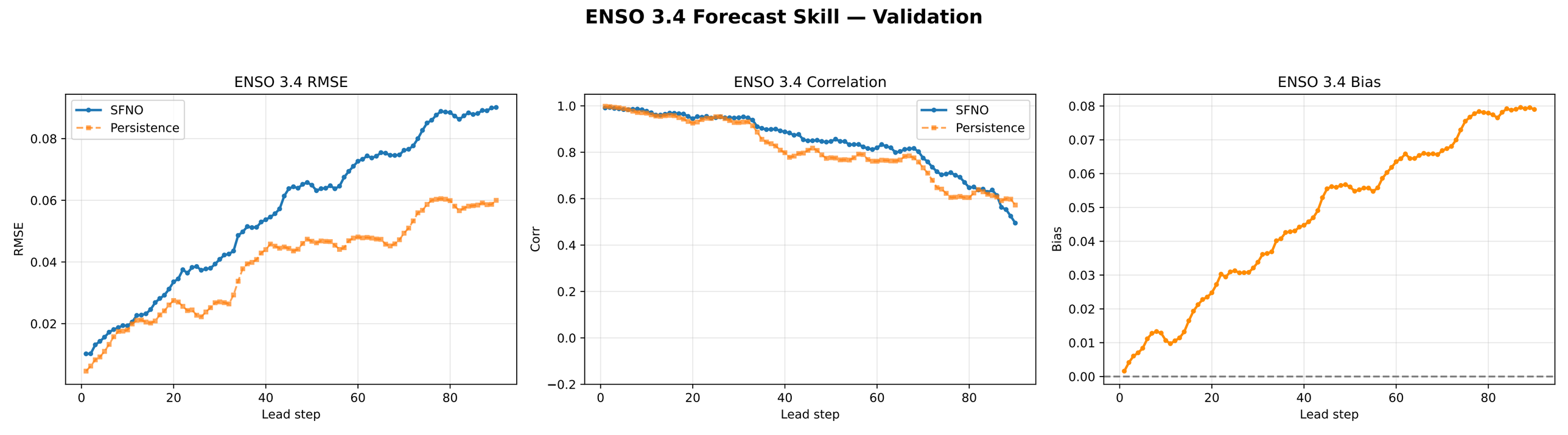


Global-Mean SST — Validation (°C)

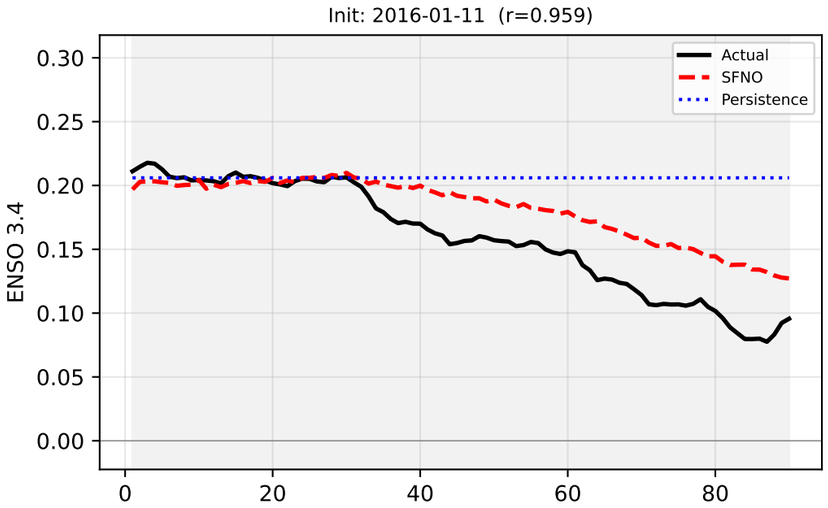


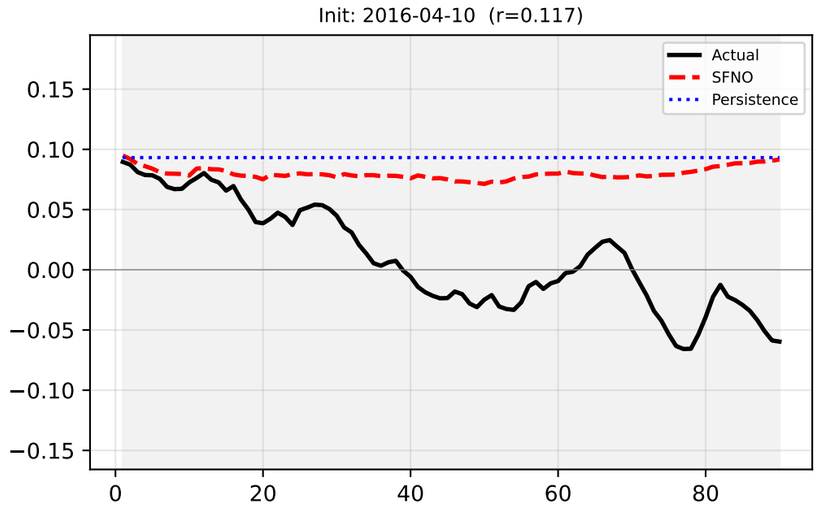
ENSO 3.4 Metrics — Validation

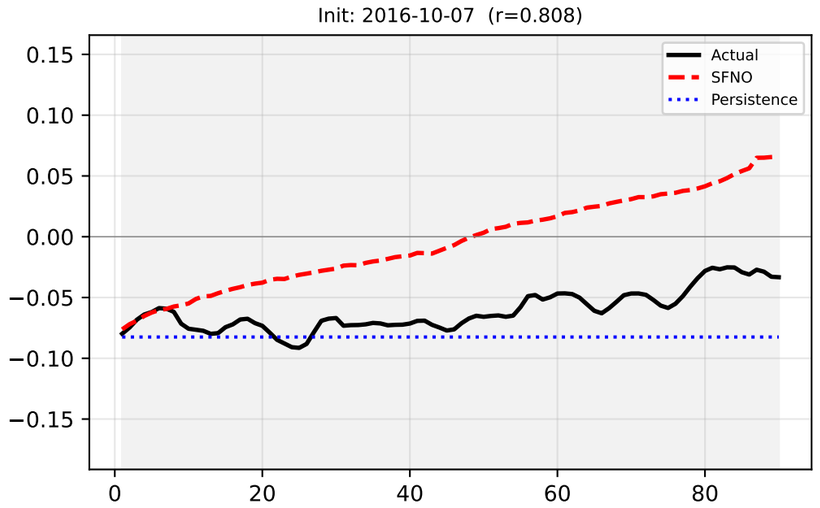
Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0102	0.0046	0.9910	0.9981	0.0016
2	0.0103	0.0063	0.9929	0.9964	0.0041
3	0.0131	0.0083	0.9890	0.9938	0.0061
4	0.0143	0.0092	0.9872	0.9920	0.0070
5	0.0156	0.0110	0.9841	0.9881	0.0084
6	0.0172	0.0133	0.9837	0.9833	0.0112
7	0.0181	0.0158	0.9852	0.9771	0.0128
8	0.0187	0.0175	0.9861	0.9714	0.0133
9	0.0194	0.0176	0.9832	0.9705	0.0129
10	0.0193	0.0180	0.9770	0.9684	0.0107
11	0.0205	0.0199	0.9703	0.9615	0.0097
12	0.0227	0.0211	0.9602	0.9562	0.0106
13	0.0228	0.0212	0.9603	0.9548	0.0114
14	0.0232	0.0205	0.9639	0.9576	0.0132
15	0.0246	0.0202	0.9687	0.9592	0.0165
16	0.0268	0.0209	0.9680	0.9576	0.0193
17	0.0282	0.0228	0.9662	0.9494	0.0212
18	0.0292	0.0241	0.9649	0.9431	0.0228
19	0.0312	0.0261	0.9546	0.9328	0.0235
20	0.0336	0.0275	0.9446	0.9257	0.0248
21	0.0345	0.0270	0.9537	0.9303	0.0272
22	0.0375	0.0256	0.9510	0.9410	0.0303
23	0.0364	0.0243	0.9548	0.9466	0.0294
24	0.0382	0.0245	0.9476	0.9458	0.0309
25	0.0385	0.0227	0.9492	0.9531	0.0313
26	0.0373	0.0222	0.9526	0.9533	0.0307
27	0.0378	0.0238	0.9486	0.9448	0.0307
28	0.0380	0.0252	0.9490	0.9366	0.0308
29	0.0393	0.0268	0.9478	0.9284	0.0321
30	0.0409	0.0271	0.9490	0.9280	0.0338
31	0.0423	0.0268	0.9525	0.9302	0.0361
32	0.0426	0.0264	0.9483	0.9312	0.0364
33	0.0436	0.0293	0.9386	0.9144	0.0369
34	0.0486	0.0338	0.9101	0.8869	0.0401
35	0.0498	0.0378	0.9033	0.8557	0.0408
36	0.0515	0.0394	0.8978	0.8439	0.0426
37	0.0512	0.0399	0.8986	0.8370	0.0428
38	0.0513	0.0408	0.8998	0.8272	0.0430
39	0.0530	0.0429	0.8922	0.8091	0.0442
40	0.0537	0.0440	0.8879	0.7981	0.0447
41	0.0546	0.0458	0.8833	0.7782	0.0458
42	0.0557	0.0451	0.8734	0.7835	0.0470
43	0.0572	0.0445	0.8770	0.7945	0.0491
44	0.0614	0.0449	0.8541	0.7965	0.0529
45	0.0638	0.0444	0.8494	0.8085	0.0555
46	0.0644	0.0436	0.8497	0.8178	0.0562
47	0.0639	0.0441	0.8518	0.8079	0.0560
48	0.0652	0.0460	0.8469	0.7888	0.0565
49	0.0658	0.0474	0.8440	0.7738	0.0568
50	0.0649	0.0467	0.8467	0.7763	0.0561
51	0.0631	0.0462	0.8564	0.7751	0.0548
52	0.0638	0.0468	0.8469	0.7672	0.0552
53	0.0639	0.0467	0.8466	0.7680	0.0557
54	0.0647	0.0466	0.8325	0.7665	0.0557
55	0.0637	0.0454	0.8335	0.7763	0.0548
56	0.0646	0.0441	0.8338	0.7919	0.0558
57	0.0675	0.0446	0.8231	0.7902	0.0586
58	0.0694	0.0469	0.8163	0.7685	0.0603
59	0.0710	0.0477	0.8119	0.7616	0.0619
60	0.0727	0.0481	0.8192	0.7614	0.0635
61	0.0733	0.0478	0.8333	0.7659	0.0644
62	0.0744	0.0480	0.8253	0.7649	0.0658
63	0.0737	0.0477	0.8195	0.7628	0.0645
64	0.0743	0.0474	0.7990	0.7625	0.0645
65	0.0755	0.0473	0.8034	0.7666	0.0654
66	0.0753	0.0458	0.8125	0.7823	0.0660
67	0.0746	0.0452	0.8148	0.7845	0.0657
68	0.0746	0.0459	0.8166	0.7758	0.0659
69	0.0747	0.0472	0.8025	0.7581	0.0656
70	0.0762	0.0493	0.7749	0.7330	0.0668
71	0.0765	0.0510	0.7587	0.7100	0.0674
72	0.0777	0.0533	0.7356	0.6791	0.0681
73	0.0800	0.0559	0.7165	0.6473	0.0700
74	0.0827	0.0567	0.7030	0.6412	0.0729
75	0.0851	0.0587	0.7062	0.6229	0.0755
76	0.0860	0.0600	0.7123	0.6055	0.0767
77	0.0876	0.0603	0.7009	0.6065	0.0777
78	0.0889	0.0605	0.6923	0.6097	0.0784
79	0.0886	0.0603	0.6699	0.6050	0.0781
80	0.0885	0.0599	0.6473	0.6048	0.0779
81	0.0873	0.0581	0.6504	0.6243	0.0774
82	0.0863	0.0566	0.6376	0.6391	0.0765
83	0.0874	0.0574	0.6413	0.6298	0.0781
84	0.0883	0.0581	0.6272	0.6190	0.0792
85	0.0878	0.0583	0.6373	0.6136	0.0788
86	0.0882	0.0585	0.6134	0.6072	0.0790
87	0.0891	0.0591	0.5626	0.5917	0.0796
88	0.0890	0.0586	0.5532	0.5994	0.0792
89	0.0900	0.0587	0.5246	0.5977	0.0795
90	0.0901	0.0600	0.4948	0.5725	0.0790

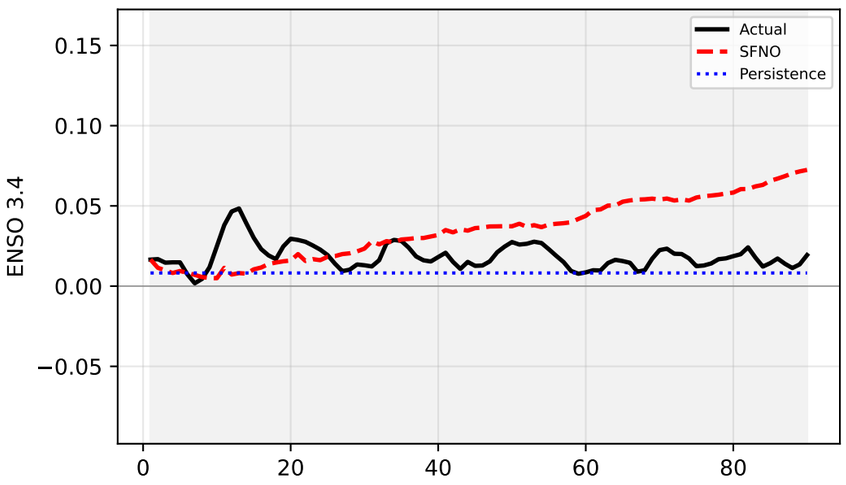

ENSO 3.4 Forecast Skill — Validation
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
SFNO
Persistence
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

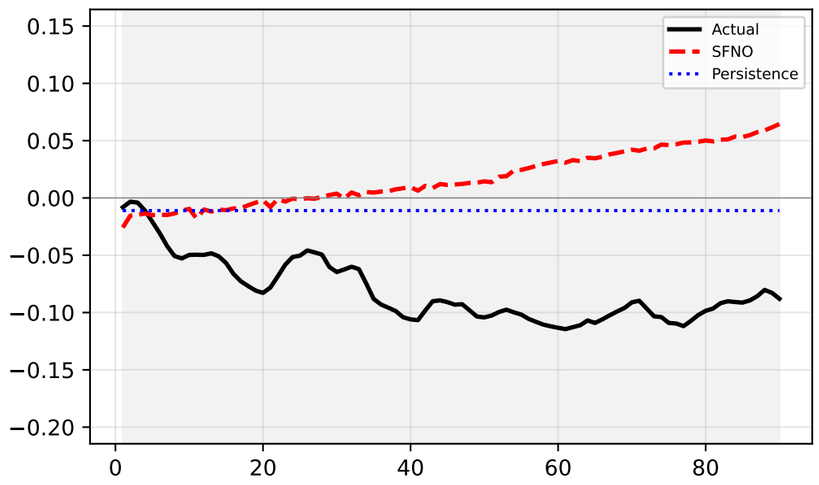
ENSO 3.4 Index — Validation

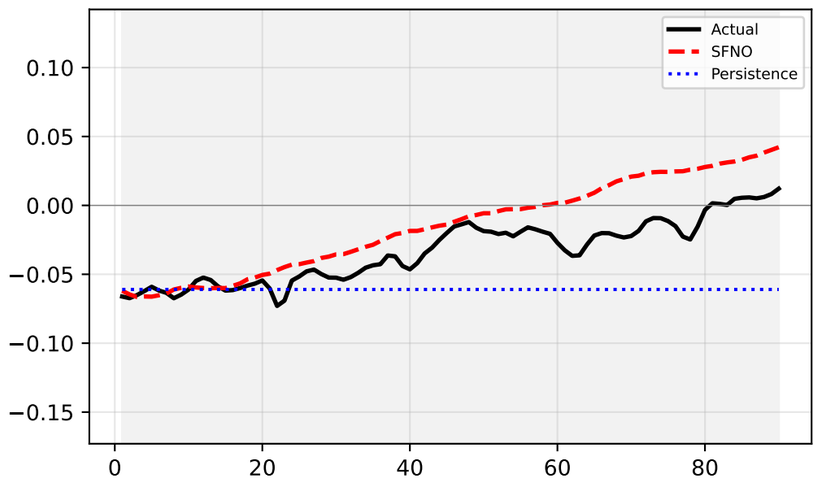


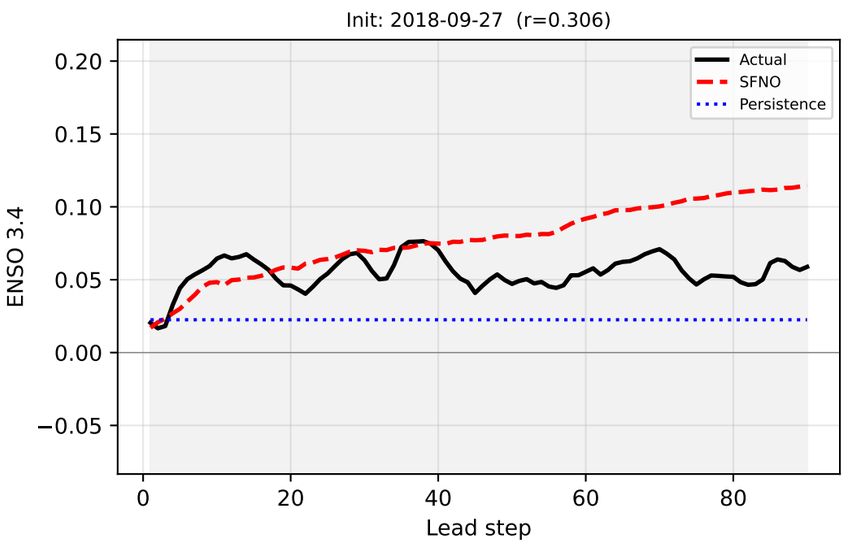


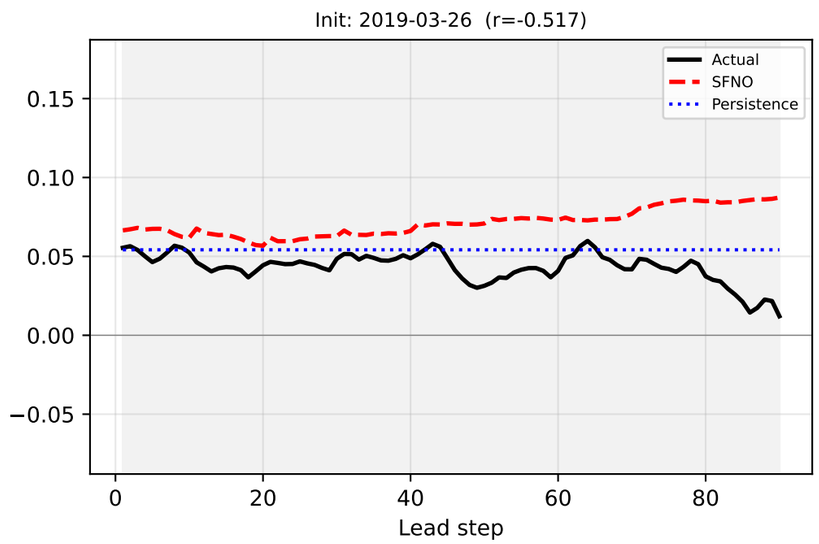


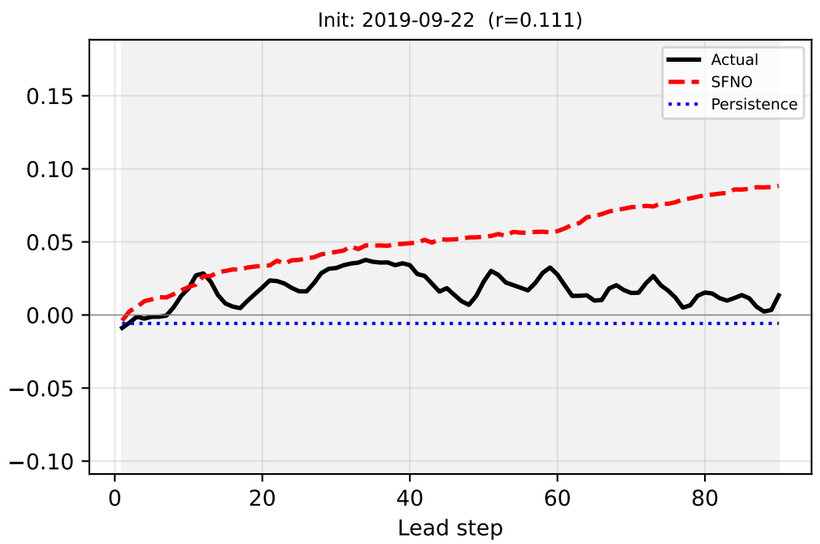




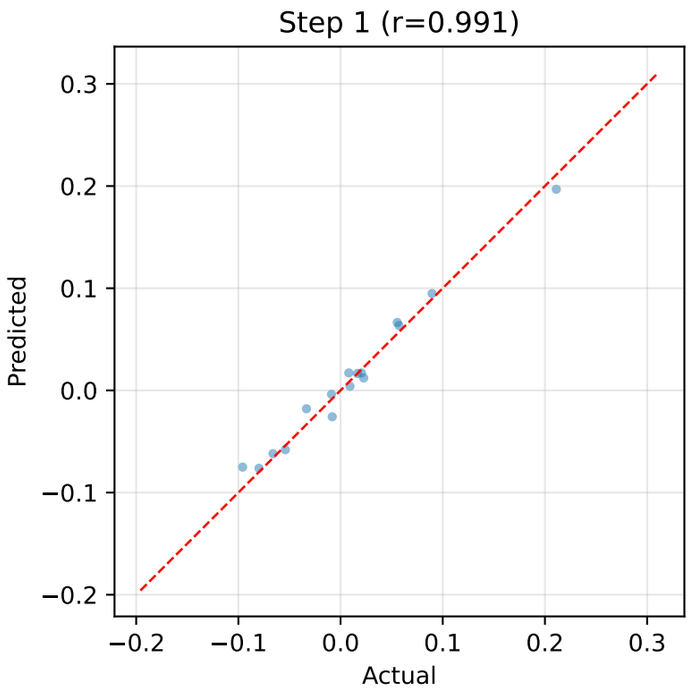


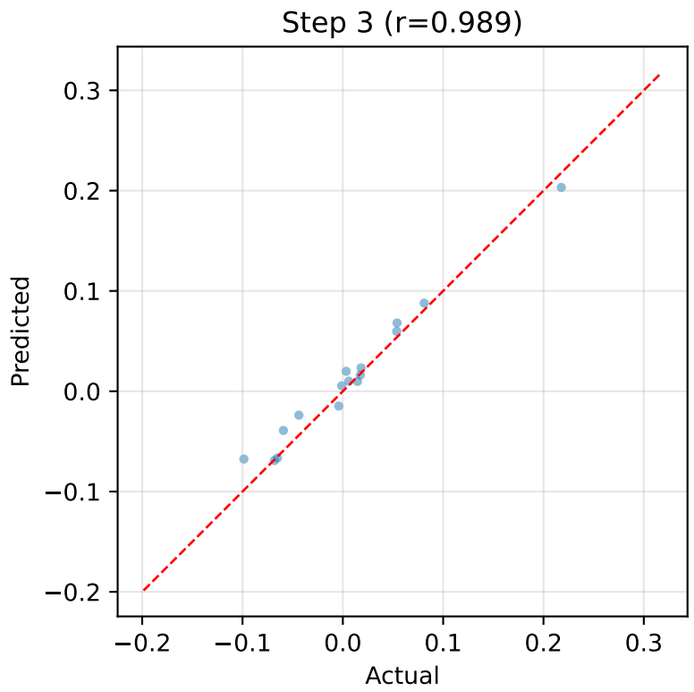


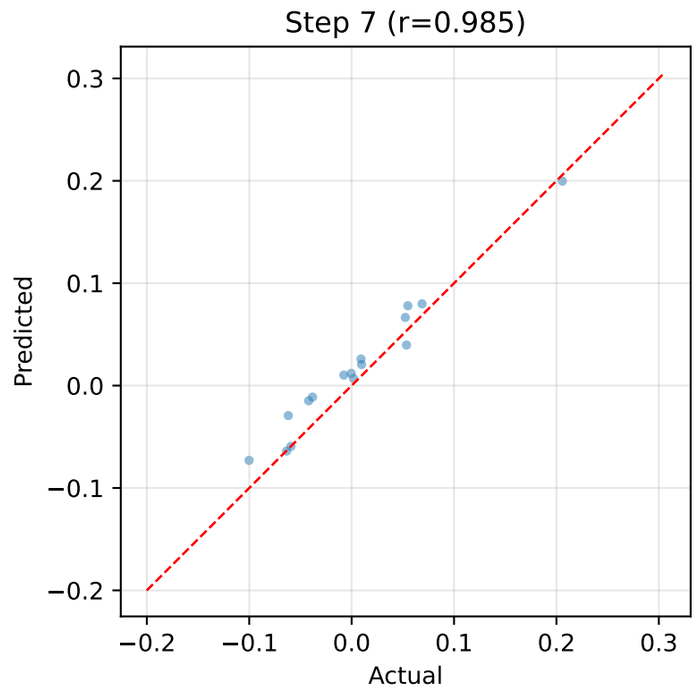


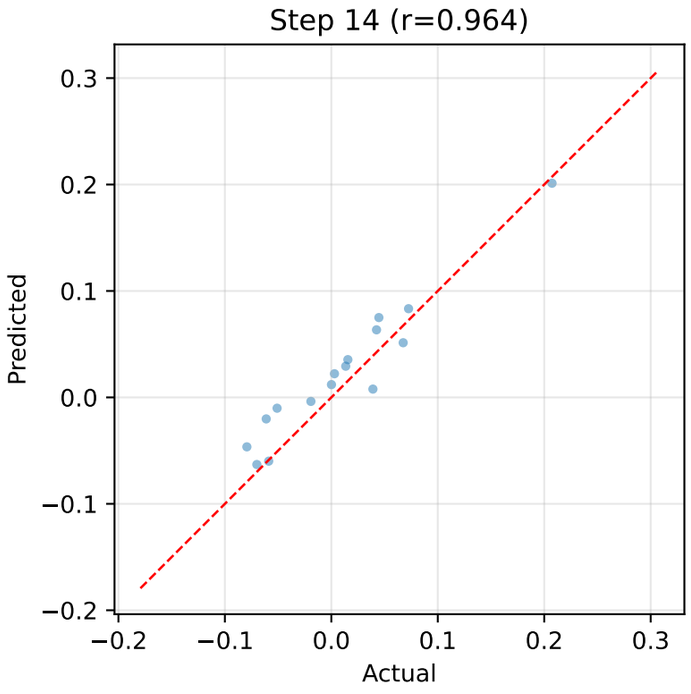


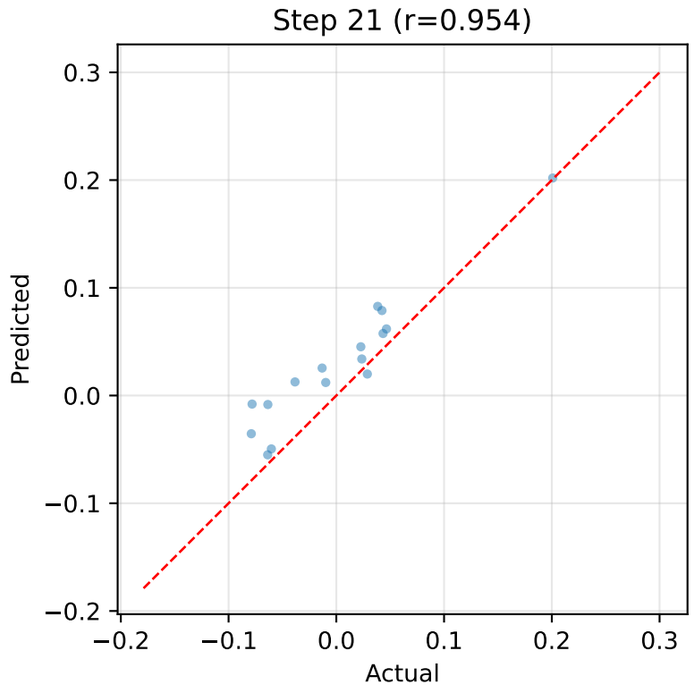
ENSO 3.4 Scatter — Validation

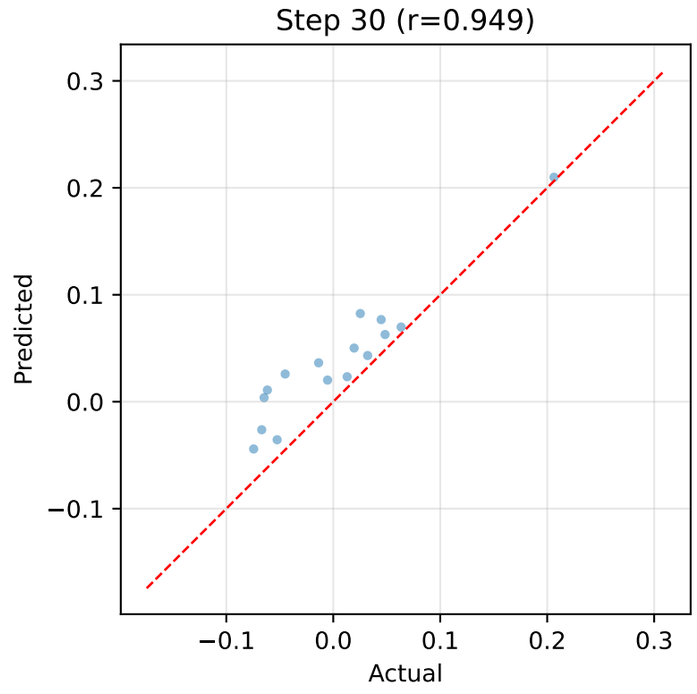


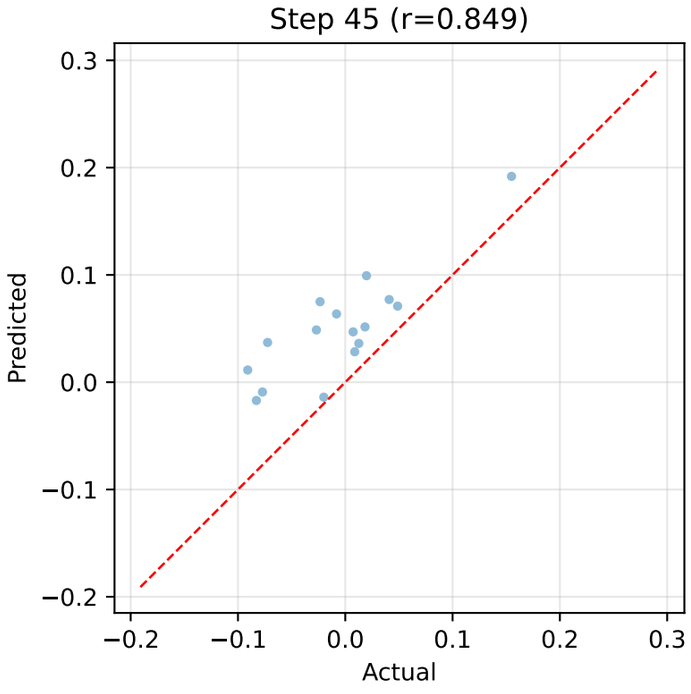


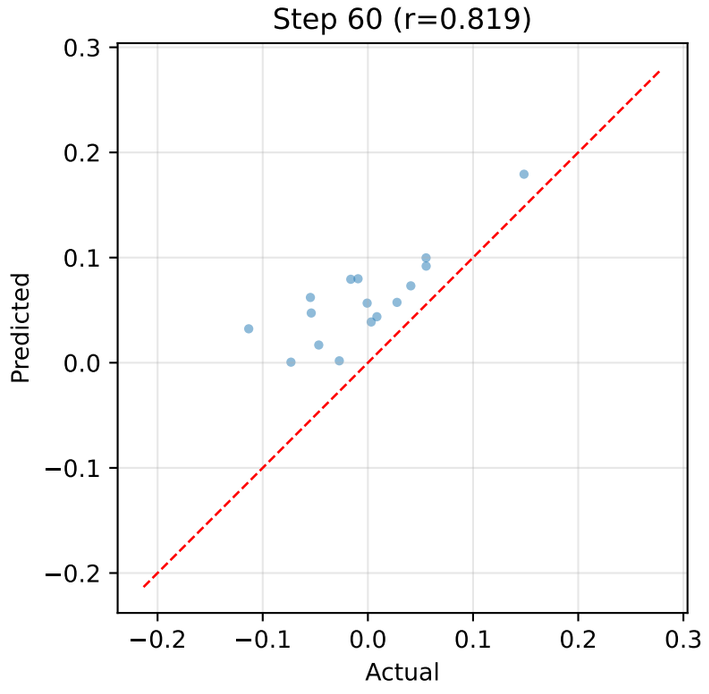


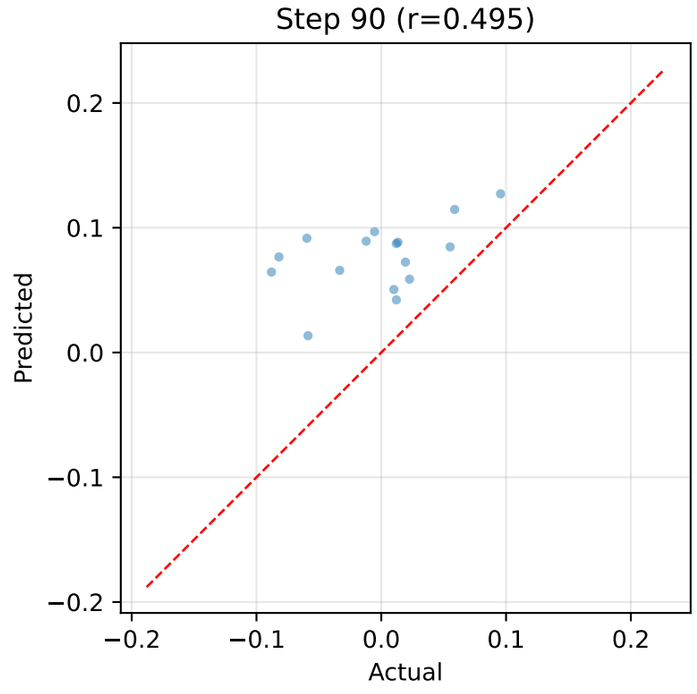








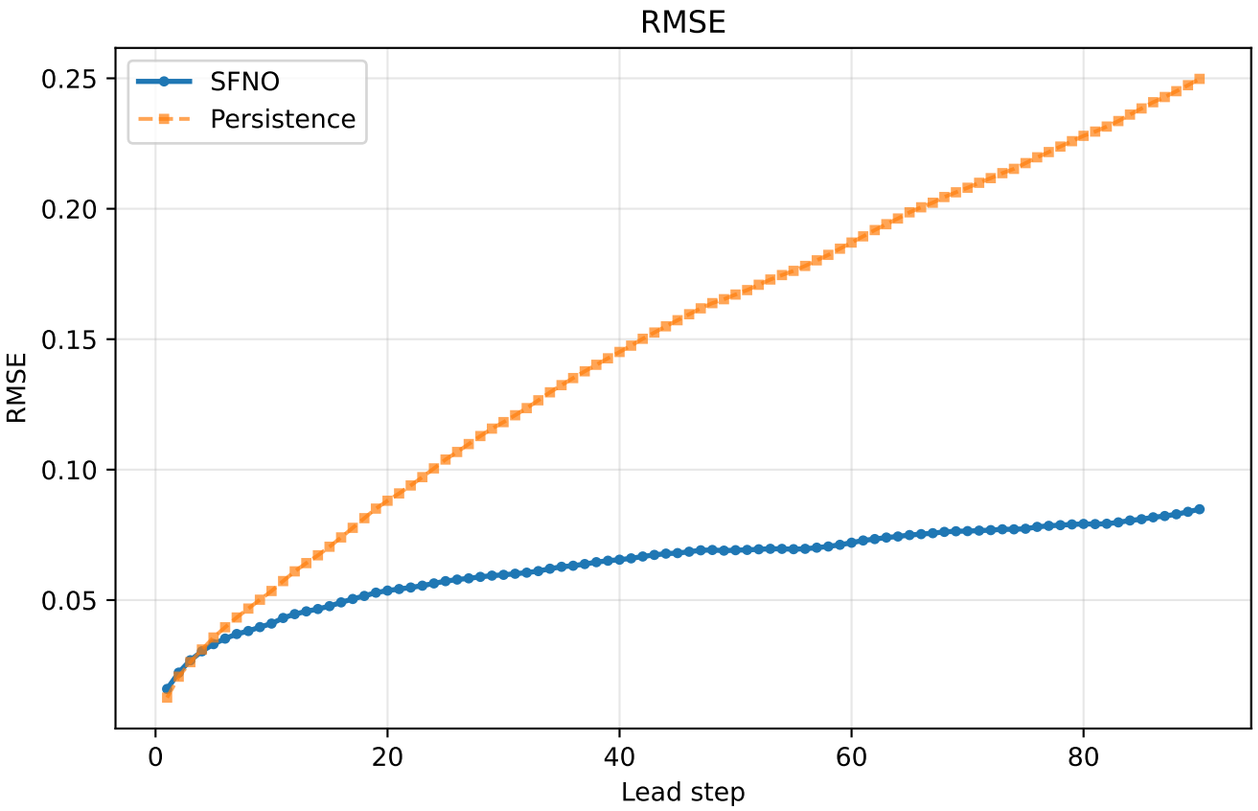


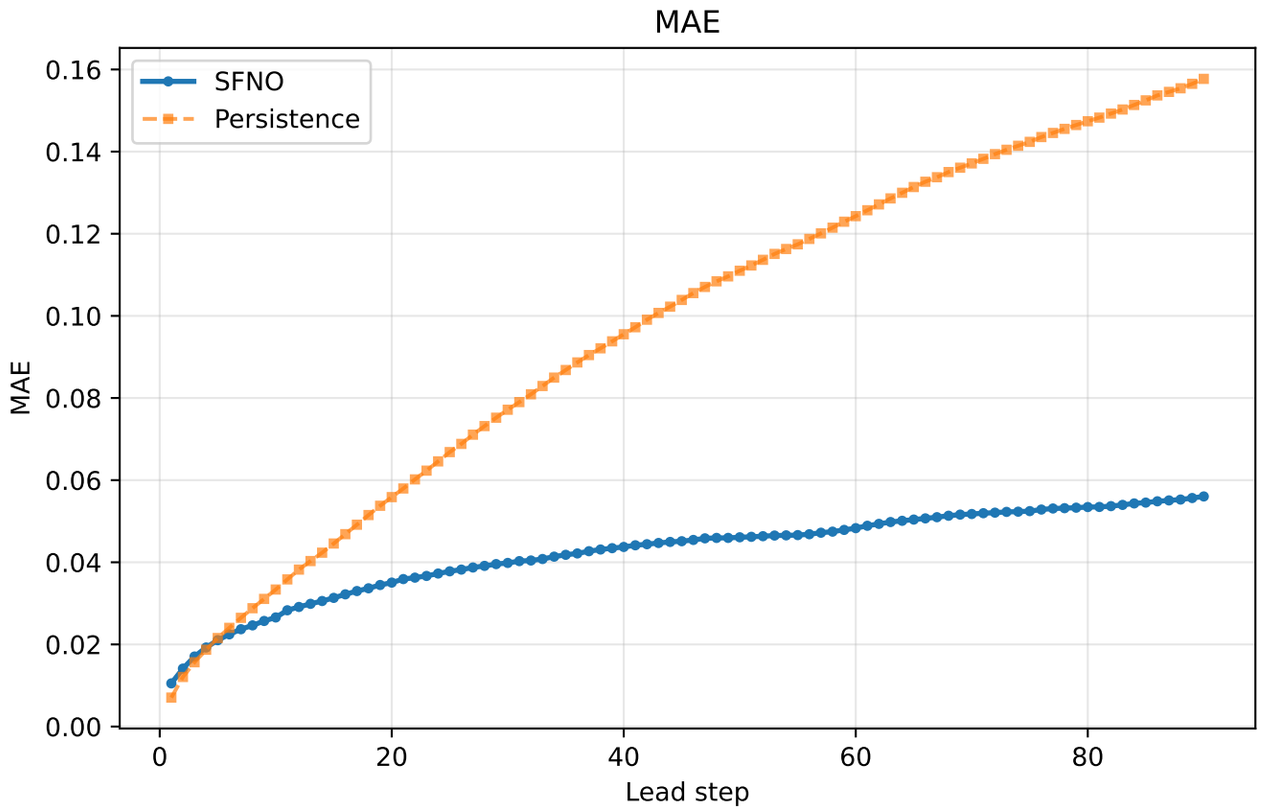


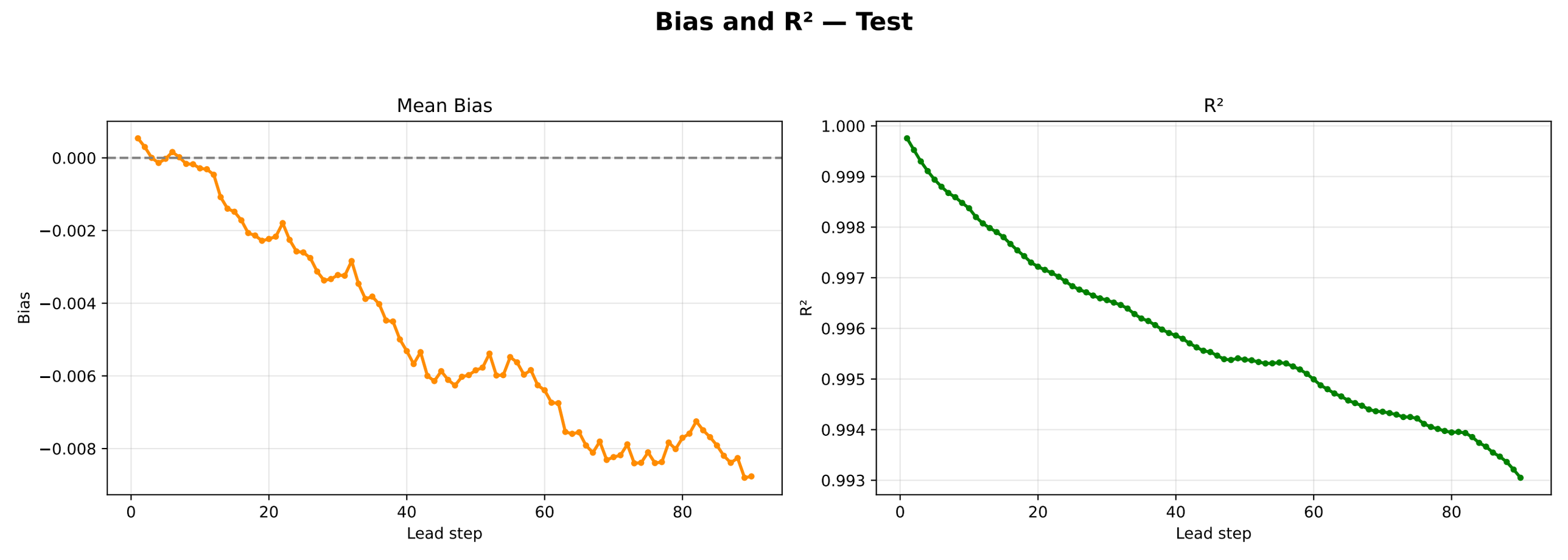
Metrics by Lead Step — Test

Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0159	0.0126	-26.2%	0.0105	0.0005	0.9706	0.9796	0.9998
2	0.0222	0.0206	-7.5%	0.0141	0.0003	0.9405	0.9462	0.9995
3	0.0269	0.0262	-2.8%	0.0171	0.0000	0.9091	0.9132	0.9993
4	0.0304	0.0311	2.2%	0.0193	-0.0001	0.8806	0.8780	0.9991
5	0.0331	0.0357	7.1%	0.0210	-0.0000	0.8546	0.8411	0.9989
6	0.0352	0.0396	11.0%	0.0225	0.0002	0.8312	0.8072	0.9988
7	0.0370	0.0433	14.6%	0.0237	0.0000	0.8104	0.7728	0.9987
8	0.0381	0.0467	18.4%	0.0246	-0.0002	0.7928	0.7389	0.9986
9	0.0396	0.0501	20.9%	0.0257	-0.0002	0.7744	0.7063	0.9985
10	0.0410	0.0535	23.3%	0.0266	-0.0003	0.7575	0.6743	0.9984
11	0.0431	0.0573	24.7%	0.0283	-0.0003	0.7341	0.6410	0.9982
12	0.0446	0.0610	26.9%	0.0291	-0.0005	0.7155	0.6068	0.9981
13	0.0457	0.0642	28.8%	0.0299	-0.0011	0.6996	0.5769	0.9980
14	0.0466	0.0672	30.7%	0.0306	-0.0014	0.6856	0.5500	0.9979
15	0.0477	0.0705	32.3%	0.0313	-0.0015	0.6699	0.5219	0.9978
16	0.0491	0.0740	33.6%	0.0322	-0.0017	0.6516	0.4938	0.9977
17	0.0504	0.0777	35.1%	0.0330	-0.0021	0.6350	0.4629	0.9975
18	0.0516	0.0814	36.6%	0.0337	-0.0021	0.6192	0.4320	0.9974
19	0.0528	0.0851	37.9%	0.0345	-0.0023	0.6020	0.4030	0.9973
20	0.0536	0.0881	39.1%	0.0351	-0.0022	0.5877	0.3801	0.9972
21	0.0542	0.0909	40.3%	0.0359	-0.0022	0.5733	0.3592	0.9972
22	0.0548	0.0939	41.6%	0.0363	-0.0018	0.5634	0.3356	0.9971
23	0.0555	0.0971	42.8%	0.0367	-0.0023	0.5539	0.3131	0.9970
24	0.0564	0.1005	43.9%	0.0373	-0.0026	0.5429	0.2889	0.9969
25	0.0573	0.1039	44.9%	0.0378	-0.0026	0.5301	0.2670	0.9968
26	0.0579	0.1068	45.8%	0.0382	-0.0028	0.5166	0.2487	0.9968
27	0.0583	0.1098	46.9%	0.0387	-0.0031	0.5040	0.2287	0.9967
28	0.0589	0.1129	47.8%	0.0391	-0.0034	0.4930	0.2138	0.9966
29	0.0594	0.1157	48.7%	0.0396	-0.0033	0.4800	0.2032	0.9966
30	0.0597	0.1182	49.5%	0.0398	-0.0032	0.4670	0.1947	0.9966
31	0.0601	0.1208	50.3%	0.0403	-0.0032	0.4523	0.1854	0.9965
32	0.0605	0.1236	51.0%	0.0404	-0.0028	0.4421	0.1758	0.9965
33	0.0611	0.1266	51.7%	0.0408	-0.0035	0.4285	0.1635	0.9964
34	0.0620	0.1296	52.2%	0.0414	-0.0039	0.4149	0.1532	0.9963
35	0.0628	0.1324	52.6%	0.0418	-0.0038	0.4007	0.1445	0.9962
36	0.0632	0.1351	53.3%	0.0422	-0.0040	0.3873	0.1347	0.9961
37	0.0638	0.1377	53.7%	0.0427	-0.0045	0.3731	0.1282	0.9961
38	0.0645	0.1402	54.0%	0.0431	-0.0045	0.3607	0.1243	0.9960
39	0.0651	0.1427	54.4%	0.0434	-0.0050	0.3521	0.1209	0.9959
40	0.0655	0.1451	54.9%	0.0437	-0.0053	0.3430	0.1145	0.9959
41	0.0660	0.1475	55.3%	0.0441	-0.0057	0.3293	0.1073	0.9958
42	0.0667	0.1502	55.6%	0.0444	-0.0053	0.3184	0.1002	0.9957
43	0.0673	0.1526	55.9%	0.0447	-0.0060	0.3068	0.0960	0.9956
44	0.0678	0.1549	56.2%	0.0450	-0.0061	0.2943	0.0912	0.9956
45	0.0680	0.1573	56.8%	0.0451	-0.0059	0.2815	0.0831	0.9955
46	0.0685	0.1596	57.0%	0.0454	-0.0061	0.2679	0.0757	0.9955
47	0.0691	0.1618	57.3%	0.0458	-0.0063	0.2569	0.0700	0.9954
48	0.0692	0.1638	57.8%	0.0459	-0.0060	0.2504	0.0685	0.9954
49	0.0690	0.1653	58.3%	0.0459	-0.0060	0.2459	0.0718	0.9954
50	0.0691	0.1671	58.6%	0.0461	-0.0058	0.2389	0.0747	0.9954
51	0.0692	0.1688	59.0%	0.0462	-0.0058	0.2302	0.0768	0.9954
52	0.0695	0.1709	59.3%	0.0464	-0.0054	0.2231	0.0753	0.9953
53	0.0697	0.1729	59.7%	0.0465	-0.0060	0.2169	0.0730	0.9953
54	0.0696	0.1746	60.1%	0.0466	-0.0060	0.2133	0.0736	0.9953
55	0.0695	0.1762	60.5%	0.0466	-0.0055	0.2097	0.0758	0.9953
56	0.0697	0.1781	60.9%	0.0468	-0.0056	0.2033	0.0736	0.9953
57	0.0701	0.1802	61.1%	0.0472	-0.0060	0.1945	0.0694	0.9952
58	0.0705	0.1823	61.3%	0.0475	-0.0058	0.1868	0.0654	0.9952
59	0.0712	0.1847	61.5%	0.0479	-0.0063	0.1786	0.0584	0.9951
60	0.0720	0.1870	61.5%	0.0483	-0.0064	0.1705	0.0535	0.9950
61	0.0728	0.1894	61.5%	0.0489	-0.0067	0.1595	0.0470	0.9949
62	0.0734	0.1918	61.7%	0.0494	-0.0068	0.1501	0.0385	0.9948
63	0.0740	0.1941	61.9%	0.0498	-0.0075	0.1388	0.0316	0.9947
64	0.0744	0.1963	62.1%	0.0501	-0.0076	0.1294	0.0248	0.9947
65	0.0749	0.1986	62.3%	0.0504	-0.0076	0.1219	0.0192	0.9946
66	0.0753	0.2006	62.5%	0.0507	-0.0079	0.1159	0.0171	0.9945
67	0.0756	0.2023	62.6%	0.0510	-0.0081	0.1119	0.0183	0.9945
68	0.0761	0.2045	62.8%	0.0513	-0.0078	0.1030	0.0135	0.9944
69	0.0764	0.2063	63.0%	0.0516	-0.0083	0.0986	0.0125	0.9944
70	0.0764	0.2080	63.3%	0.0518	-0.0082	0.0951	0.0122	0.9944
71	0.0766	0.2099	63.5%	0.0519	-0.0082	0.0902	0.0110	0.9943
72	0.0768	0.2117	63.7%	0.0521	-0.0079	0.0877	0.0120	0.9943
73	0.0771	0.2136	63.9%	0.0523	-0.0084	0.0848	0.0109	0.9942
74	0.0771	0.2153	64.2%	0.0523	-0.0084	0.0861	0.0133	0.9942
75	0.0773	0.2175	64.4%	0.0525	-0.0081	0.0861	0.0086	0.9942
76	0.0781	0.2197	64.5%	0.0528	-0.0084	0.0800	0.0050	0.9941
77	0.0785	0.2217	64.6%	0.0531	-0.0084	0.0769	0.0049	0.9941
78	0.0787	0.2238	64.8%	0.0532	-0.0078	0.0740	0.0037	0.9940
79	0.0790	0.2259	65.0%	0.0533	-0.0080	0.0720	0.0019	0.9940
80	0.0792	0.2279	65.3%	0.0535	-0.0077	0.0677	-0.0015	0.9939
81	0.0791	0.2296	65.5%	0.0535	-0.0076	0.0668	-0.0020	0.9940
82	0.0793	0.2315	65.8%	0.0537	-0.0073	0.0626	-0.0056	0.9939
83	0.0798	0.2336	65.9%	0.0540	-0.0075	0.0583	-0.0076	0.9939
84	0.0805	0.2361	65.9%	0.0543	-0.0077	0.0557	-0.0090	0.9937
85	0.0810	0.2385	66.0%	0.0545	-0.0079	0.0541	-0.0119	0.9937
86	0.0817	0.2408	66.1%	0.0549	-0.0082	0.0507	-0.0147	0.9935
87	0.0822	0.2428	66.1%	0.0551	-0.0084	0.0483	-0.0142	0.9935
88	0.0829	0.2451	66.2%	0.0553	-0.0083	0.0453	-0.0133	0.9934
89	0.0838	0.2474	66.1%	0.0556	-0.0088	0.0426	-0.0151	0.9932
90	0.0848	0.2498	66.0%	0.0560	-0.0088	0.0370	-0.0180	0.9930

Forecast Skill vs Lead Step — Test

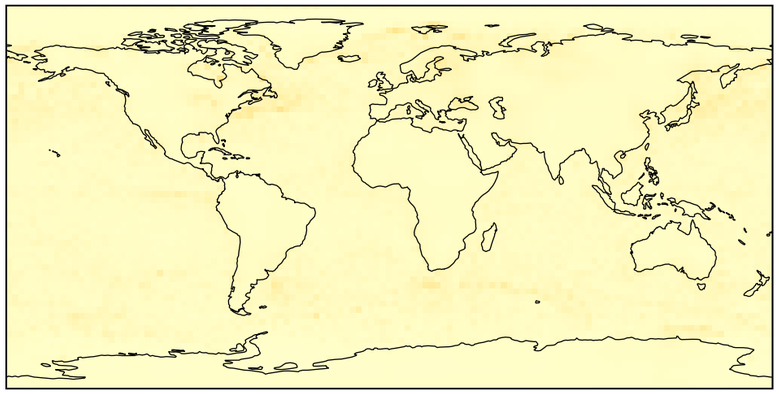




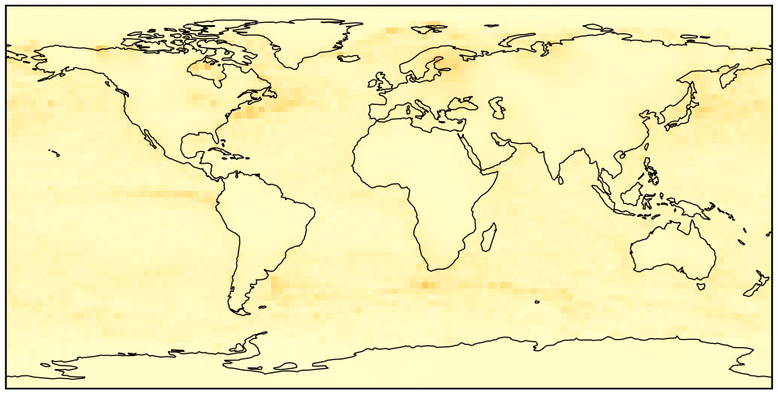

Bias and R² — Test
Mean Bias
Bias
0.000
−0.002
−0.004
−0.006
−0.008
0
20
40
60
80
Lead step
R²
R²
1.000
0.999
0.998
0.997
0.996
0.995
0.994
0.993
0
20
40
60
80
Lead step

Spatial RMSE (all grid) — Test

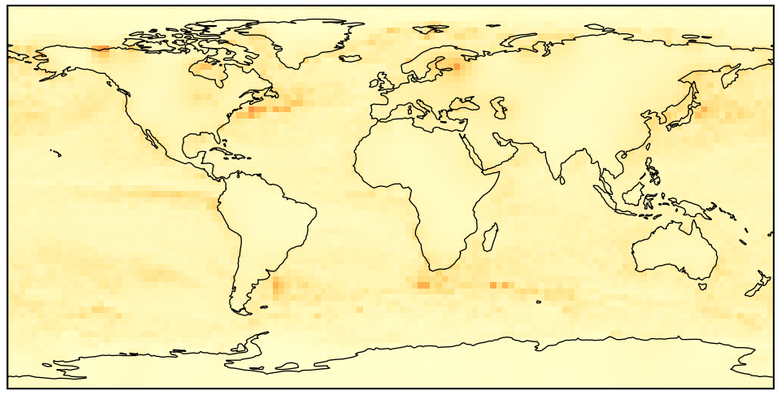
Step 1



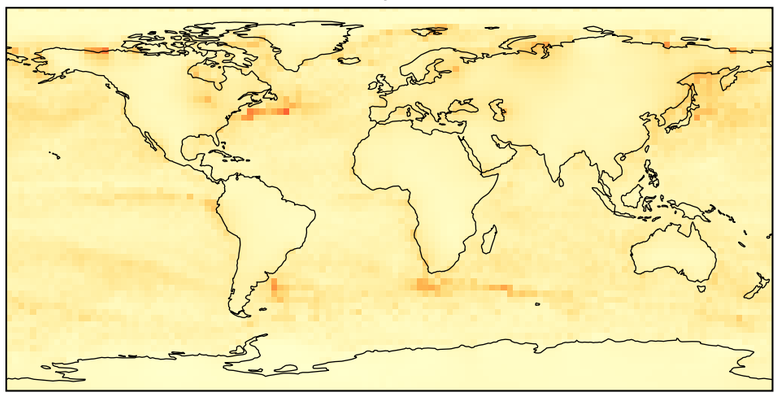
Step 3



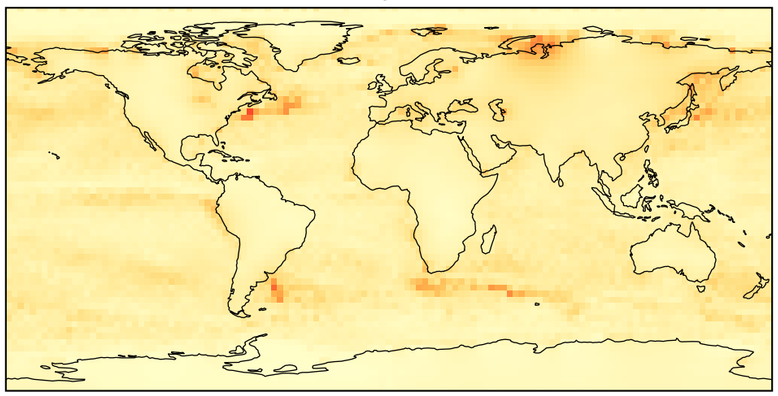
Step 7



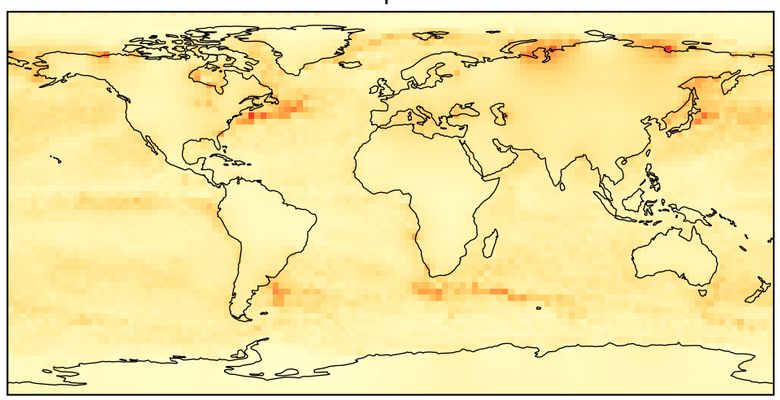
Step 14



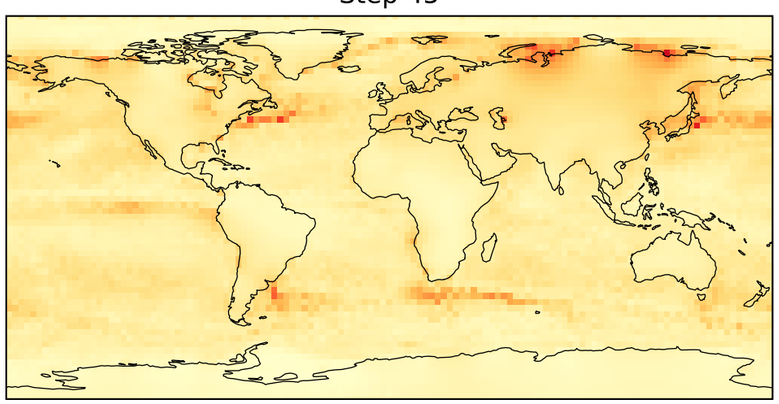
Step 21



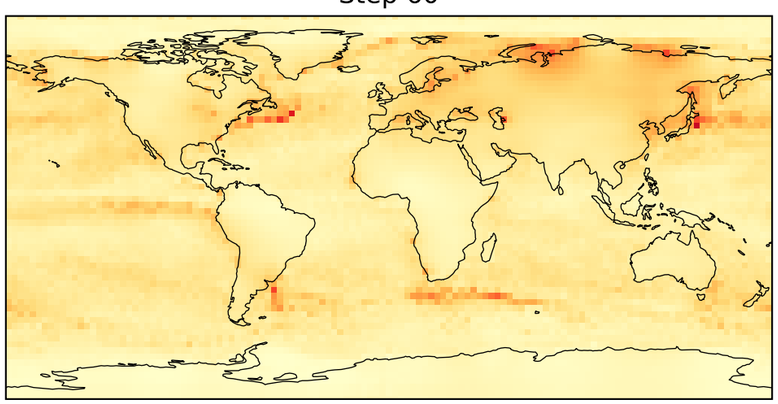
Step 30



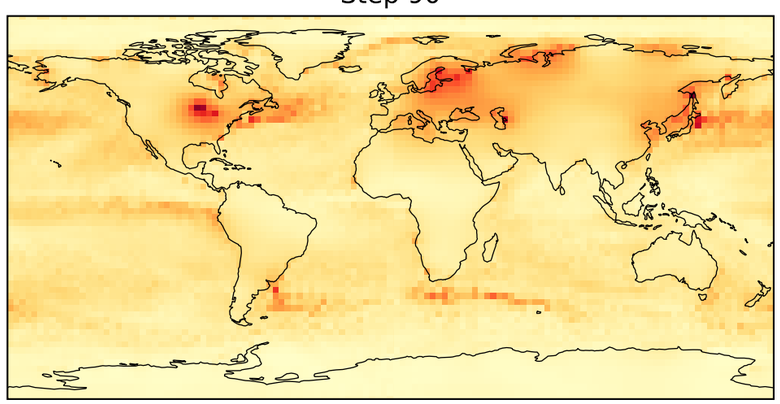
Step 45



Step 60

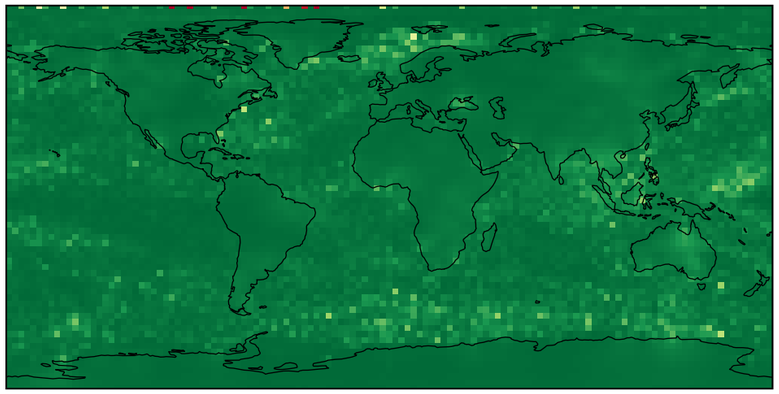


Step 90

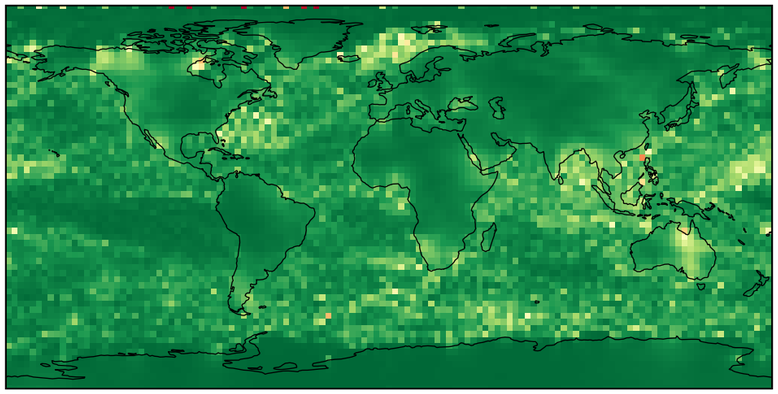


Spatial ACC (all grid) — Test

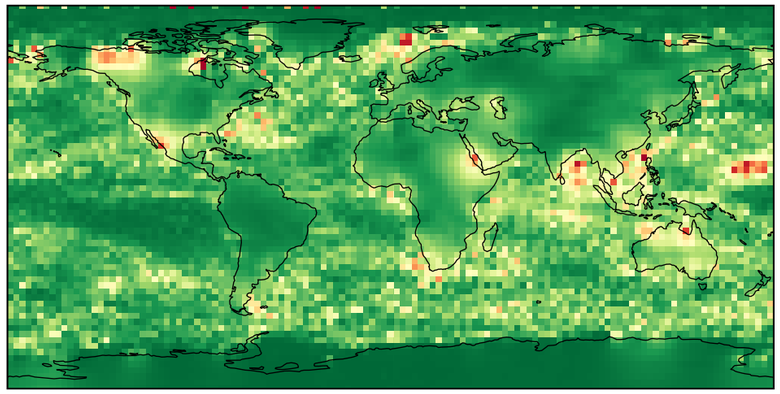
Step 1



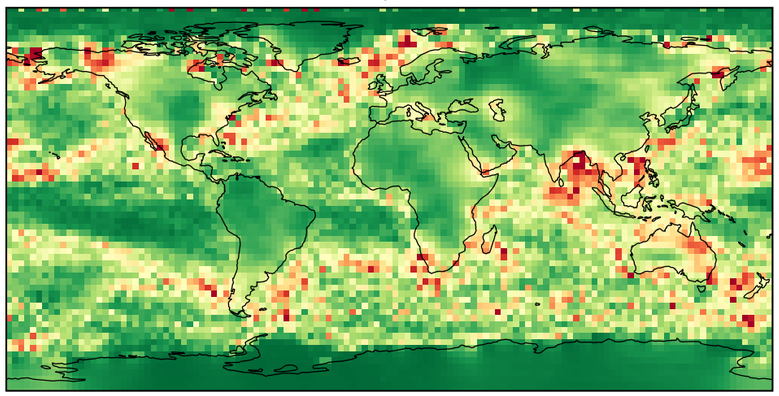
Step 3



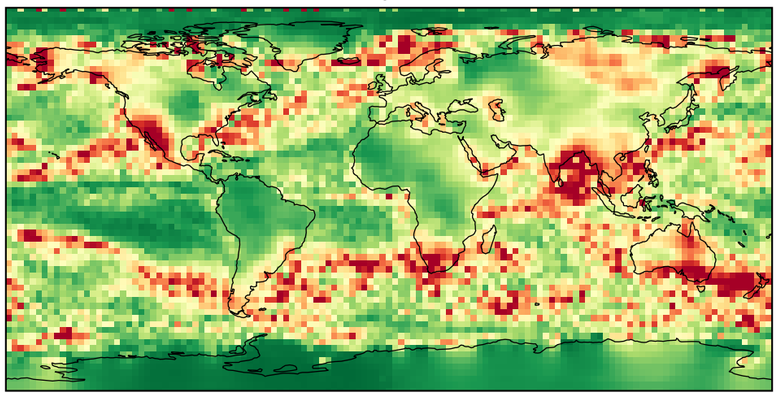
Step 7



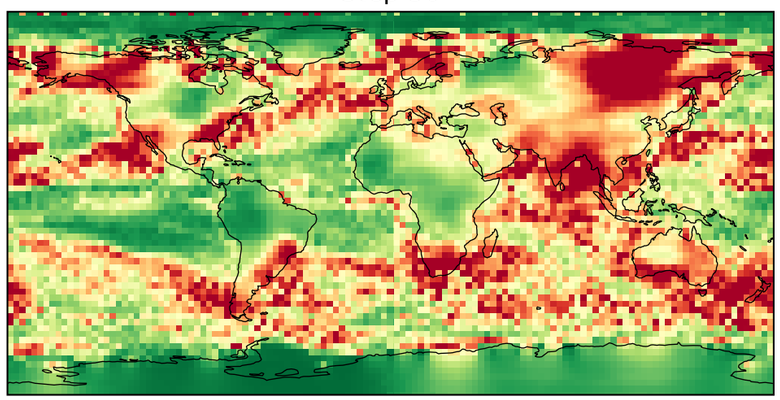
Step 14



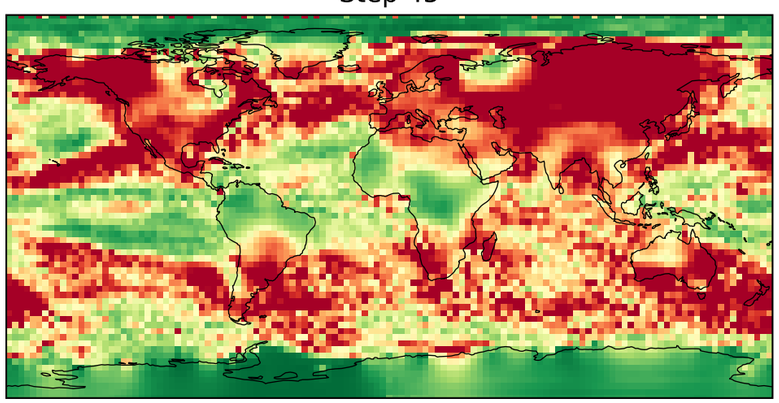
Step 21



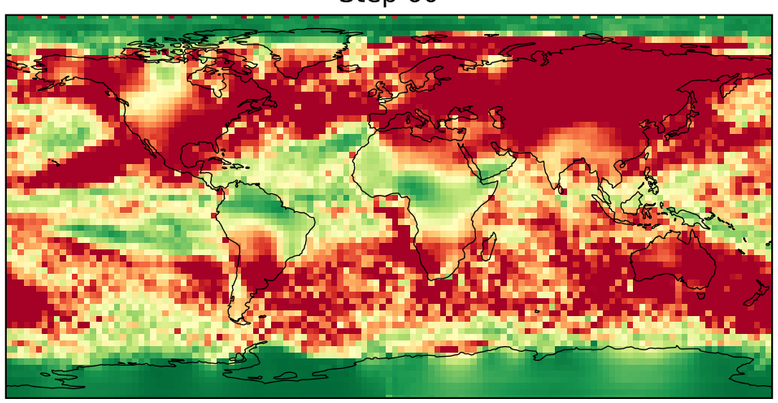
Step 30



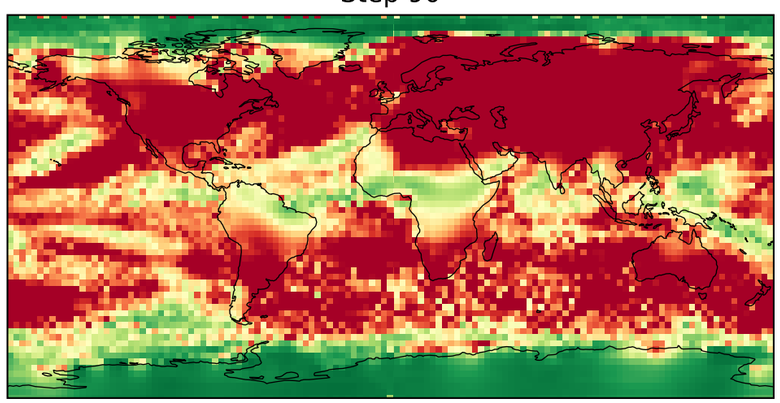
Step 45

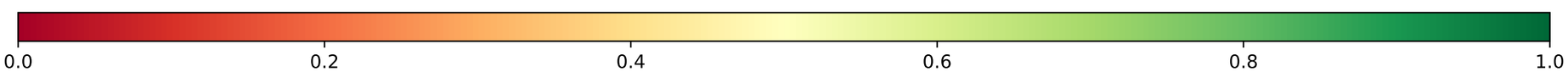


Step 60



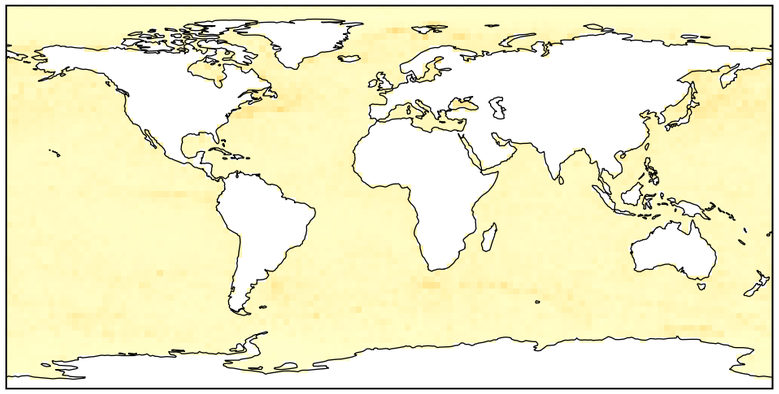
Step 90



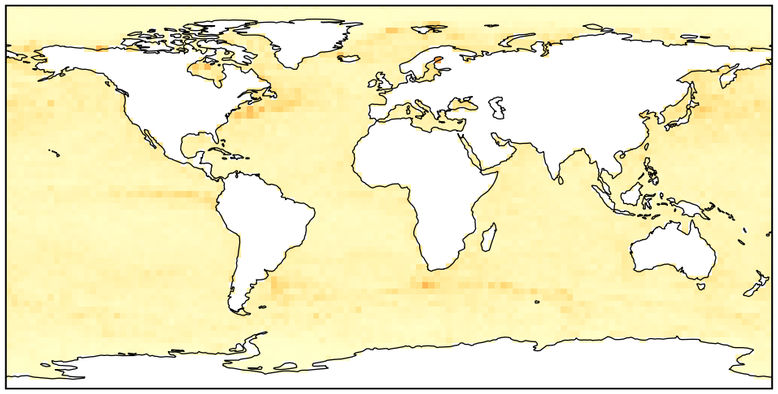


Spatial RMSE (ocean only) — Test

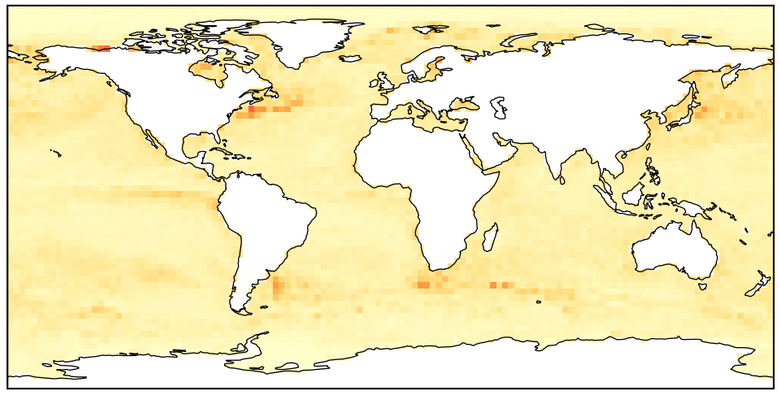
Step 1



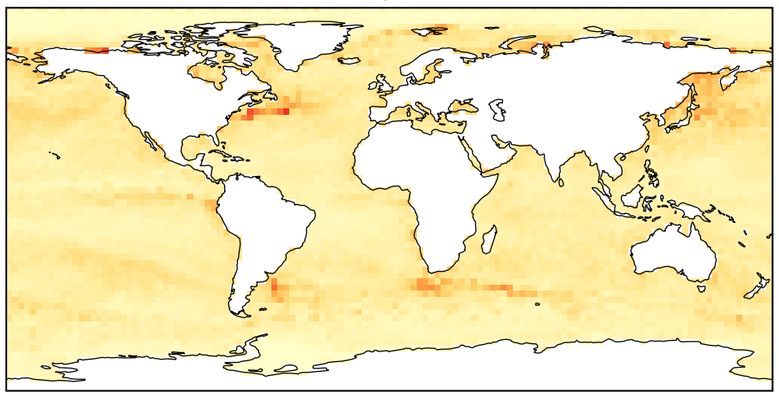
Step 3



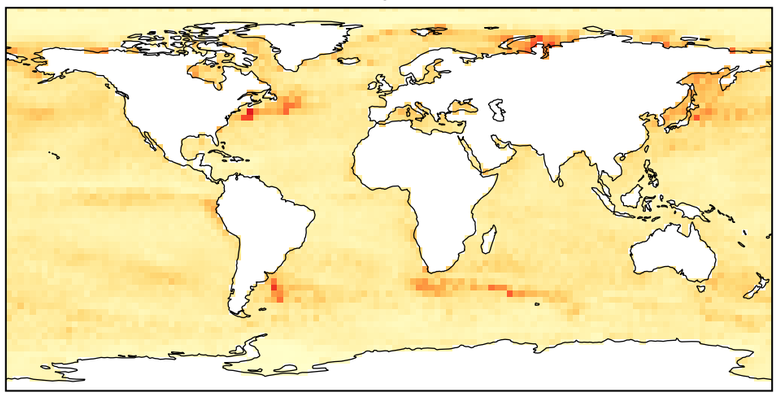
Step 7



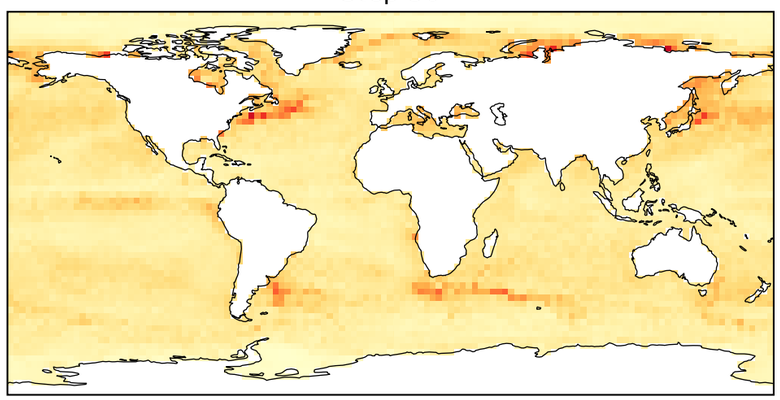
Step 14



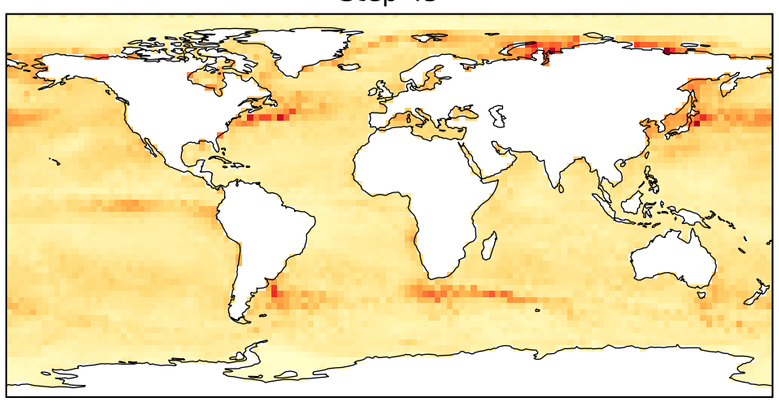
Step 21



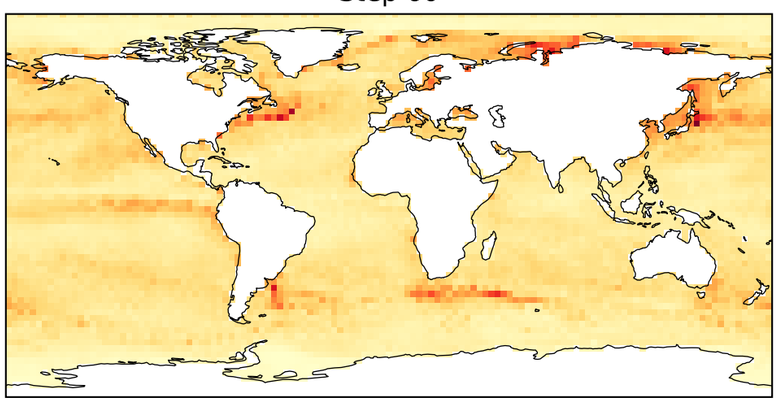
Step 30



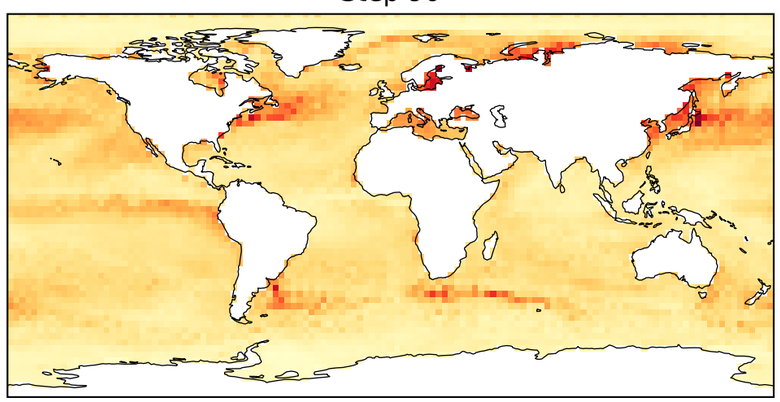
Step 45



Step 60

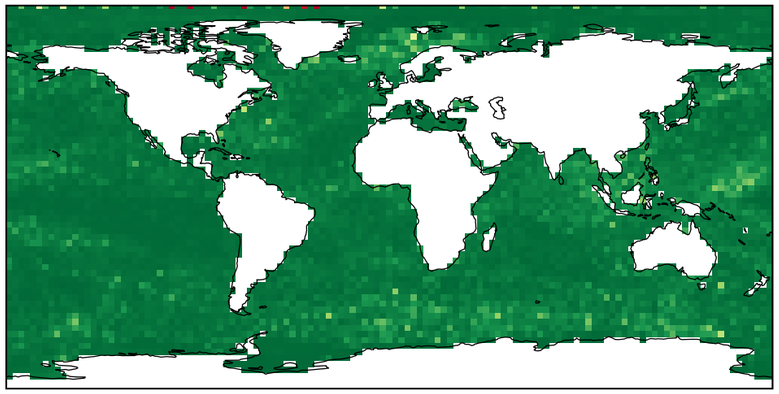


Step 90

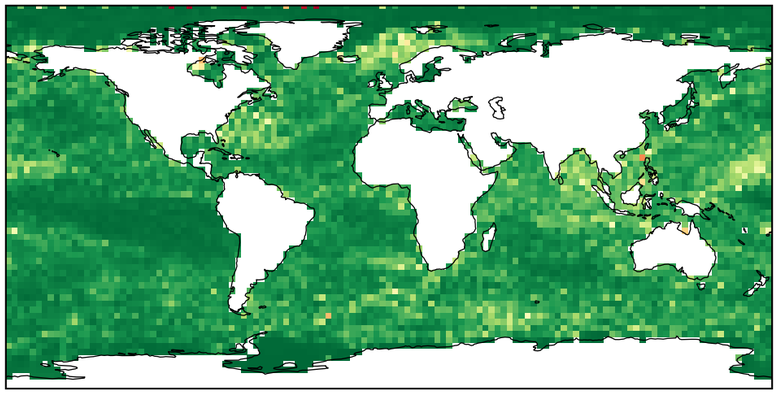


Spatial ACC (ocean only) — Test

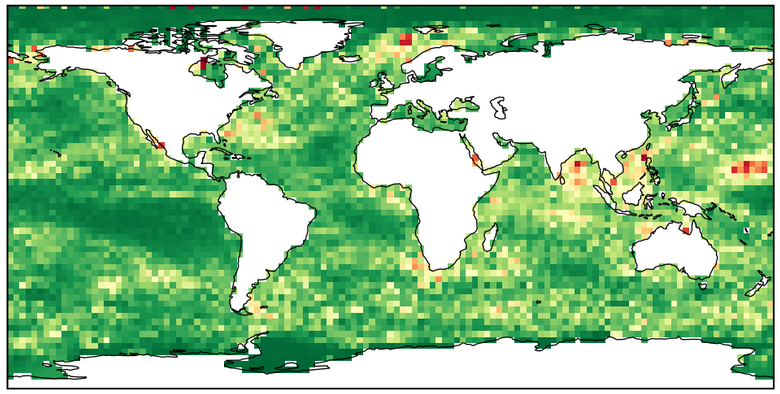
Step 1



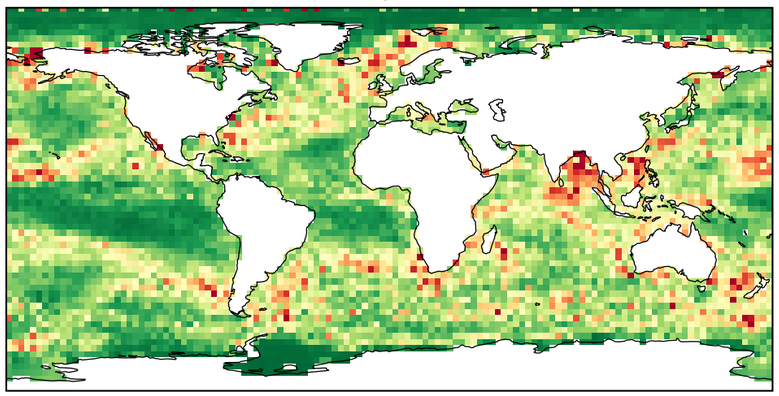
Step 3



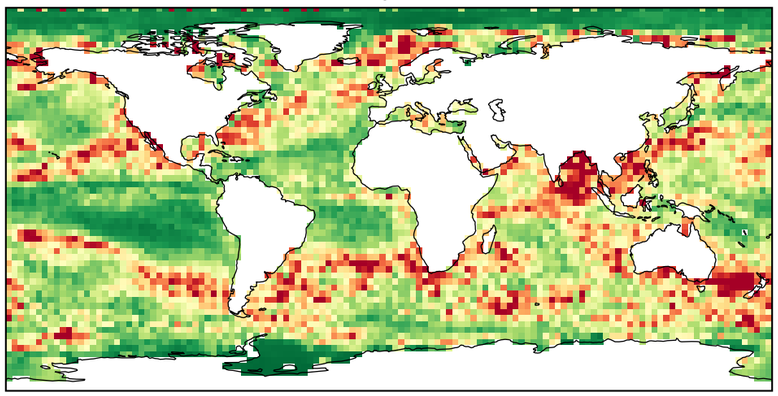
Step 7



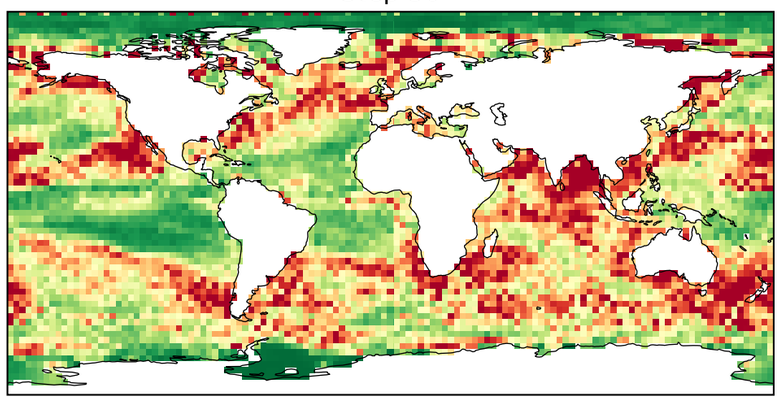
Step 14



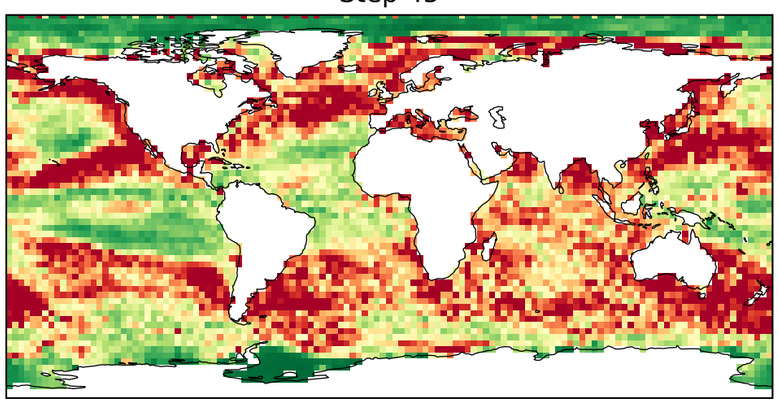
Step 21



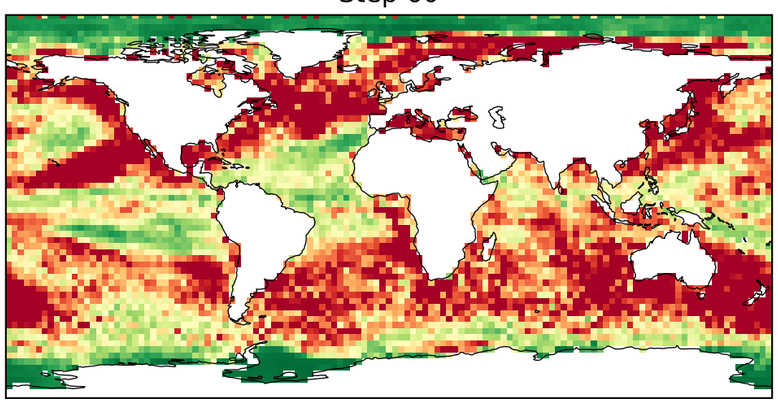
Step 30



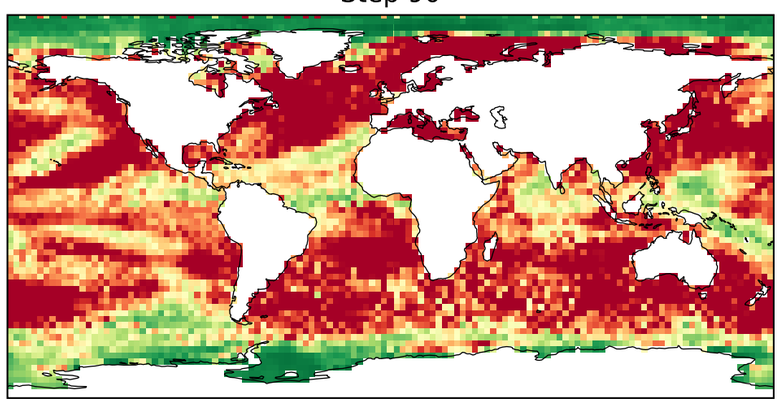
Step 45

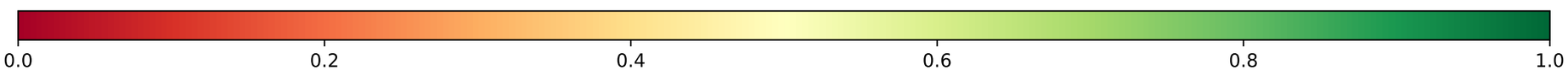


Step 60



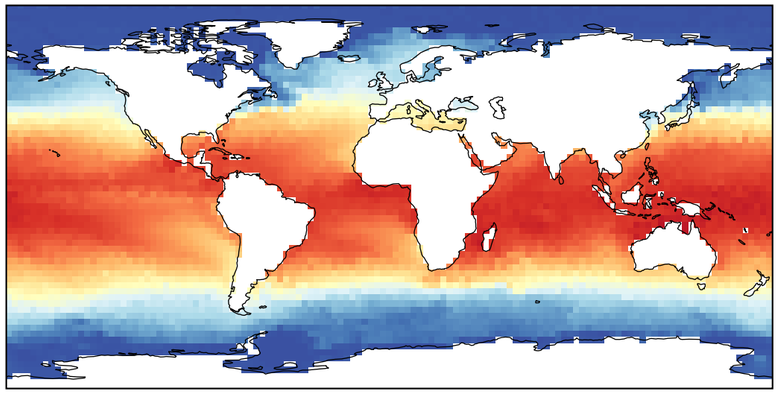
Step 90



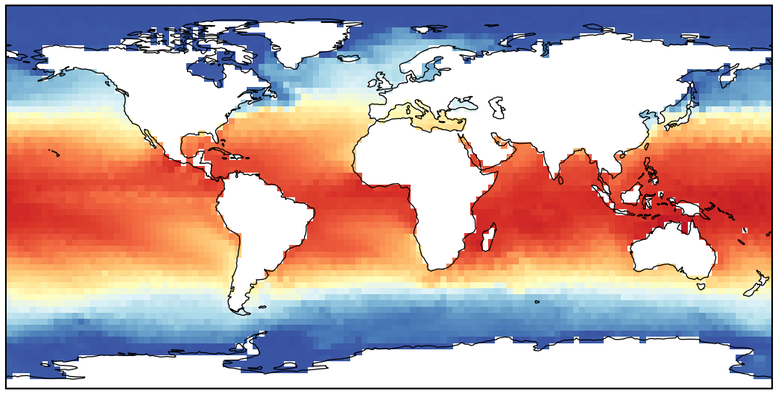


Chain init: 2020-01-11 — Test (°C)

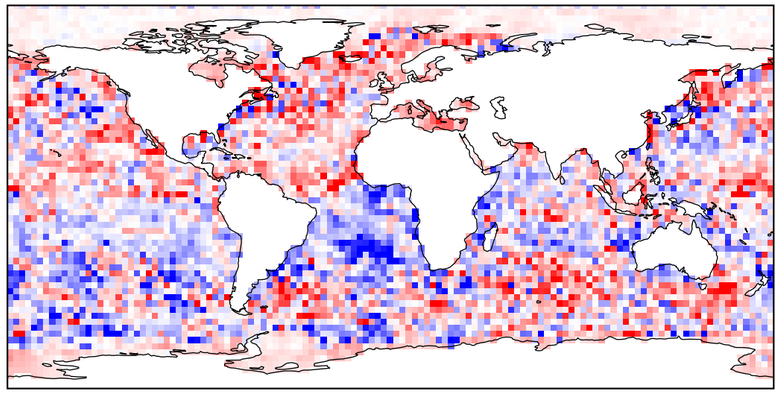
Actual — Step 1 (°C)



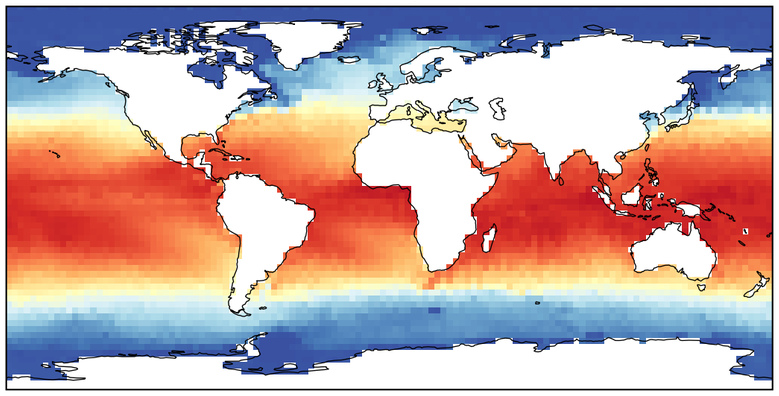
SFNO — Step 1 (°C)



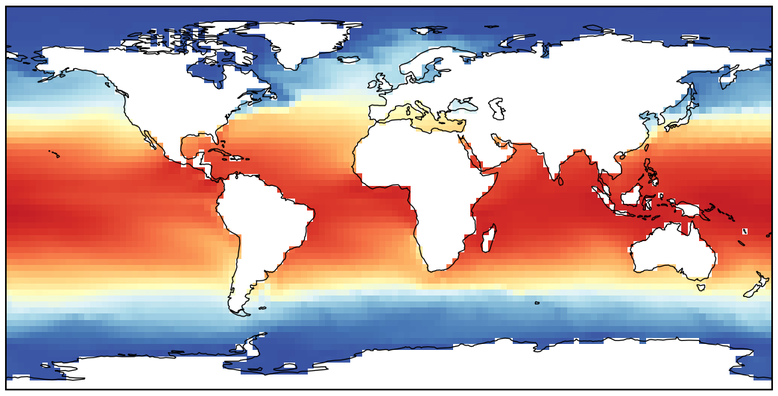
Error — Step 1 (°C)



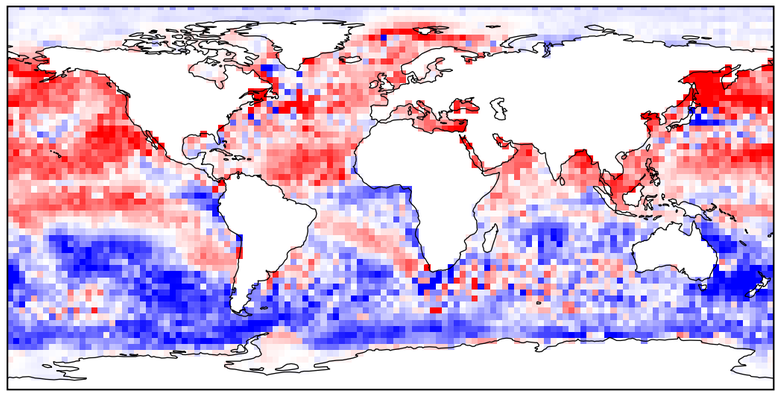
Actual — Step 46 (°C)



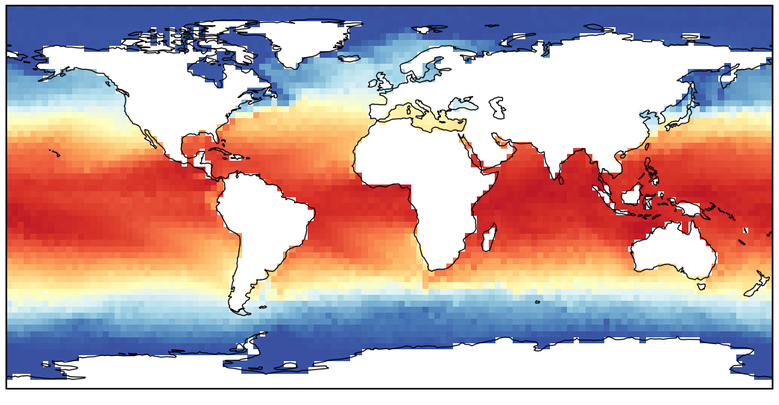
SFNO — Step 46 (°C)



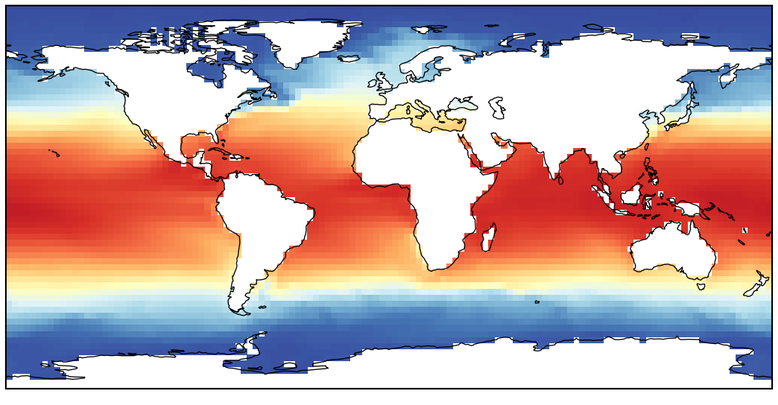
Error — Step 46 (°C)



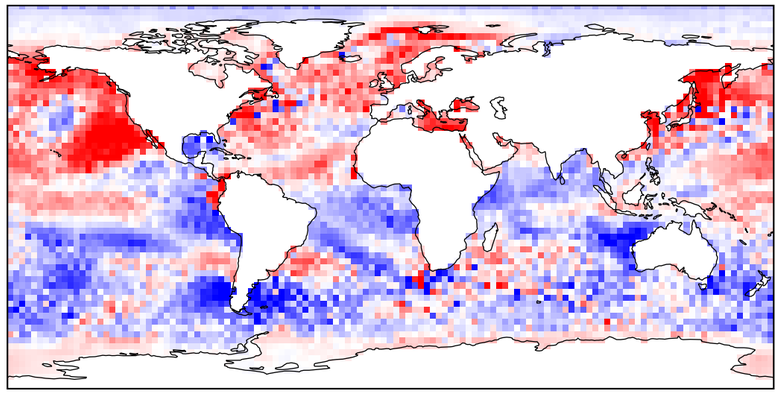
Actual — Step 90 (°C)

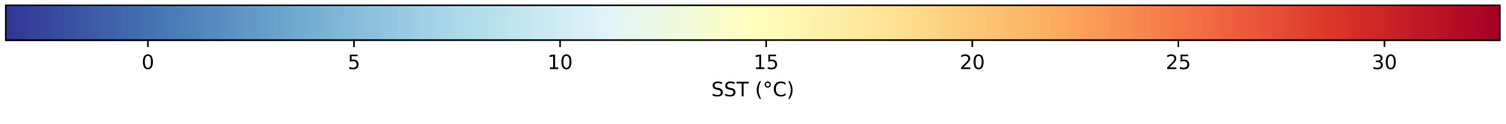


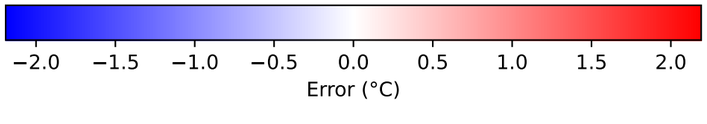
SFNO — Step 90 (°C)



Error — Step 90 (°C)

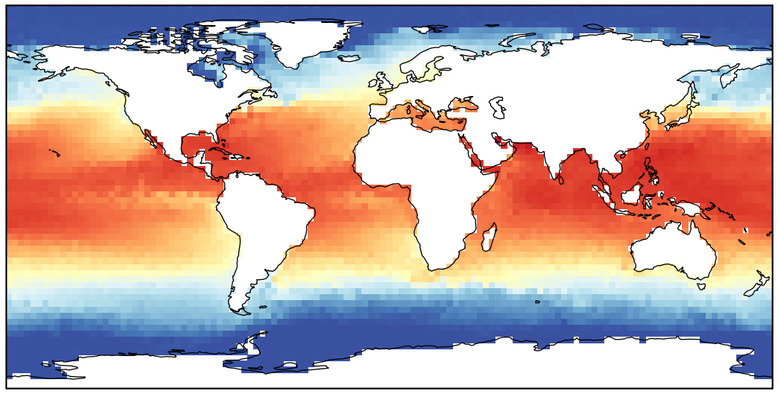




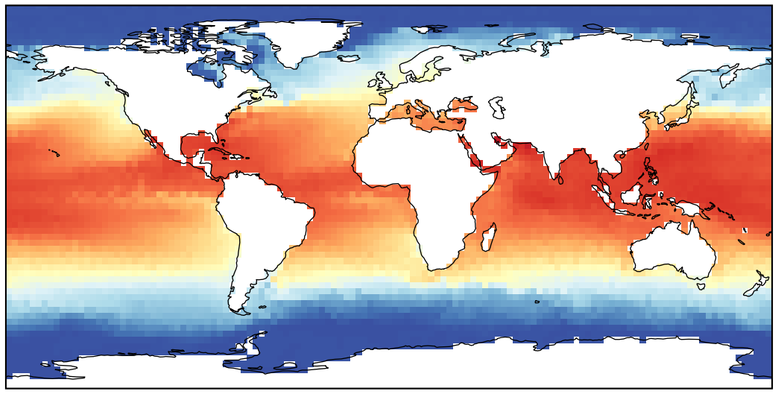


Chain init: 2020-07-09 — Test (°C)

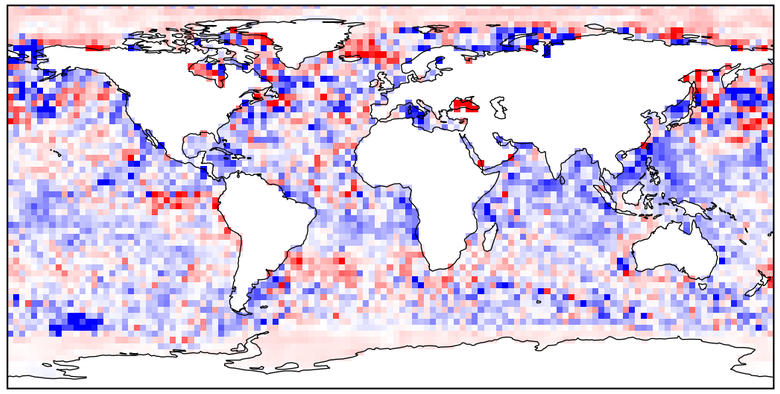
Actual — Step 1 (°C)



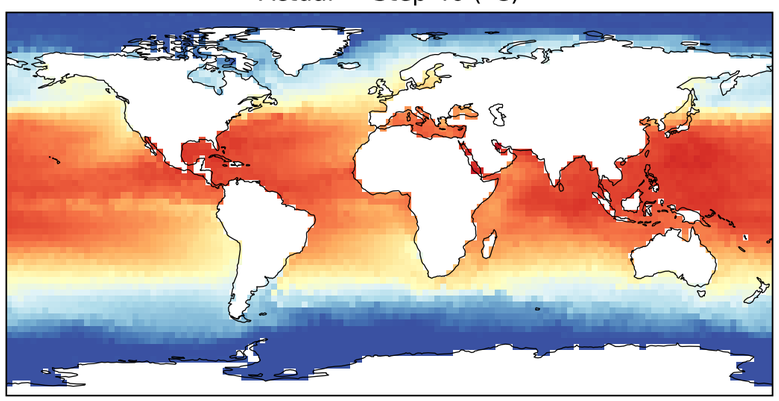
SFNO — Step 1 (°C)



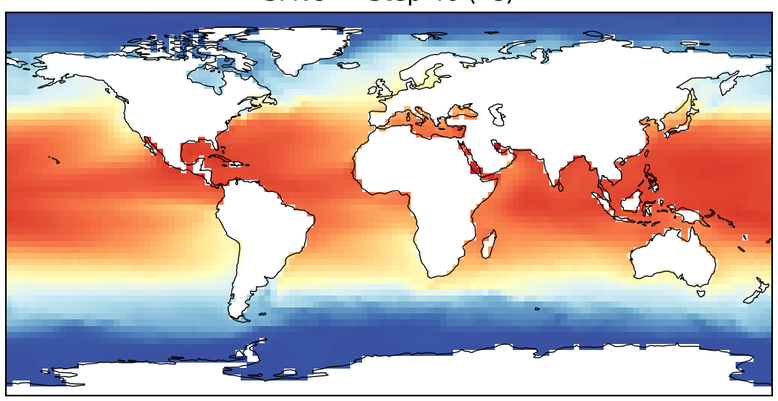
Error — Step 1 (°C)



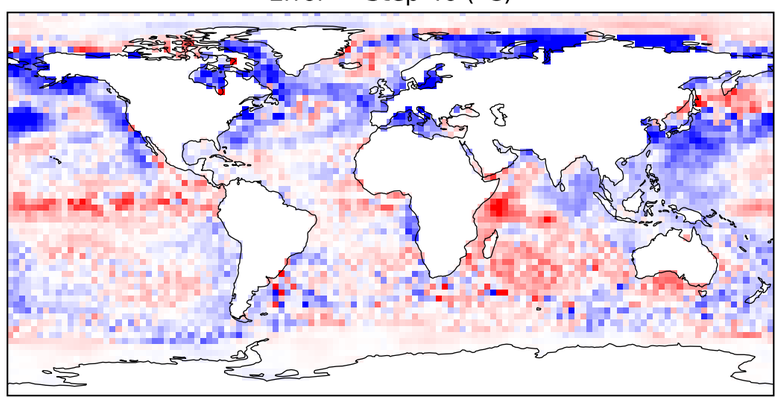
Actual — Step 46 (°C)



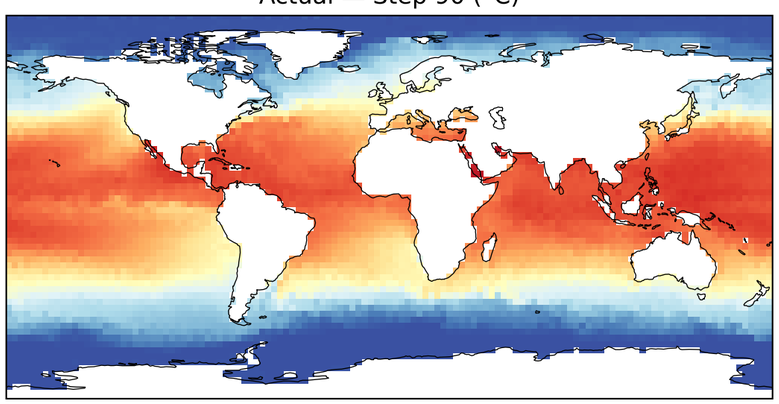
SFNO — Step 46 (°C)



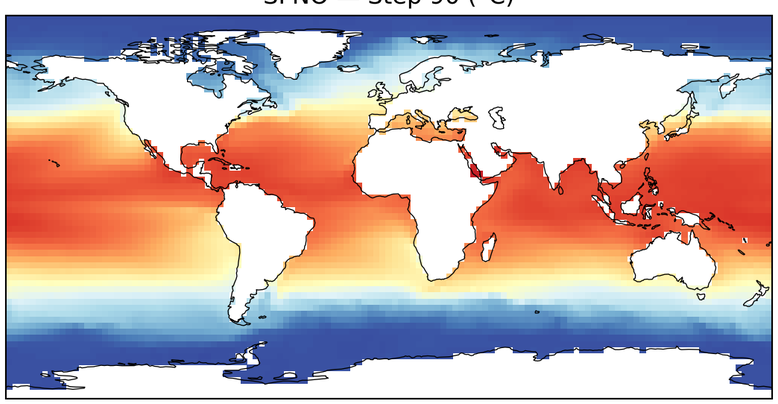
Error — Step 46 (°C)



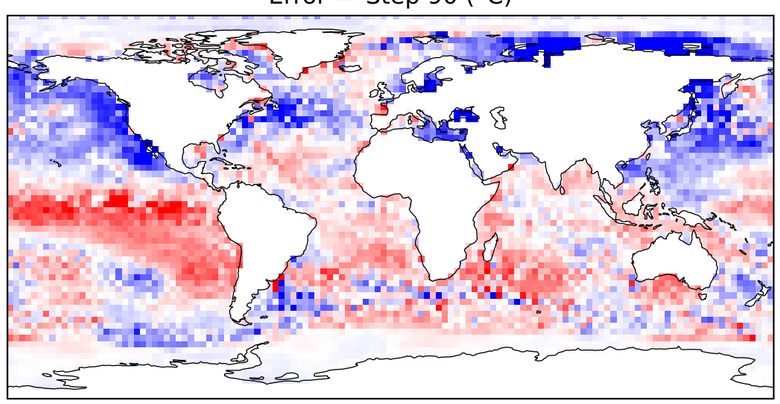
Actual — Step 90 (°C)

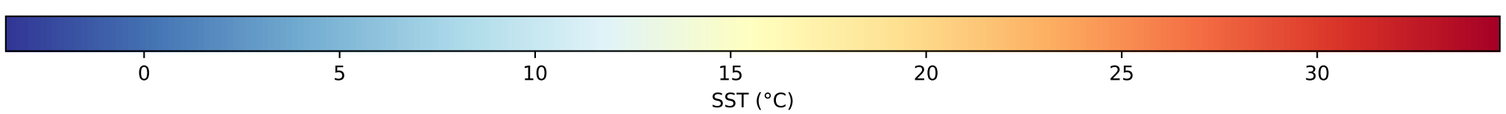


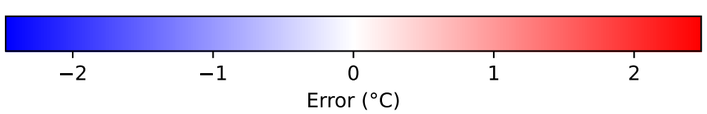
SFNO — Step 90 (°C)



Error — Step 90 (°C)

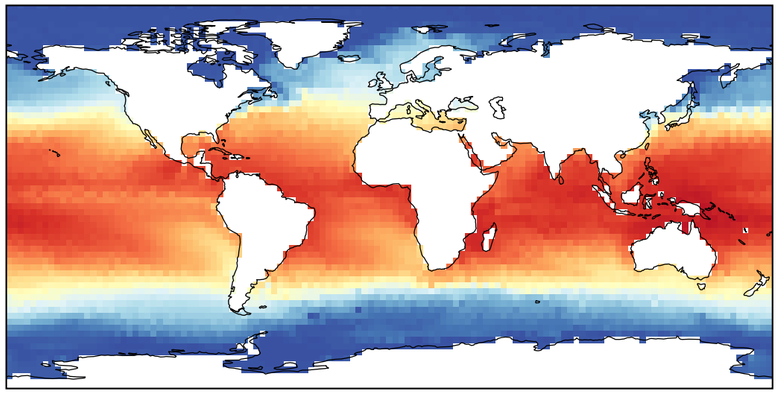




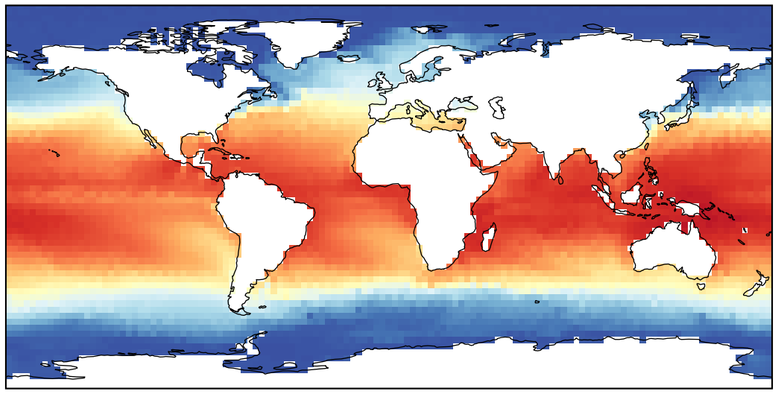


Chain init: 2021-01-06 — Test (°C)

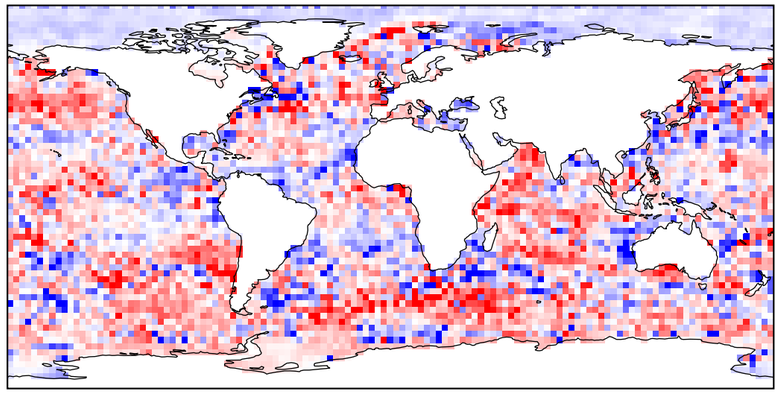
Actual — Step 1 (°C)



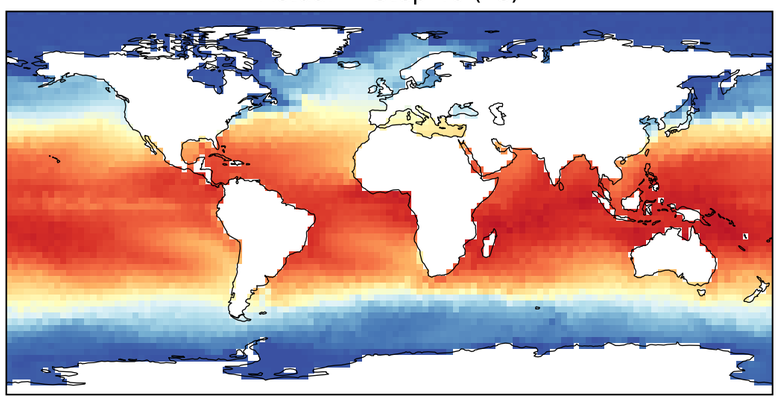
SFNO — Step 1 (°C)



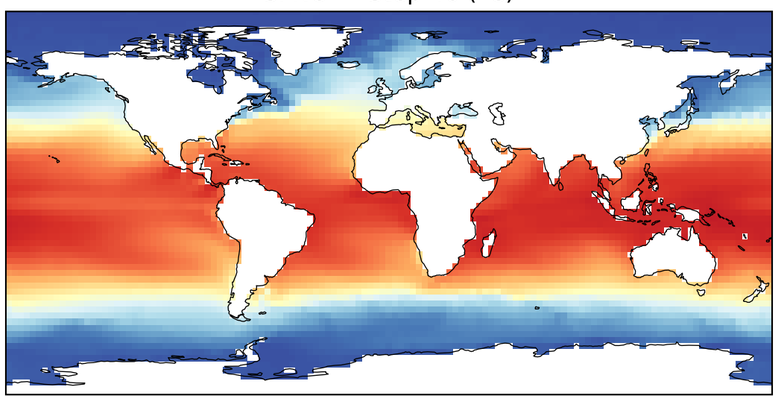
Error — Step 1 (°C)



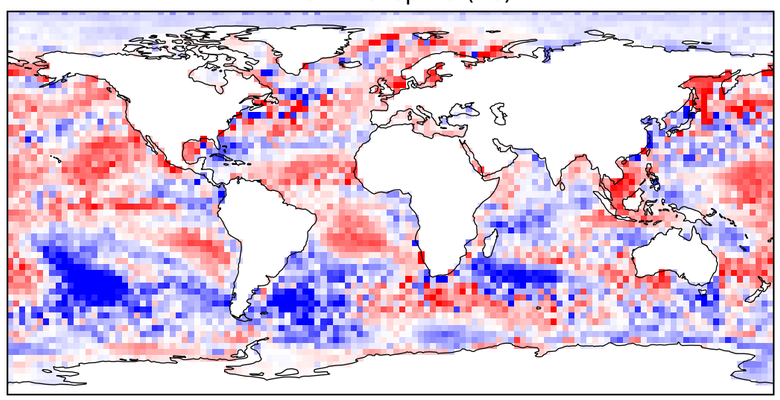
Actual — Step 46 (°C)



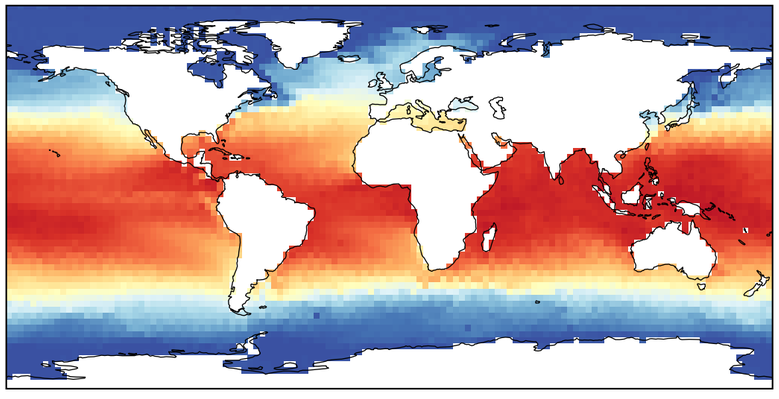
SFNO — Step 46 (°C)



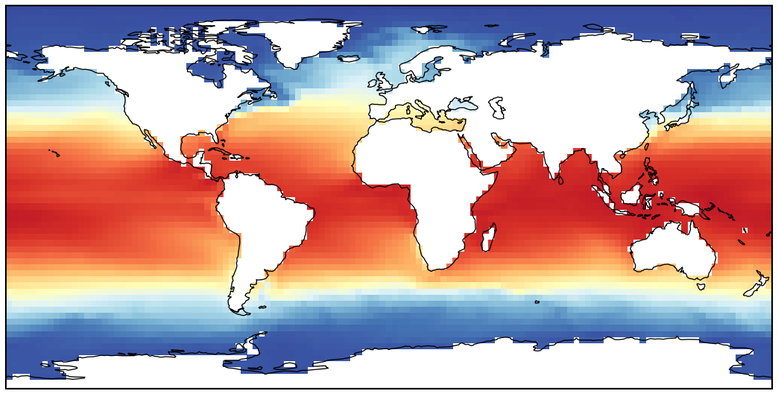
Error — Step 46 (°C)



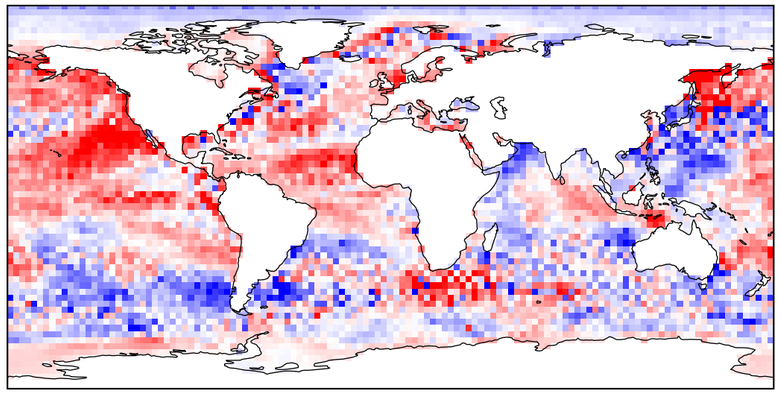
Actual — Step 90 (°C)

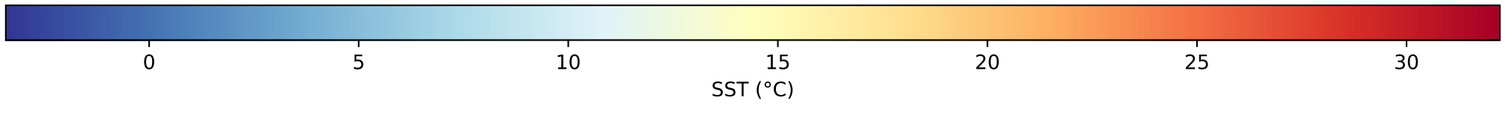


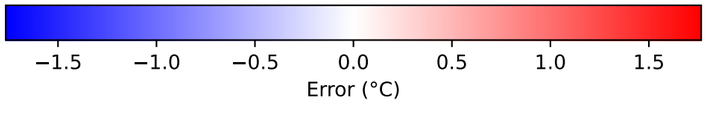
SFNO — Step 90 (°C)



Error — Step 90 (°C)

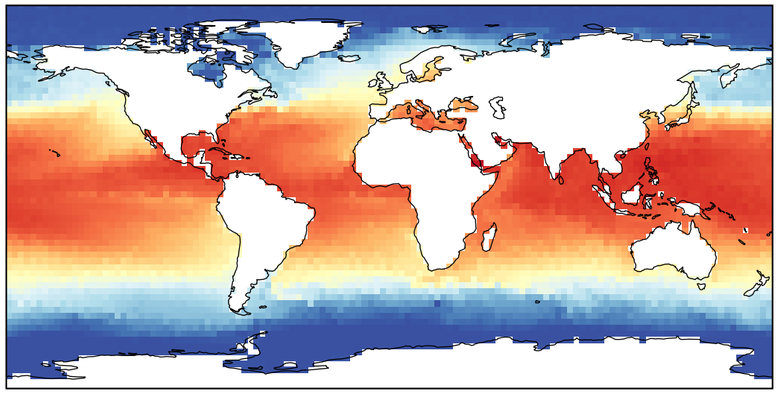




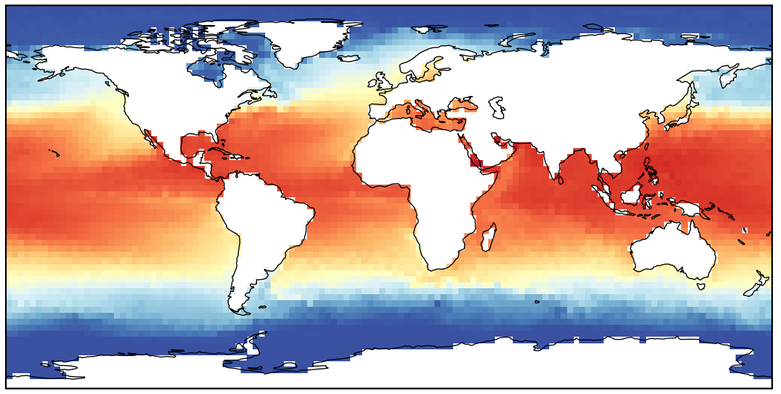


Chain init: 2021-07-05 — Test (°C)

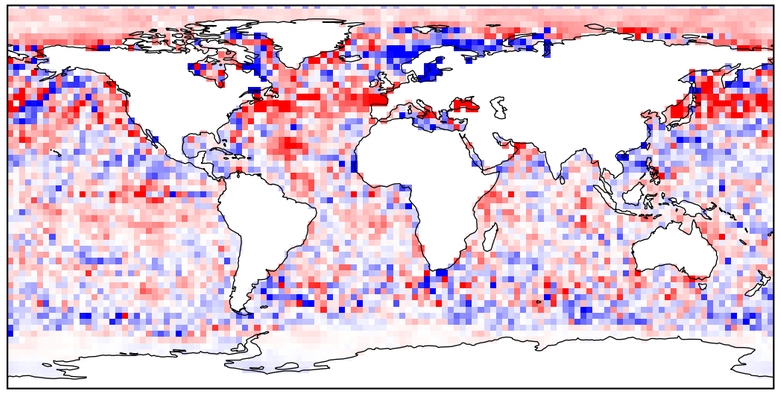
Actual — Step 1 (°C)



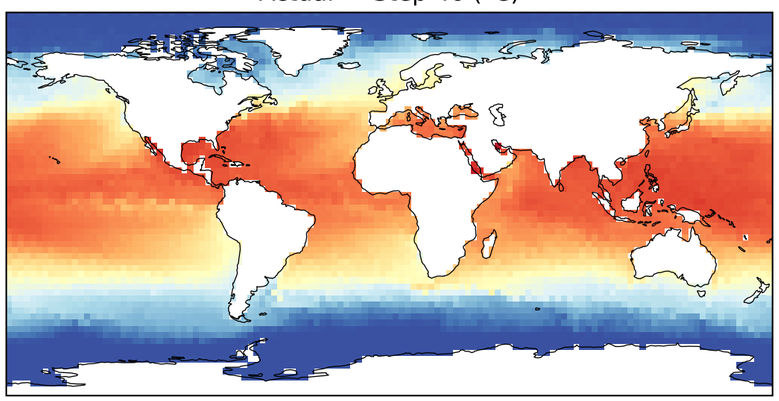
SFNO — Step 1 (°C)



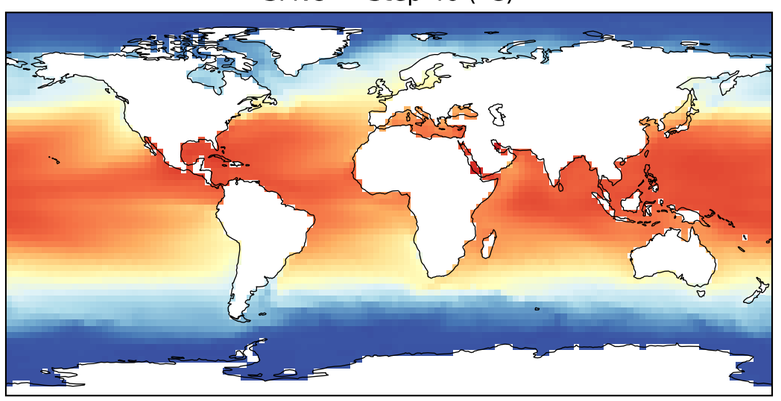
Error — Step 1 (°C)



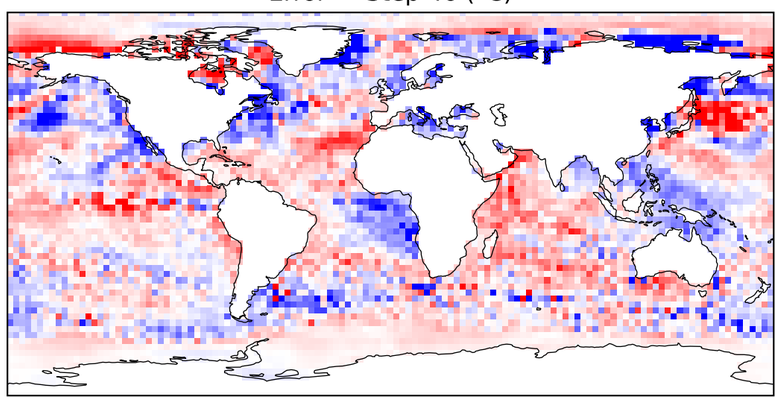
Actual — Step 46 (°C)



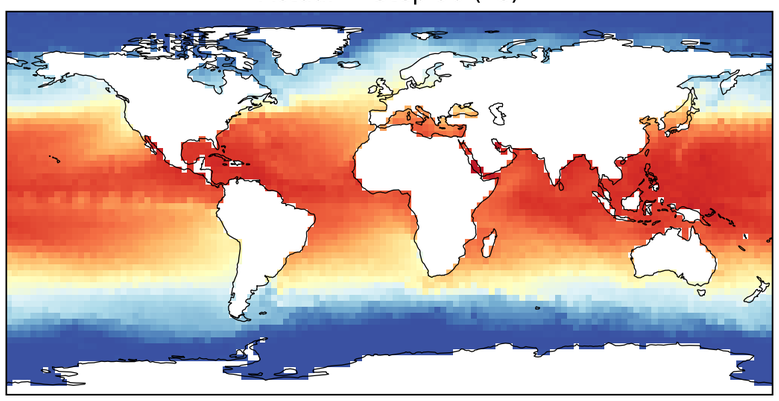
SFNO — Step 46 (°C)



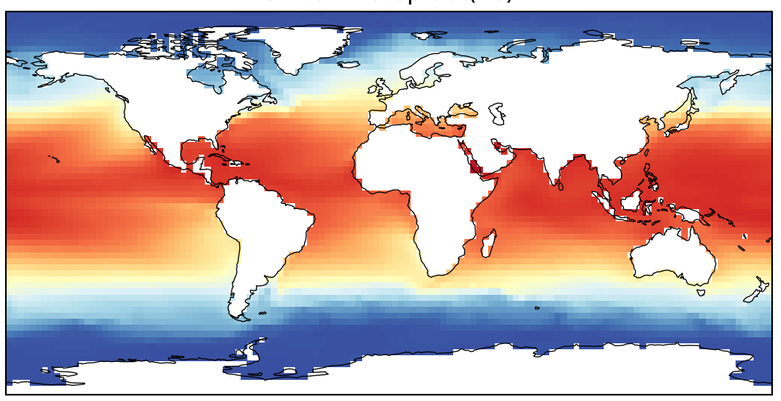
Error — Step 46 (°C)



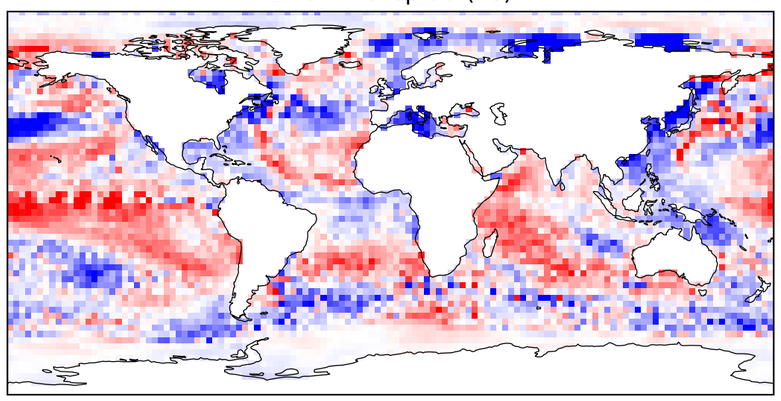
Actual — Step 90 (°C)

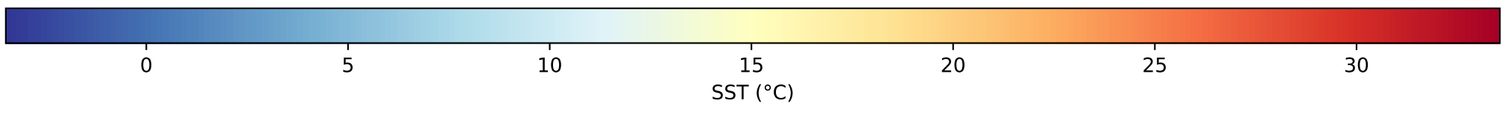


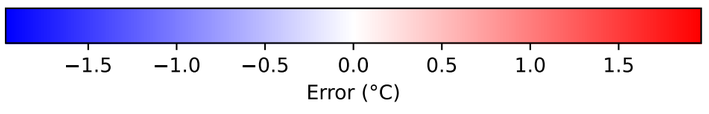
SFNO — Step 90 (°C)



Error — Step 90 (°C)

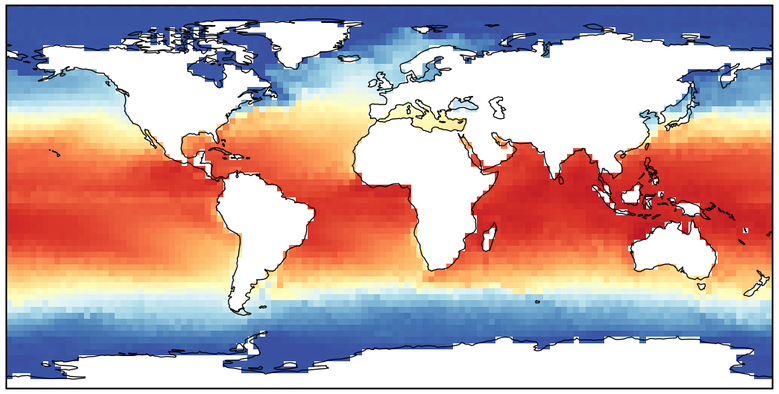




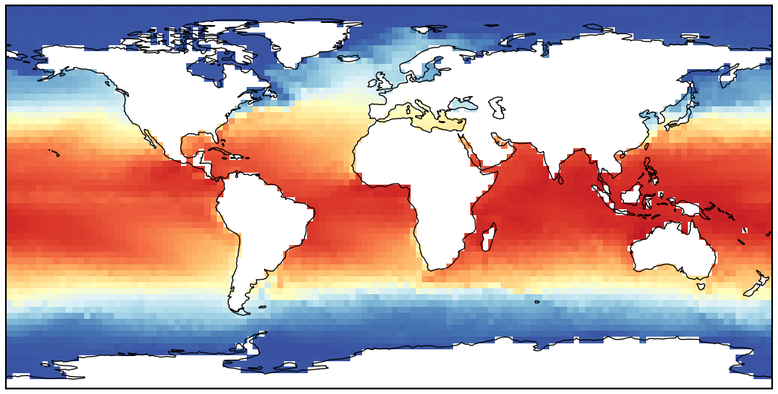


Chain init: 2022-04-01 — Test (°C)

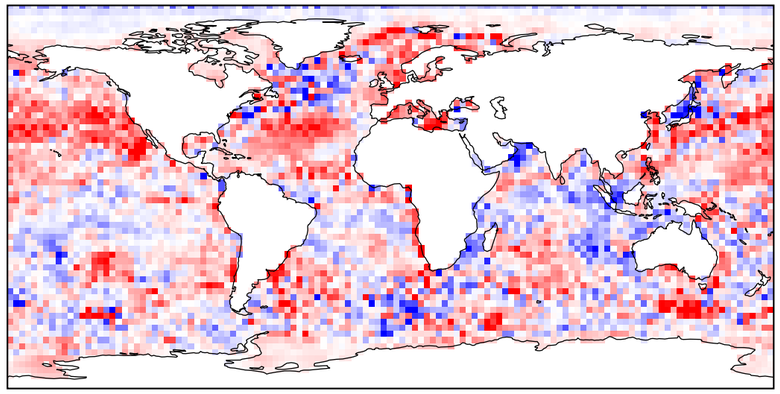
Actual — Step 1 (°C)



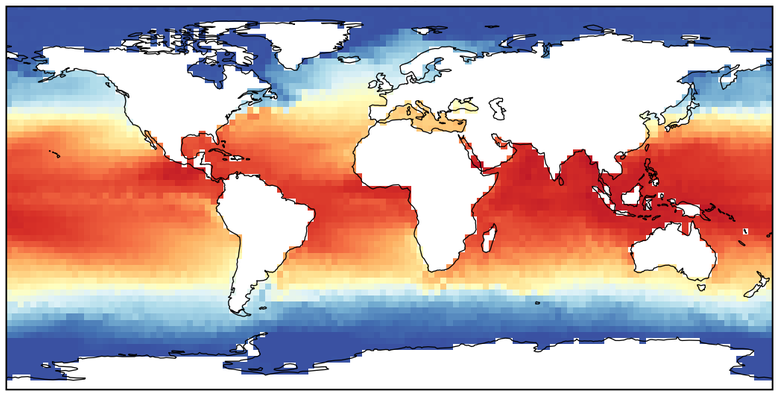
SFNO — Step 1 (°C)



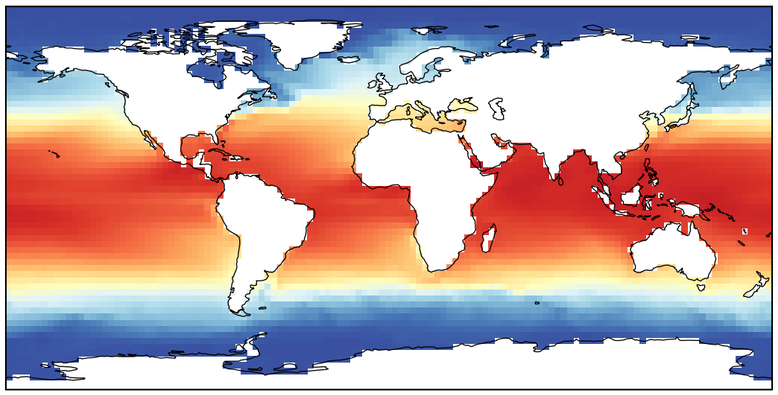
Error — Step 1 (°C)



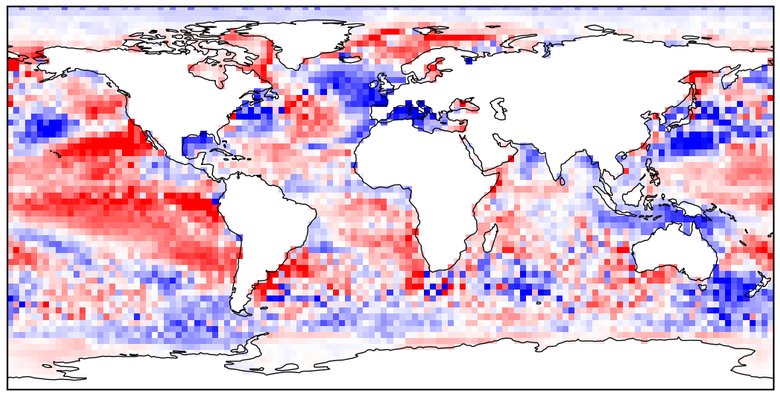
Actual — Step 46 (°C)



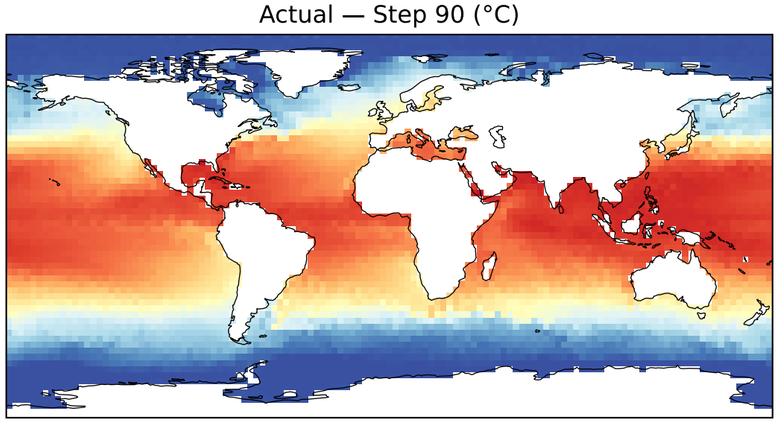
SFNO — Step 46 (°C)



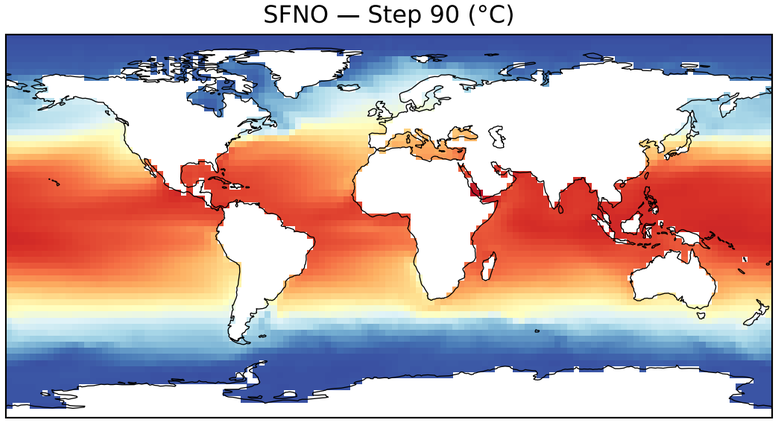
Error — Step 46 (°C)



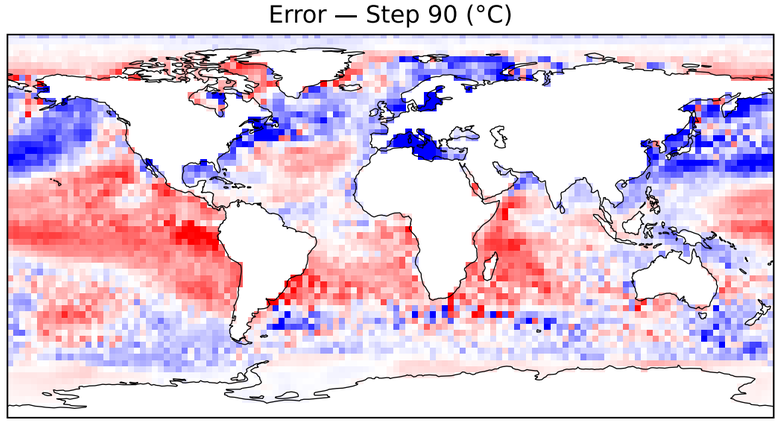
Actual — Step 90 (°C)

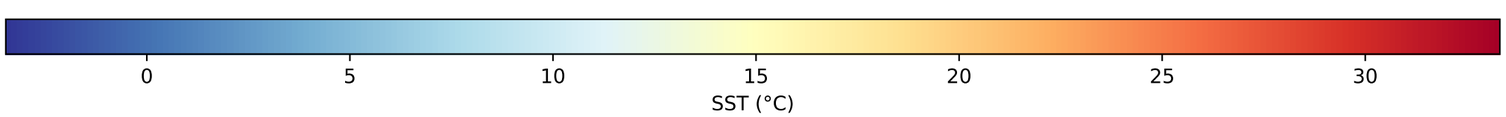


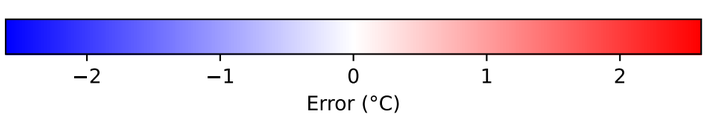
SFNO — Step 90 (°C)



Error — Step 90 (°C)

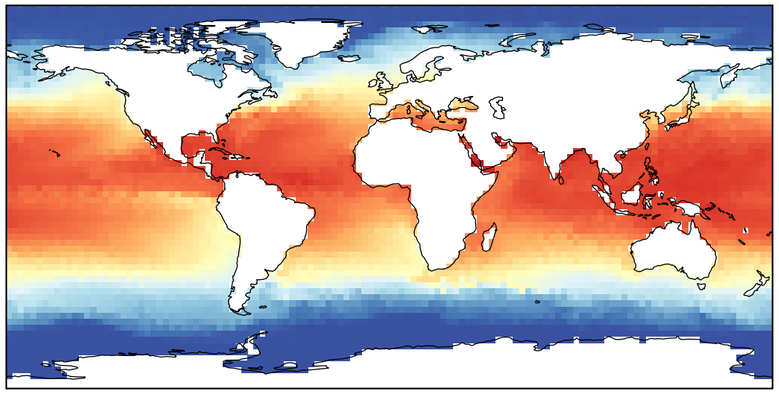




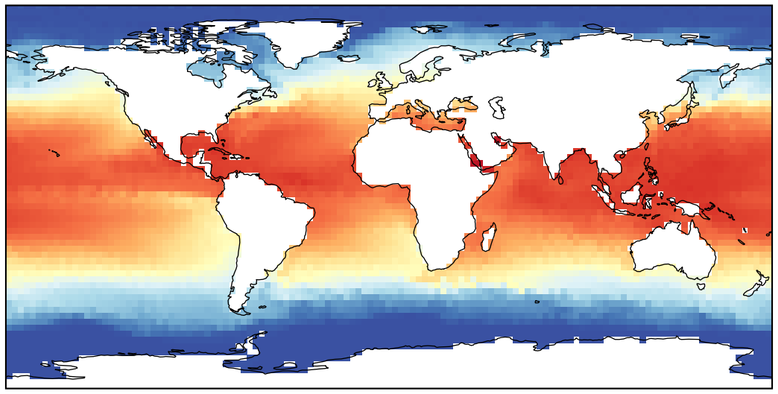


Chain init: 2022-09-28 — Test (°C)

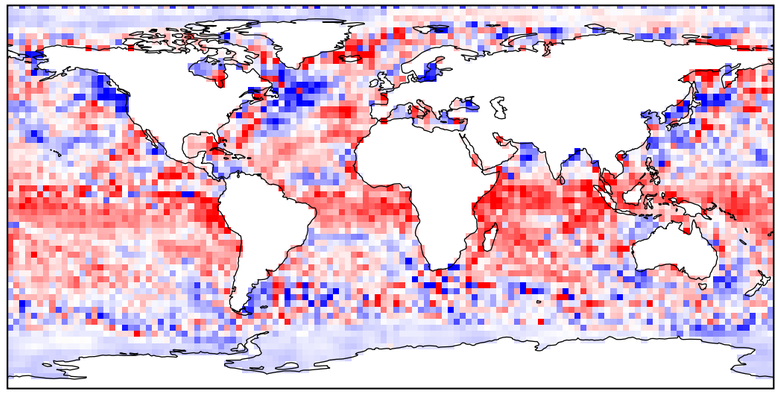
Actual — Step 1 (°C)



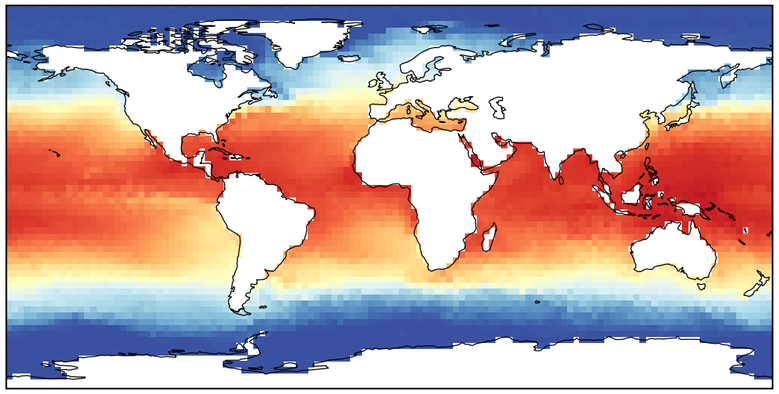
SFNO — Step 1 (°C)



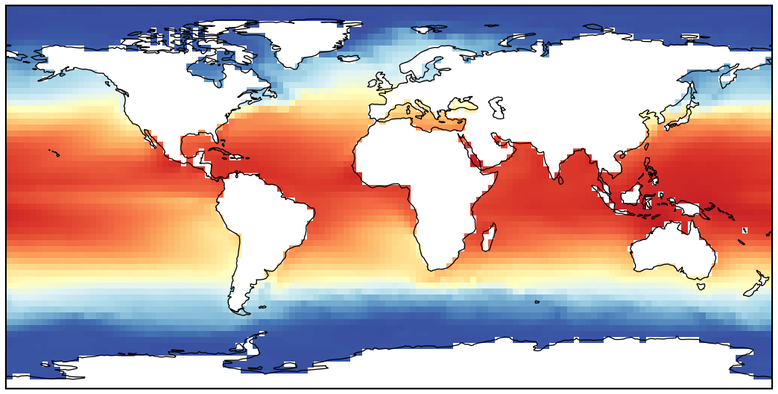
Error — Step 1 (°C)



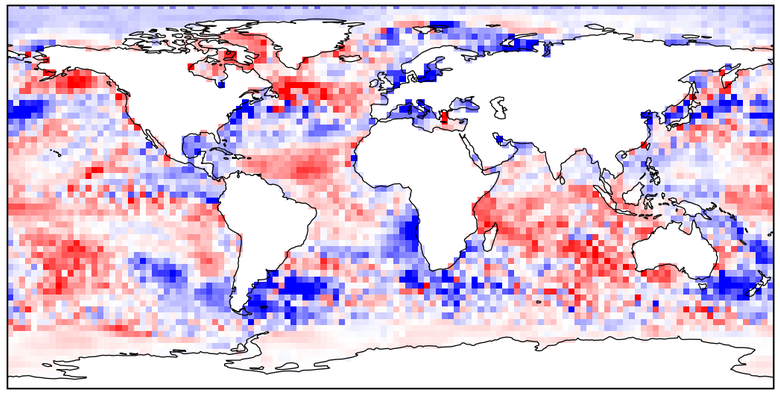
Actual — Step 46 (°C)



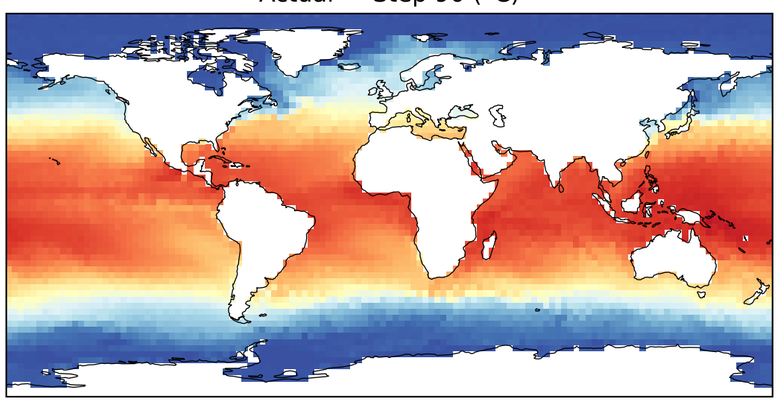
SFNO — Step 46 (°C)



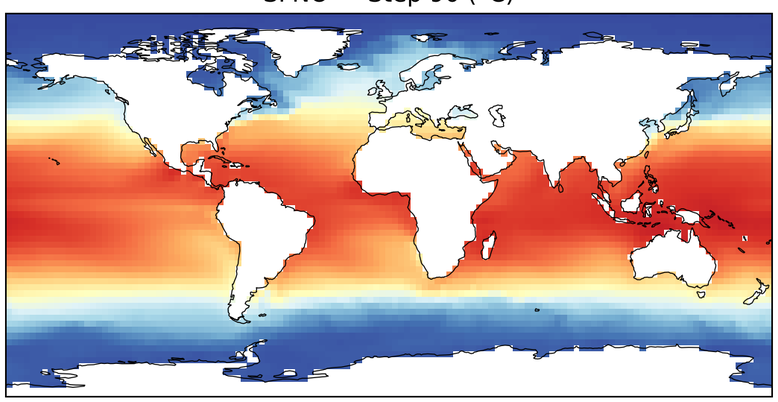
Error — Step 46 (°C)



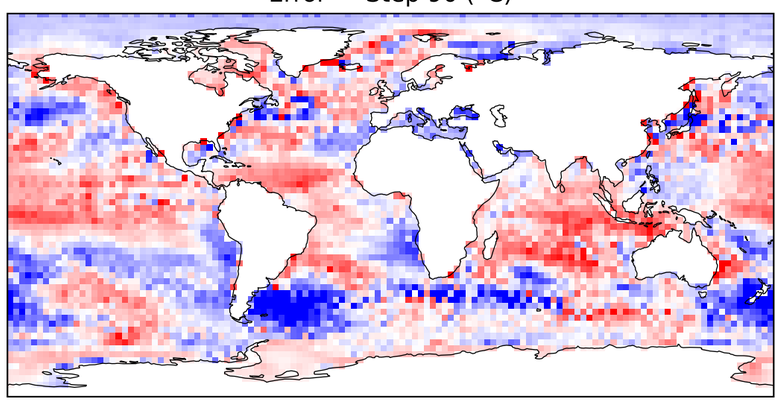
Actual — Step 90 (°C)

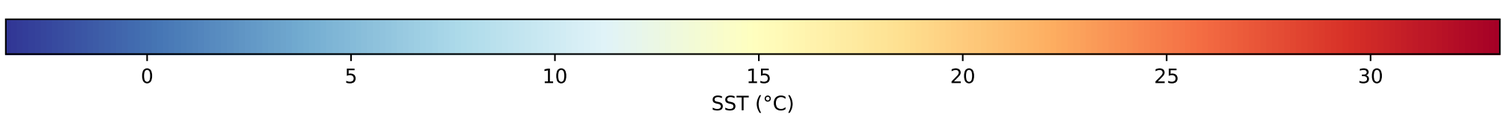


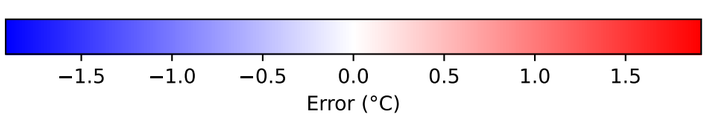
SFNO — Step 90 (°C)



Error — Step 90 (°C)

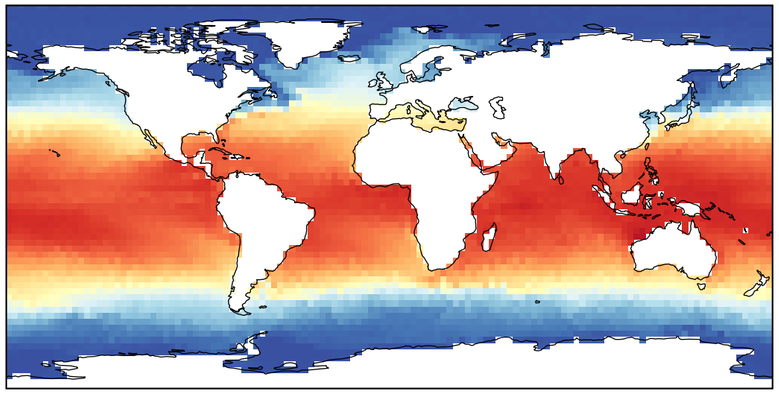




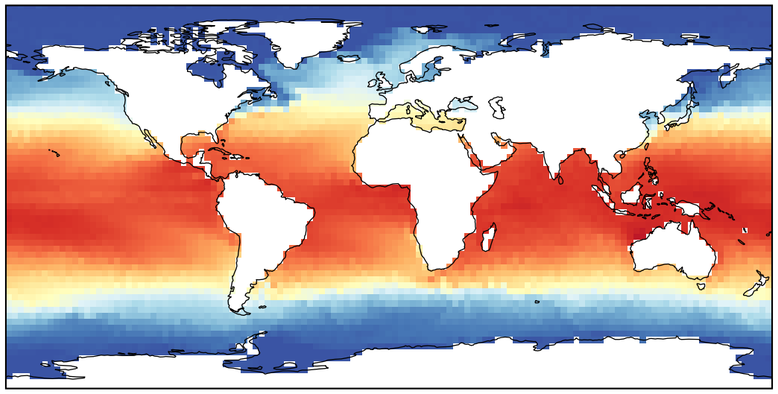


Chain init: 2023-03-27 — Test (°C)

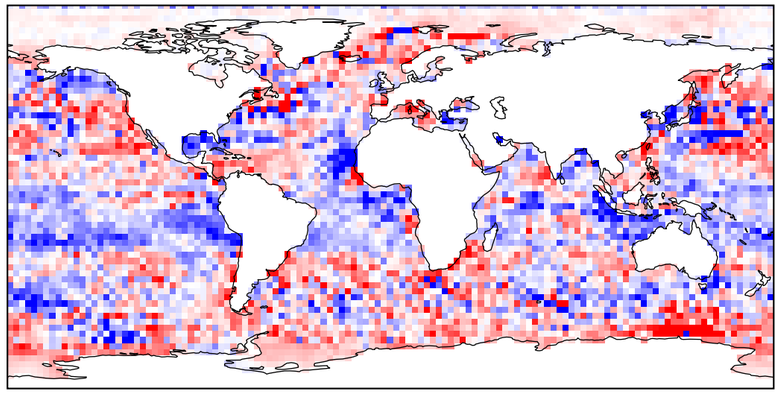
Actual — Step 1 (°C)



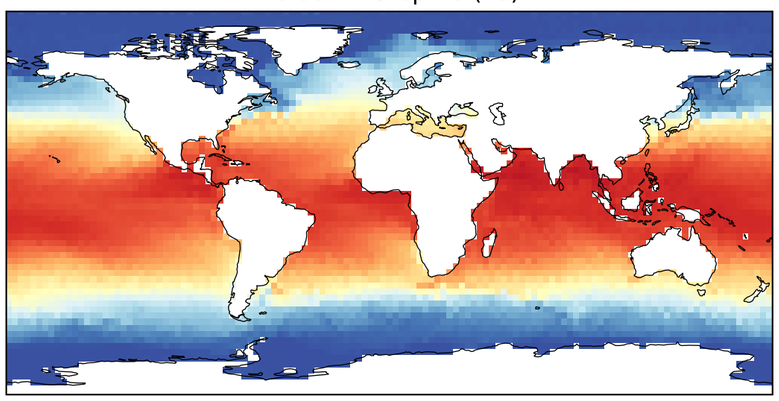
SFNO — Step 1 (°C)



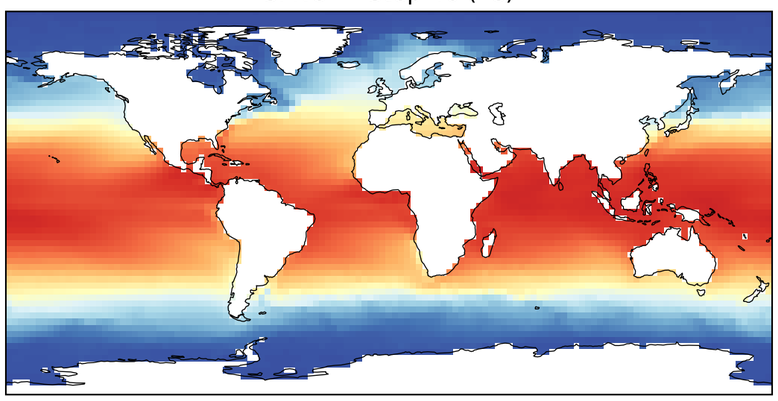
Error — Step 1 (°C)



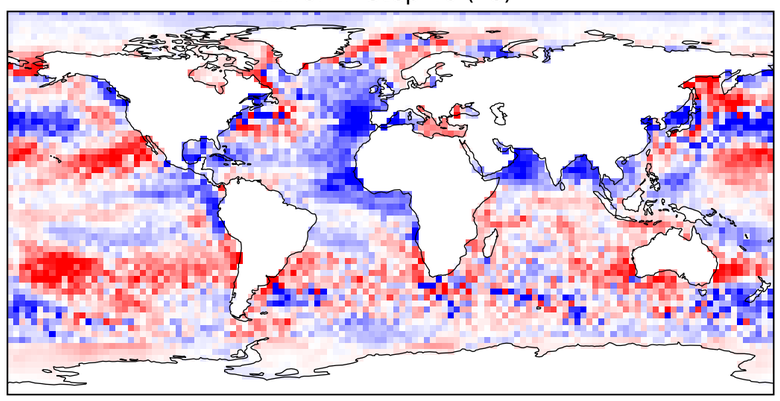
Actual — Step 46 (°C)



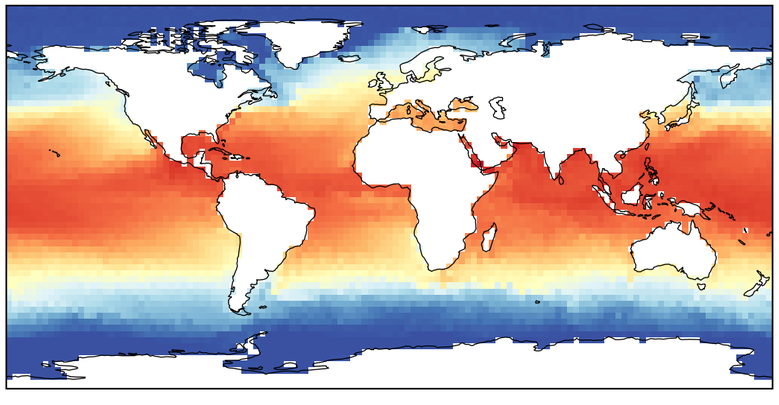
SFNO — Step 46 (°C)



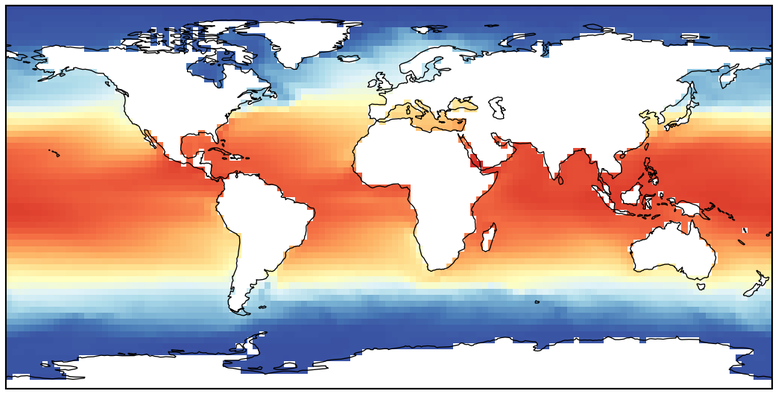
Error — Step 46 (°C)



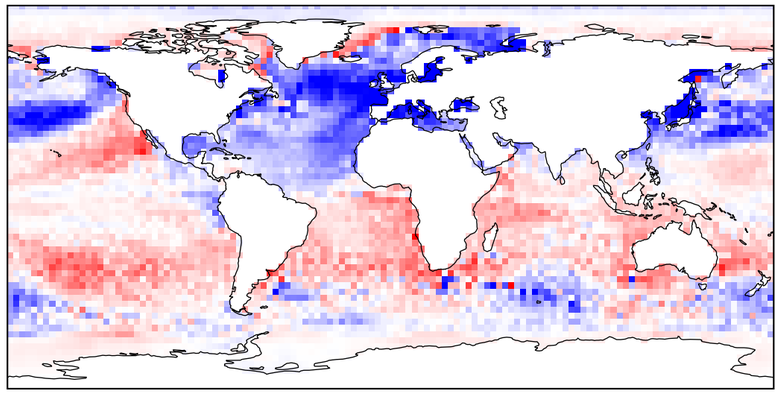
Actual — Step 90 (°C)

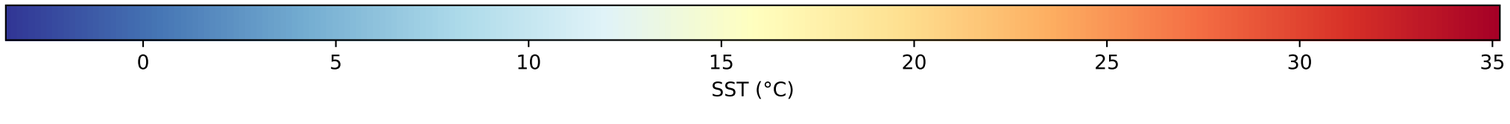


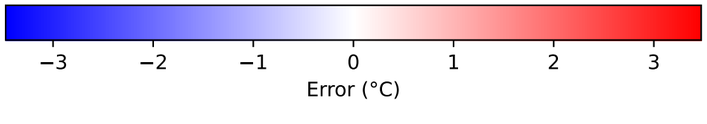
SFNO — Step 90 (°C)



Error — Step 90 (°C)

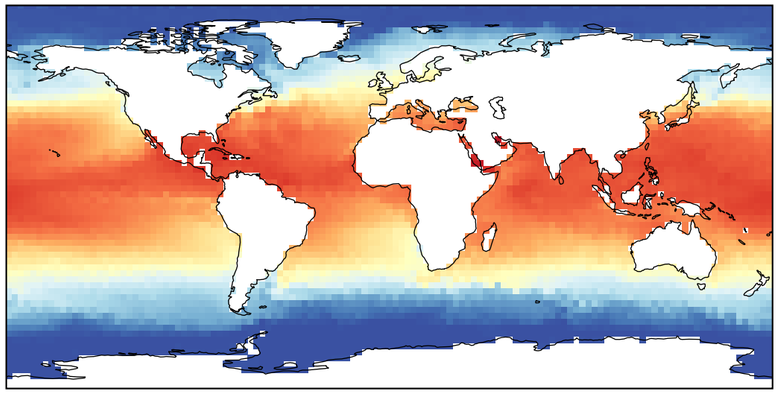




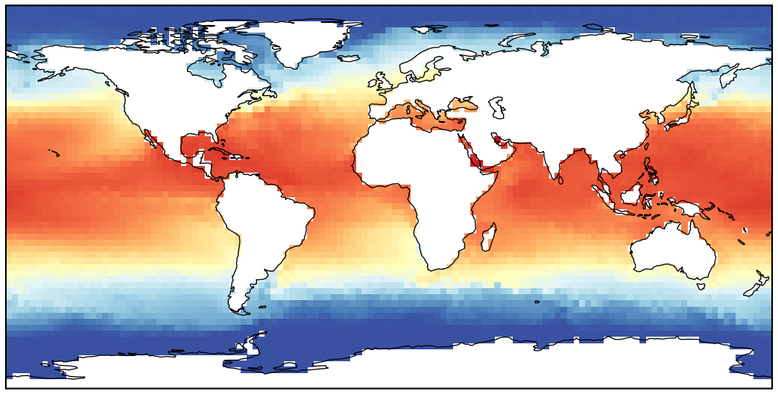


Chain init: 2023-09-23 — Test (°C)

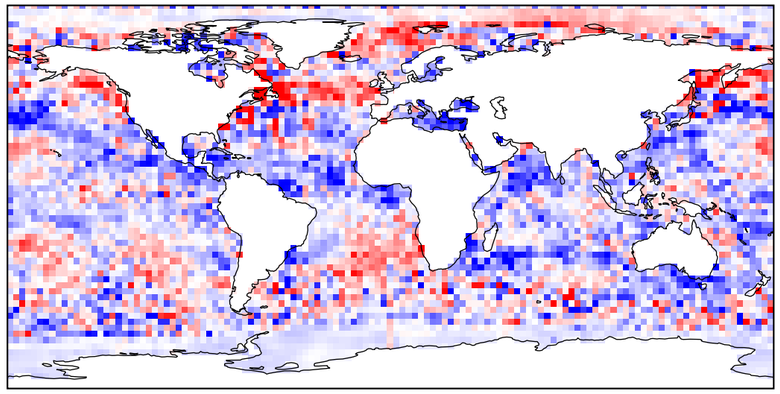
Actual — Step 1 (°C)



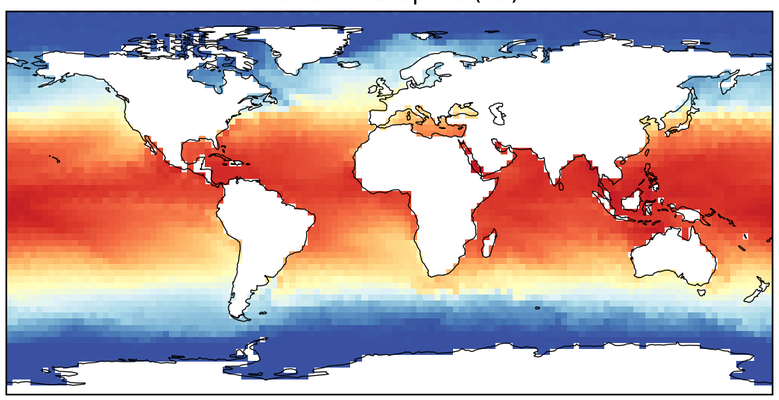
SFNO — Step 1 (°C)



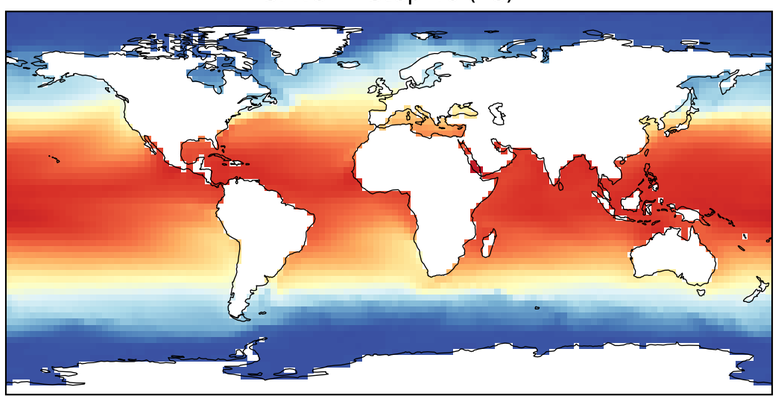
Error — Step 1 (°C)



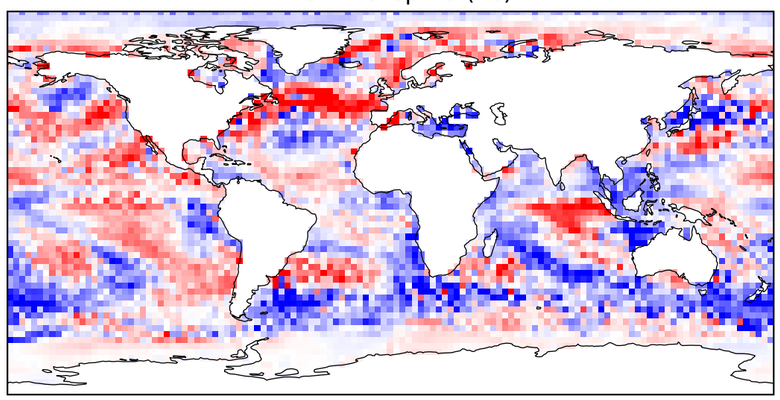
Actual — Step 46 (°C)



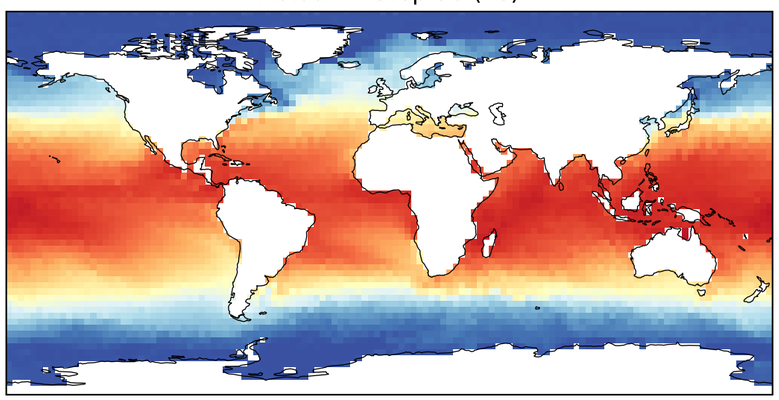
SFNO — Step 46 (°C)



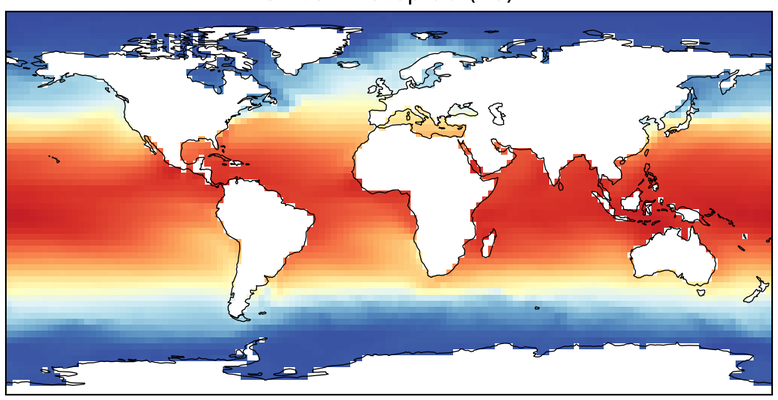
Error — Step 46 (°C)



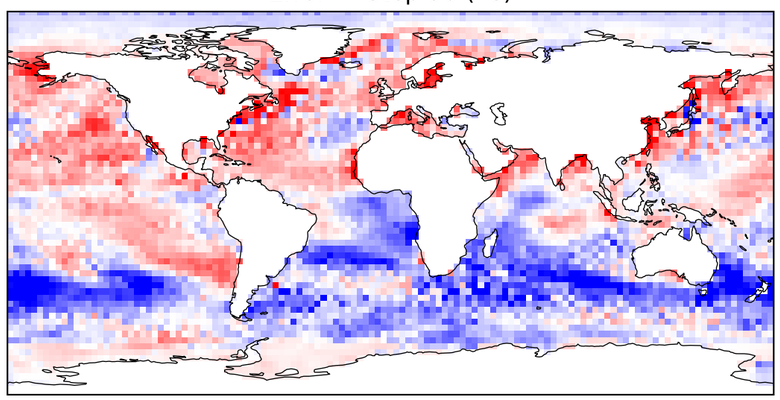
Actual — Step 90 (°C)

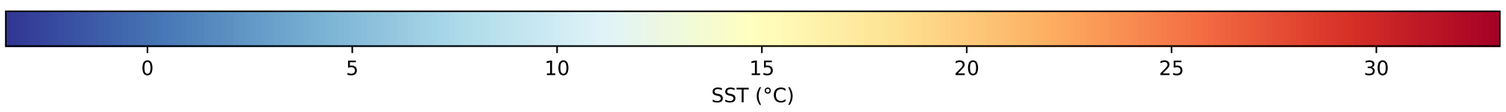


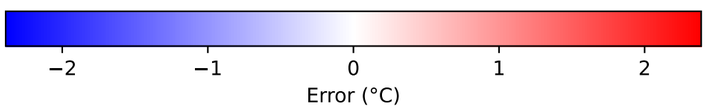
SFNO — Step 90 (°C)



Error — Step 90 (°C)

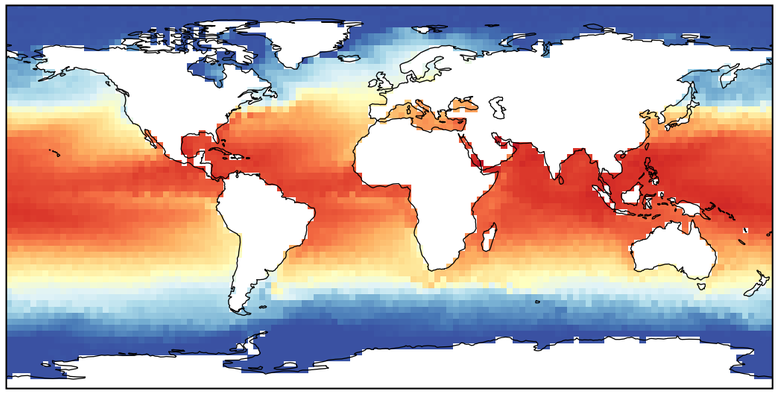




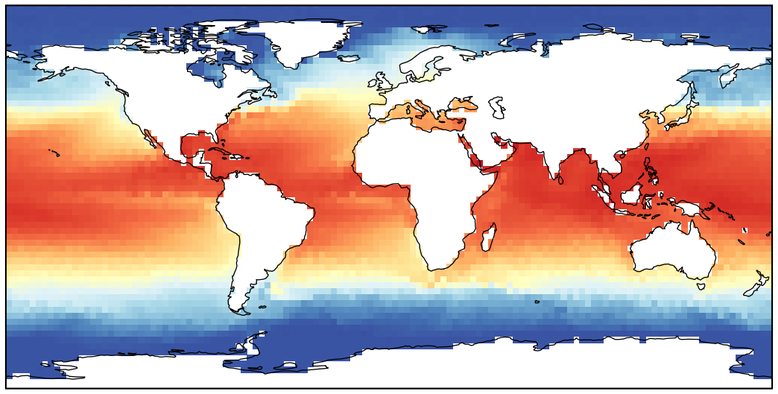


Chain init: 2024-06-19 — Test (°C)

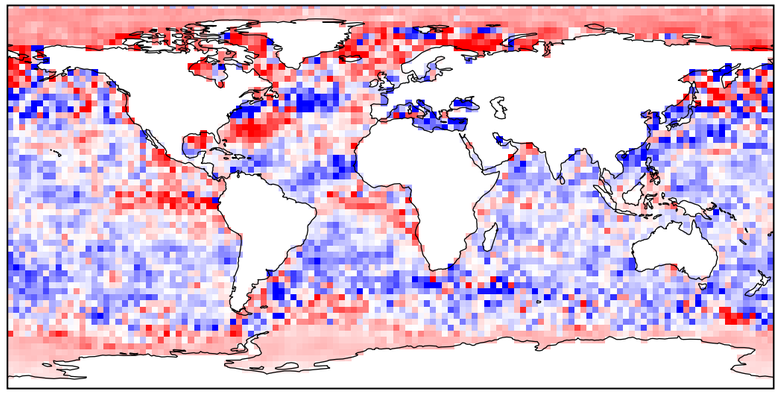
Actual — Step 1 (°C)



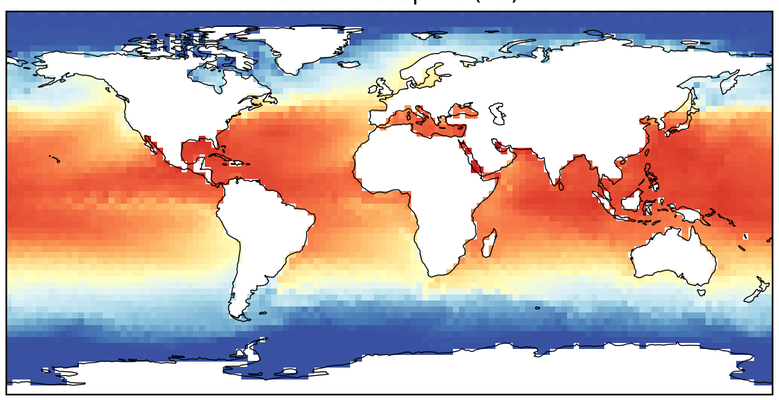
SFNO — Step 1 (°C)



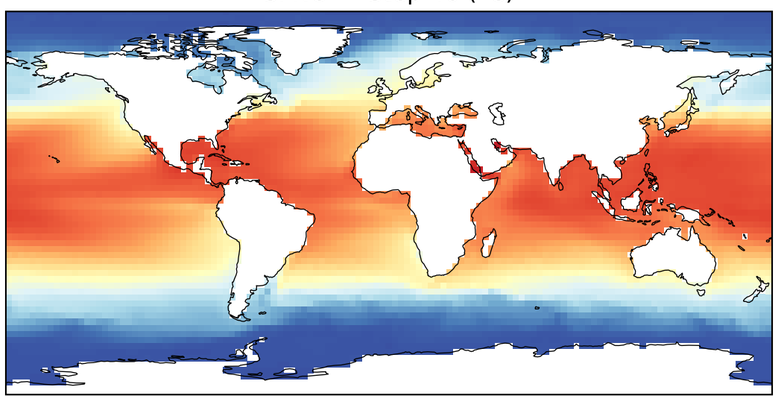
Error — Step 1 (°C)



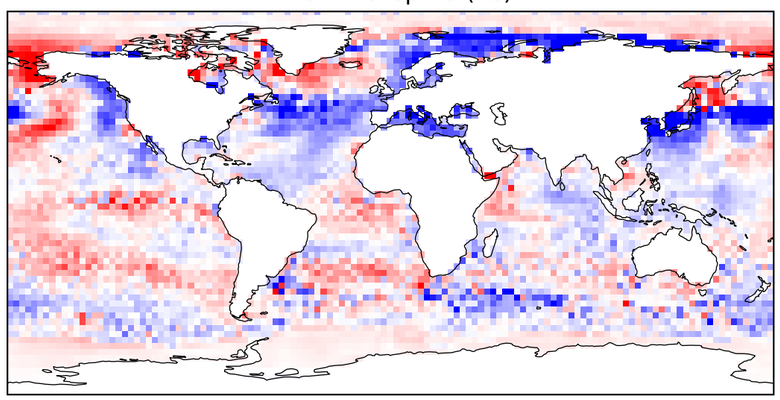
Actual — Step 46 (°C)



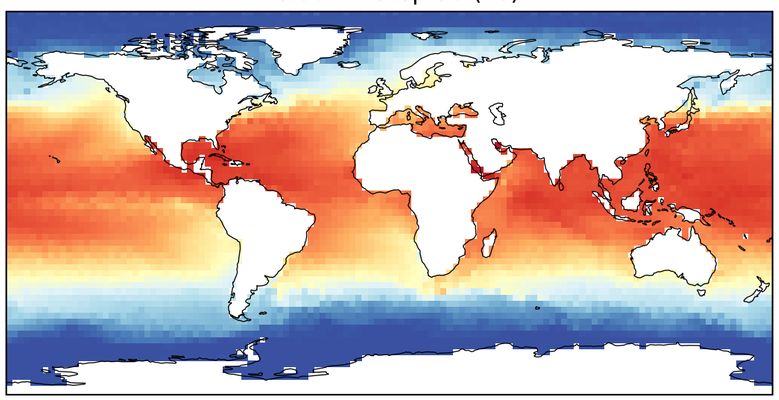
SFNO — Step 46 (°C)



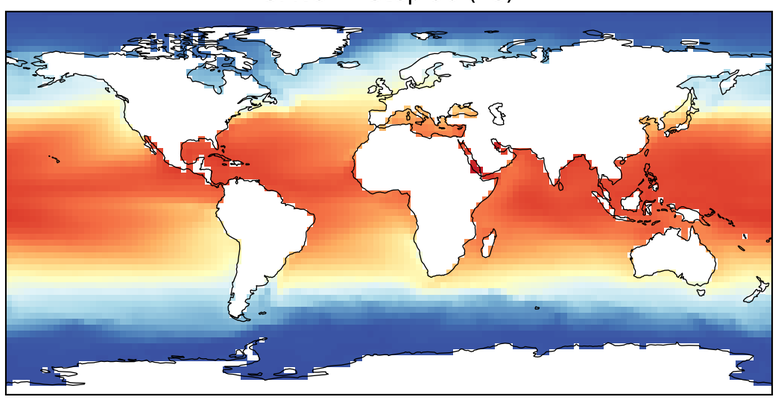
Error — Step 46 (°C)



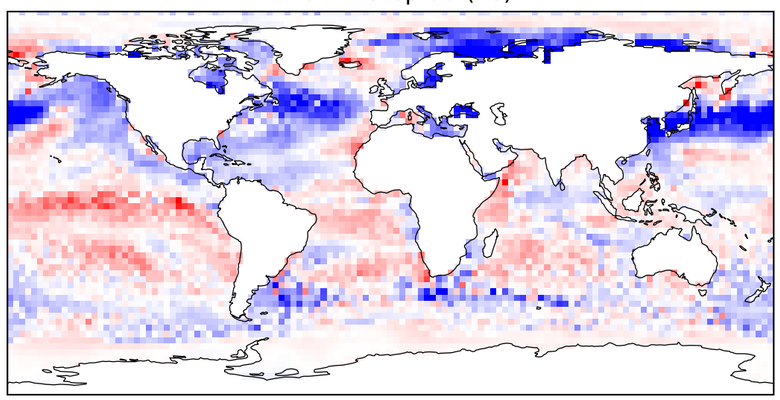
Actual — Step 90 (°C)

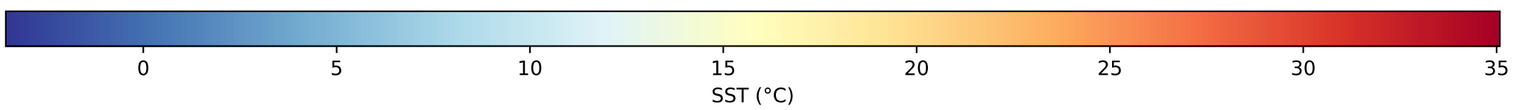


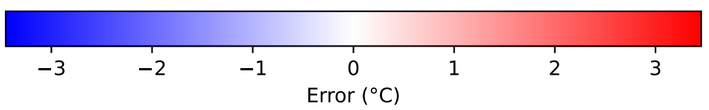
SFNO — Step 90 (°C)



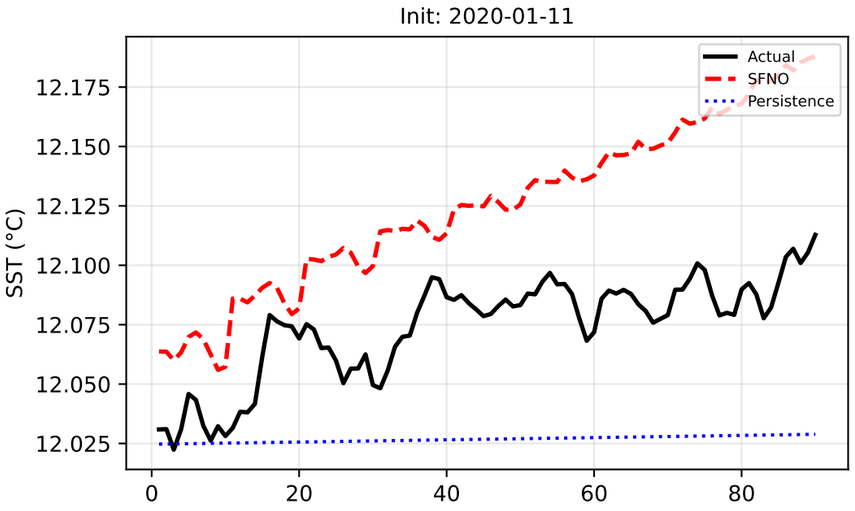
Error — Step 90 (°C)

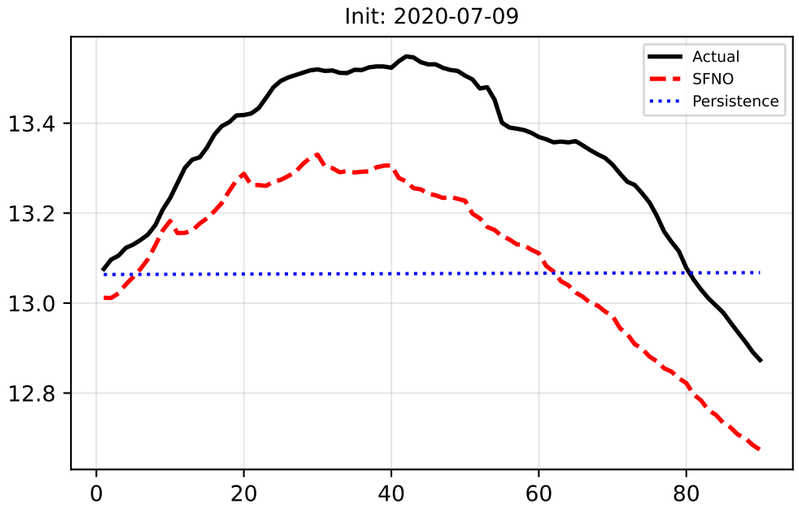


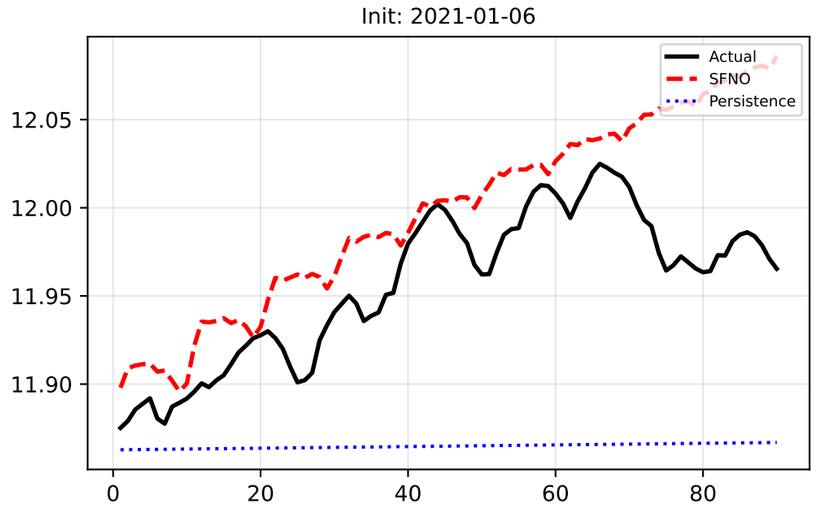


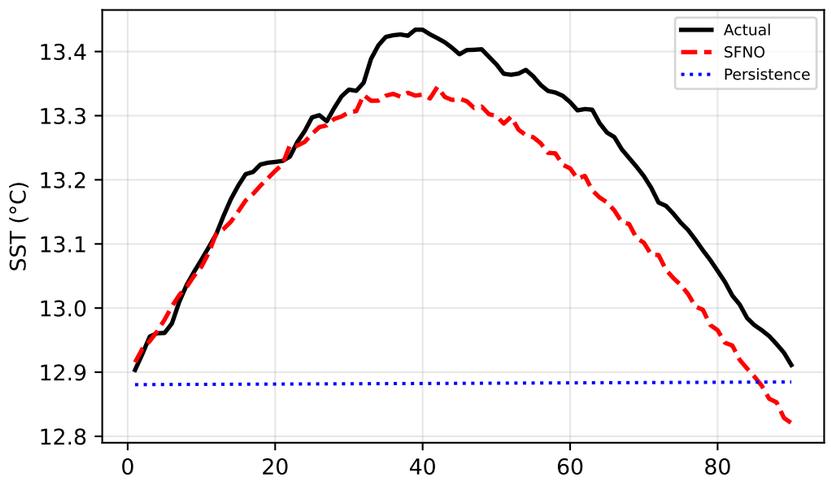


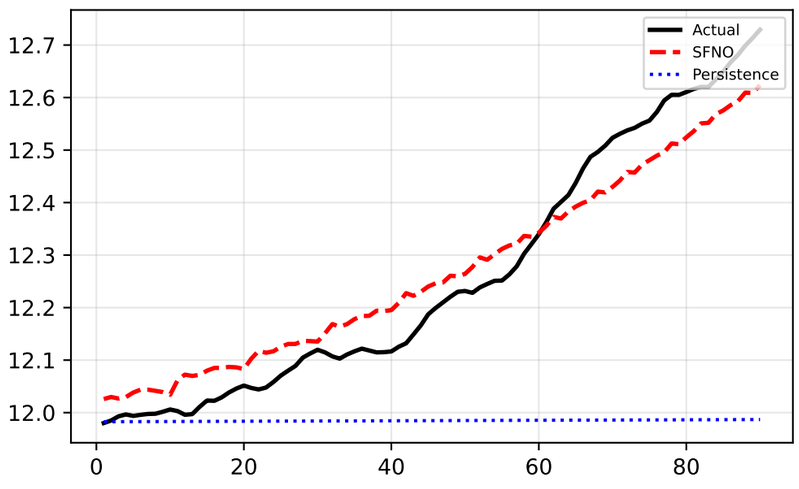
Global-Mean SST — Test (°C)

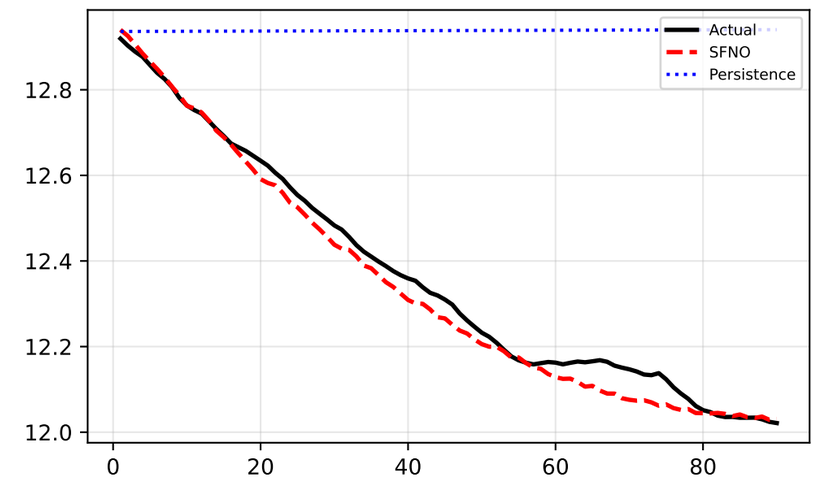


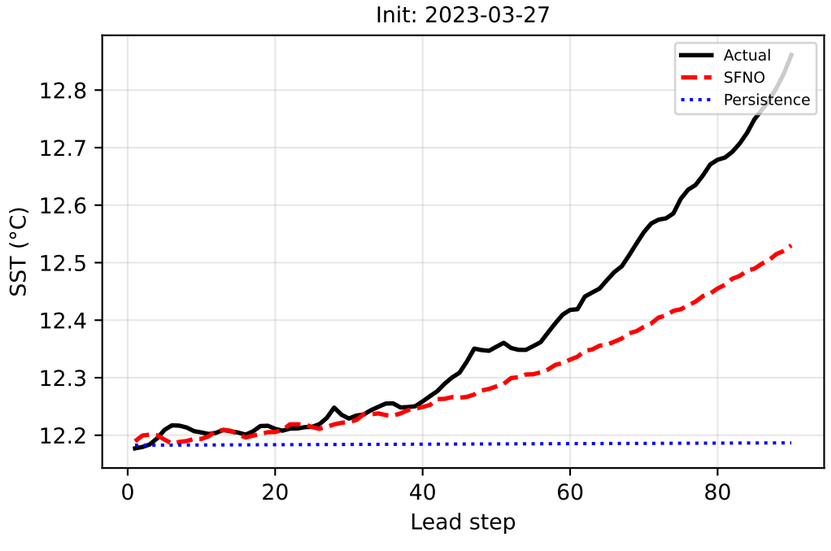


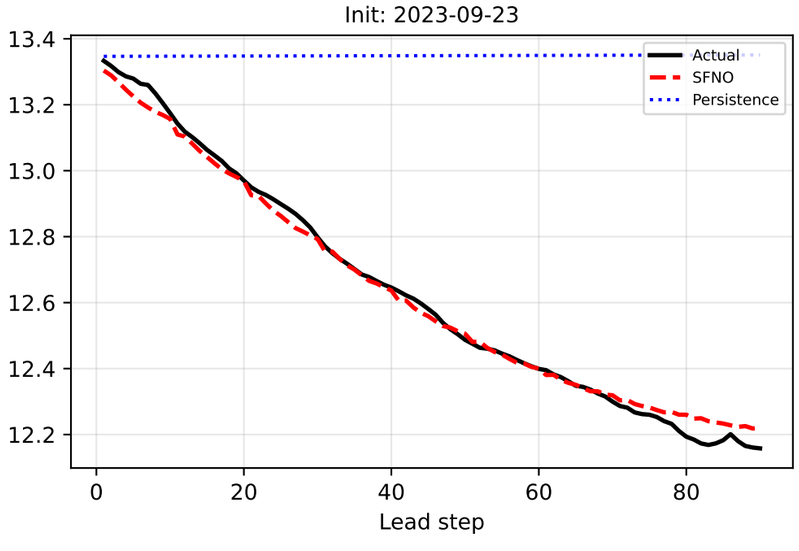


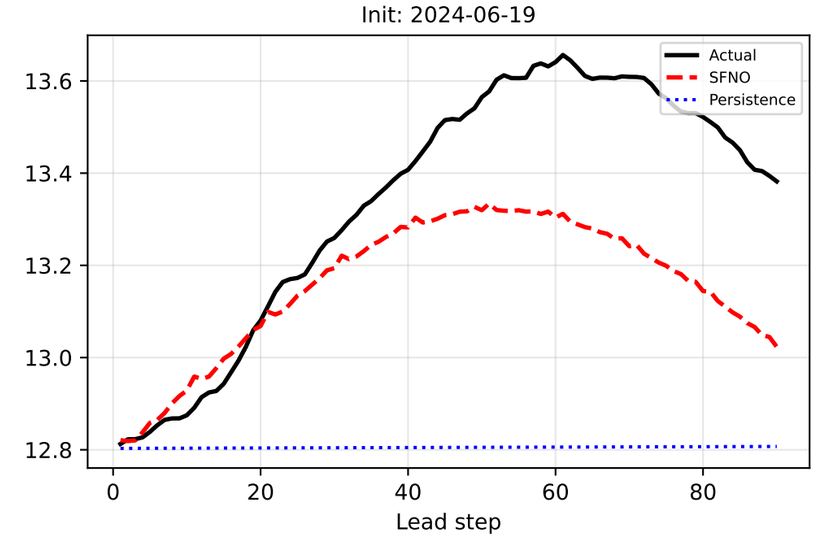






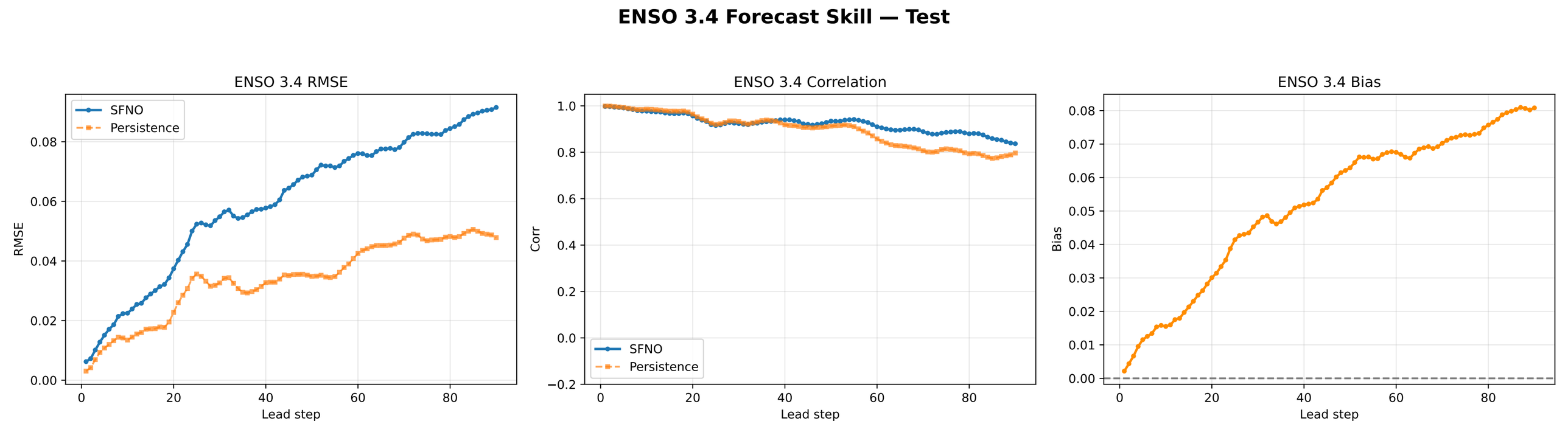






ENSO 3.4 Metrics — Test

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0062	0.0031	0.9973	0.9992	0.0022
2	0.0072	0.0042	0.9969	0.9989	0.0044
3	0.0101	0.0069	0.9944	0.9972	0.0066
4	0.0128	0.0093	0.9930	0.9950	0.0095
5	0.0151	0.0108	0.9909	0.9927	0.0115
6	0.0171	0.0120	0.9872	0.9896	0.0125
7	0.0186	0.0133	0.9842	0.9867	0.0134
8	0.0214	0.0144	0.9794	0.9842	0.0154
9	0.0223	0.0142	0.9775	0.9843	0.0158
10	0.0225	0.0135	0.9770	0.9857	0.0155
11	0.0239	0.0145	0.9752	0.9850	0.0160
12	0.0254	0.0155	0.9739	0.9833	0.0175
13	0.0258	0.0160	0.9732	0.9817	0.0179
14	0.0277	0.0171	0.9693	0.9784	0.0197
15	0.0289	0.0172	0.9673	0.9772	0.0213
16	0.0301	0.0173	0.9662	0.9774	0.0230
17	0.0314	0.0178	0.9667	0.9770	0.0248
18	0.0321	0.0177	0.9689	0.9777	0.0262
19	0.0343	0.0195	0.9657	0.9736	0.0282
20	0.0374	0.0227	0.9566	0.9644	0.0301
21	0.0403	0.0260	0.9459	0.9526	0.0314
22	0.0431	0.0285	0.9384	0.9440	0.0334
23	0.0456	0.0308	0.9325	0.9369	0.0353
24	0.0500	0.0341	0.9187	0.9251	0.0387
25	0.0524	0.0356	0.9154	0.9197	0.0414
26	0.0528	0.0349	0.9176	0.9226	0.0427
27	0.0521	0.0332	0.9236	0.9287	0.0431
28	0.0518	0.0315	0.9283	0.9351	0.0435
29	0.0536	0.0319	0.9253	0.9350	0.0453
30	0.0548	0.0326	0.9235	0.9322	0.0467
31	0.0565	0.0342	0.9207	0.9251	0.0482
32	0.0571	0.0344	0.9190	0.9217	0.0486
33	0.0551	0.0325	0.9242	0.9261	0.0469
34	0.0543	0.0308	0.9252	0.9305	0.0461
35	0.0546	0.0295	0.9295	0.9367	0.0468
36	0.0555	0.0293	0.9316	0.9386	0.0481
37	0.0565	0.0297	0.9333	0.9370	0.0495
38	0.0573	0.0304	0.9365	0.9336	0.0509
39	0.0574	0.0314	0.9397	0.9265	0.0514
40	0.0578	0.0327	0.9396	0.9182	0.0518
41	0.0582	0.0329	0.9399	0.9161	0.0521
42	0.0589	0.0329	0.9362	0.9151	0.0524
43	0.0605	0.0339	0.9324	0.9113	0.0536
44	0.0637	0.0353	0.9225	0.9066	0.0561
45	0.0644	0.0351	0.9202	0.9069	0.0570
46	0.0656	0.0355	0.9174	0.9049	0.0584
47	0.0671	0.0355	0.9202	0.9067	0.0602
48	0.0682	0.0355	0.9232	0.9080	0.0614
49	0.0685	0.0352	0.9284	0.9099	0.0621
50	0.0688	0.0349	0.9343	0.9127	0.0629
51	0.0706	0.0349	0.9336	0.9148	0.0645
52	0.0722	0.0352	0.9339	0.9157	0.0661
53	0.0719	0.0346	0.9385	0.9176	0.0660
54	0.0719	0.0345	0.9400	0.9155	0.0662
55	0.0714	0.0348	0.9413	0.9104	0.0655
56	0.0719	0.0362	0.9379	0.9010	0.0656
57	0.0734	0.0378	0.9332	0.8912	0.0669
58	0.0744	0.0390	0.9285	0.8831	0.0674
59	0.0754	0.0408	0.9189	0.8706	0.0677
60	0.0760	0.0425	0.9099	0.8575	0.0675
61	0.0760	0.0435	0.9052	0.8473	0.0669
62	0.0754	0.0441	0.9005	0.8398	0.0661
63	0.0754	0.0448	0.8971	0.8324	0.0658
64	0.0767	0.0452	0.8948	0.8290	0.0673
65	0.0776	0.0452	0.8953	0.8279	0.0685
66	0.0776	0.0452	0.8978	0.8256	0.0689
67	0.0778	0.0453	0.8991	0.8230	0.0692
68	0.0774	0.0457	0.8995	0.8187	0.0687
69	0.0781	0.0462	0.8967	0.8146	0.0692
70	0.0798	0.0476	0.8890	0.8065	0.0702
71	0.0814	0.0486	0.8827	0.8014	0.0711
72	0.0825	0.0490	0.8780	0.8005	0.0718
73	0.0828	0.0487	0.8777	0.8030	0.0720
74	0.0828	0.0474	0.8825	0.8113	0.0726
75	0.0827	0.0468	0.8855	0.8146	0.0728
76	0.0825	0.0470	0.8875	0.8124	0.0726
77	0.0825	0.0471	0.8887	0.8097	0.0729
78	0.0824	0.0472	0.8892	0.8062	0.0732
79	0.0838	0.0480	0.8837	0.7976	0.0748
80	0.0844	0.0482	0.8793	0.7935	0.0757
81	0.0851	0.0479	0.8813	0.7956	0.0765
82	0.0858	0.0482	0.8802	0.7931	0.0774
83	0.0874	0.0492	0.8747	0.7855	0.0788
84	0.0885	0.0500	0.8651	0.7786	0.0794
85	0.0892	0.0506	0.8590	0.7738	0.0798
86	0.0897	0.0500	0.8549	0.7761	0.0803
87	0.0903	0.0492	0.8521	0.7818	0.0810
88	0.0906	0.0490	0.8450	0.7847	0.0807
89	0.0908	0.0487	0.8397	0.7888	0.0802
90	0.0915	0.0478	0.8368	0.7967	0.0808


ENSO 3.4 Forecast Skill — Test
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

ENSO 3.4 Index — Test

ENSO 3.4 Scatter — Test